

Tensor Omics (TOX)

Analyze gene expression data and evolutionary relationships between genes

User's Guide

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1 Tensor Omics Methods and Theory

The Tensor Omics (TOX) tool implements Geometric Learning in a semantic vector space. We represent the data in its natural space, where each measurement is its own axis. We use model-free and minimum-parameter geometric methods to describe the data, find meaningful patterns, identify significant contributions of factors to dependent variables, analyse trajectories, and learn fields of vectors with intrinsic similarity.

We rely exclusively on pure linear algebra to characterize these structures, using distances, angles, projections, and linear mapping.

Tensor Omics differs from conventional geometric learning tensor decomposition in that it operates in empirically defined semantic vector spaces rather than abstract latent dimensions — a demonstration (See Figure 1):

(A) Medical omics vectors in semantic space. Each gene, protein or clinical patient measure is placed as a point in multi-dimensional space, with axes representing gene expression in a tissue or clinical patient measures. Arrows denote shift vectors from gene family centroids and enable the explainable detection of changes in gene expression from healthy control. Grid points support interpolation and smoothing. Patient time trajectories are shown in the lower left (healthy controls in green, mean control trajectory in fat light green, and a cancer trajectory in orange). The disease specific signals can directly be obtained by geometric trajectory comparison (violet time resolved shift arrows $\vec{s}_{t_i}$). Significant signals are detected empirically as outliers in the upper 5%-ile.

(B) Angle to the space diagonal as a measure of omics specificity: low angles indicate e.g. broad tissue expression, high angles signal strong tissue preference.

(C) Clock hand angle in the Relative Axes Plane (RAP), capturing changes in tissue preference with robust correction for the overall expression magnitude.

(D) Causal and explainable omics learning by selecting factors showing geometric similarities in predictive properties. Tensor Omics supports three independent strategies: radial subset growing until only background signal remains (orange), metadata-based selection (e.g. inflammatory response, green), and directional filtering by alignment with a user-defined direction of interest (DOI, e.g. median survival time after treatment or distance of patient trajectory to the healthy control, blue). Methods can be combined to define coherent subfields for geometric analysis.

(E) Explainable analysis of patient time trajectories (as shown in A). Changes in tissue preference (angles ϕ) relative to the anchor axis (AA). Inset plots show the distance to the healthy control trajectory (top) and the angle between consecutive shift vectors (bottom).

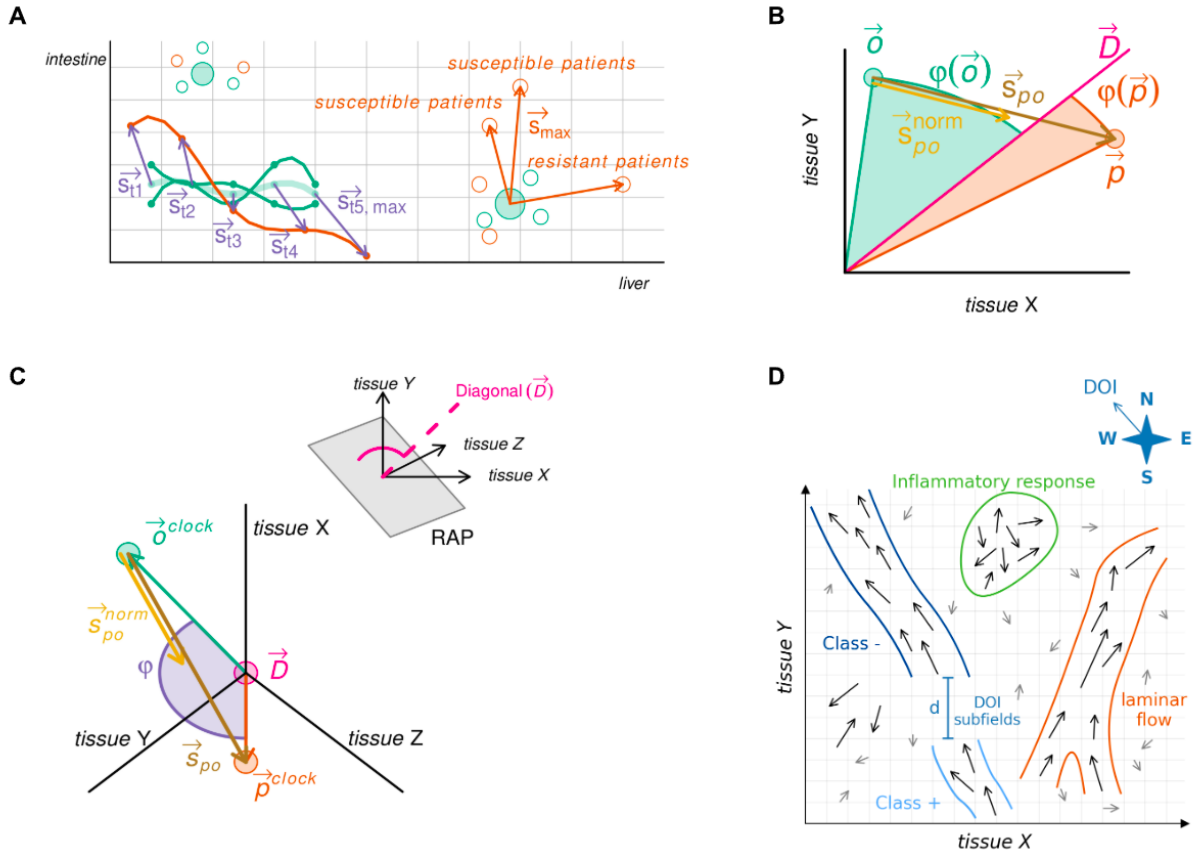
(F) Learned subfield (see D) characterization by emulating the aggregated effect of vectors on a virtual particle. Measures include signal strength (Frobenius norm $\|M\|_F$), disease subtyping as directional spread (rank), potential disease specific biomarkers as principal flow direction (first eigenvector of covariance matrix C), and disease specific changes in omics factors as rotation axis and magnitude (from the antisymmetric component $0.5 \times (C - C^T)$).

(G) Gradient-following kernel traversal to elucidate disease trajectory analysis in early, mid, and

late stages: subfield (see D) structure is decomposed via Singular-Value-Decomposition (SVD) into dominant axes (E1, E2 top left) which are smoothed into a single semantic axis (pink line), along which kernel profiles (pink bell top right) reveal flow, rotation, spread, and signal strength (as in F; lower left). Embedding of subfields for machine learning (lower right): The semantic axis is used to orient the subfield with the major flow of the vectors ($\oplus$) and split into five-percentile windows pushed along the semantic axis (dotted lines). Each section is represented by an aggregated vector of field strength (norm), directional spread (rank), principal direction (eigenvector), and rotational moment.

(H) The semantic axis (see G) is used to push a higher dimensional object through a lower dimensional space for visualization in the “Data-Vis” interactive graphical user interface (Tensor Omics package). This enables direct visualization of higher dimensions and can be used to show for example the developmental hourglass.

Panels A–H together illustrate how Tensor Omics unifies geometric learning, empirical explainability, and visualization within a single analytical framework.



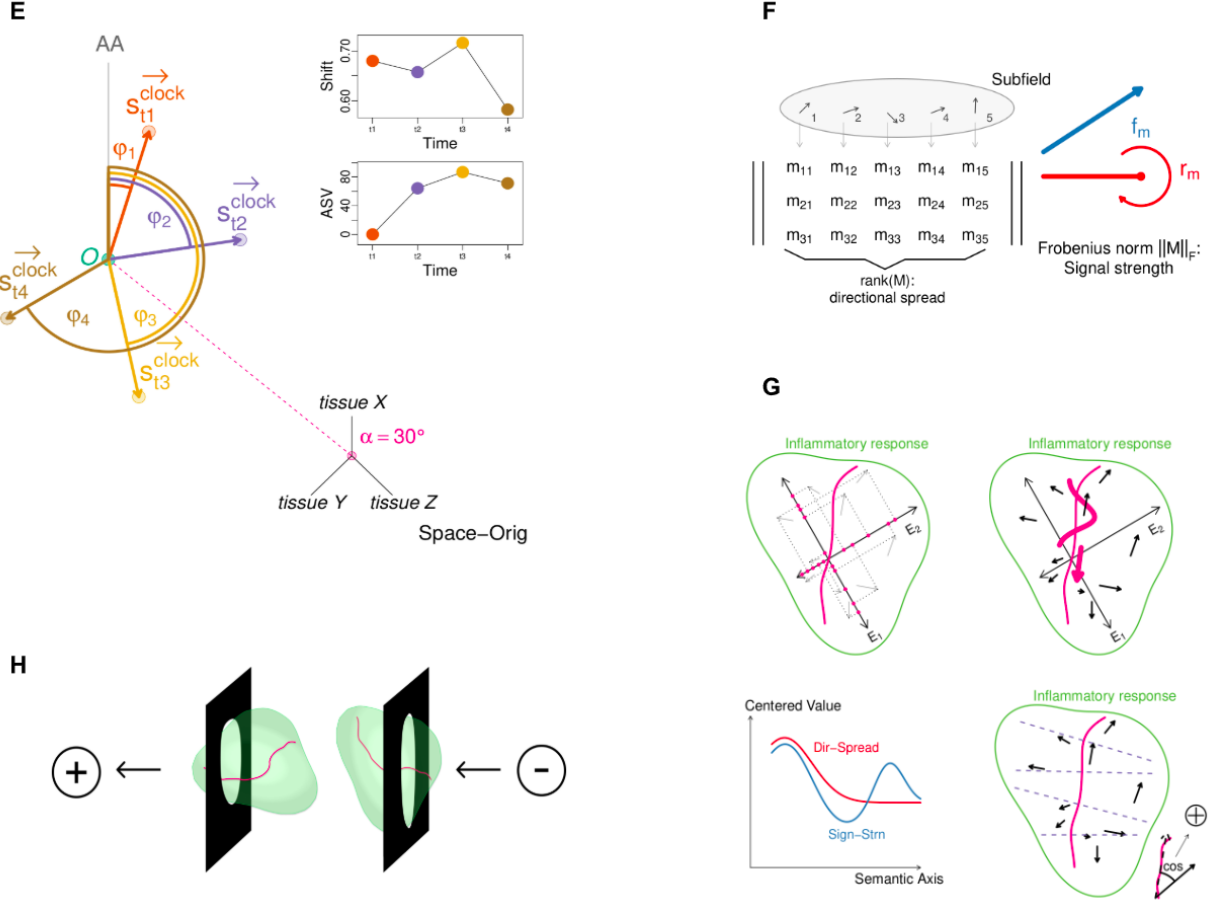


Figure 1: Overview of the Tensor Omics

See more details in the [Tensor Omics Methods file](#).

2 Error Handling

In case you encounter errors while using TOX, the following error codes may help you identify the issue. The error codes are grouped into categories based on their nature, such as I/O errors, format validation errors, memory errors, and internal errors.

Table 1: TOX Error Codes Reference

Error Code	Description
0	Success
1xx: I/O / File Reading Errors	
101	Could not open file.
102	Could not read magic number.
103	Could not read array type code.
104	Could not read number of dimensions.

Continued on next page

Table 1 – continued from previous page

Error Code	Description
105	Could not read array dimensions.
106	Could not read character length.
107	Could not read array data.
112	Could not write magic number
113	Could not write array type code
114	Could not write number of dimensions
115	Could not write dimensions
116	Could not write character length
117	Could not write array data
2xx: Format / Input Validation Errors	
200	Invalid format detected (e.g., magic mismatch or corrupt format).
201	Invalid input provided.
202	Empty input arrays provided (e.g., zero-length arrays).
203	Dimension mismatch detected (shape inconsistencies).
204	NaN or Inf found in input data where not allowed.
205	Unsupported data type encountered (e.g., complex types).
206	Array size mismatch
208	Array index out of bounds
209	Division by zero detected
3xx: Memory Errors	
301	Memory allocation failed.
302	Null pointer reference encountered.
5xxx: Fortran Runtime / Unit State Errors	
5002	Fortran runtime error: unit not open or not connected.
9xxx: Internal / Unknown Errors	
9001	Internal error: unexpected state or invariant violated.
9999	Unknown error (fallback for unexpected errors).

3 Implementation Notes

This section details the practical pipeline for using the TOX library to read and process tabular data. The workflow is consistent across Fortran, Python, and R, starting with reading a CSV file into a staging area of strings and then converting columns to their proper data types.

3.1 Fortran Pipeline

3.1.1 Overview

1. Data dependencies and relationships All arrays read have dependencies to each other. The gene ids array stores the ids, column number seven of the expression vectors belongs to entry number seven in the gene ids. Entry number nine in the gene to family mapping array, belongs to entry nine in the gene ids. These dependencies must be given at any time, therefore, only the provided

functions shall be used to modify arrays in any form. For error code details please check [error handling section](#).

2. Error handling reference (placeholder for now)
3. General usage principle (Placeholder for now)

3.1.2 Data Management

1. Data Reading and Processing

(a) Reading Data

• Read Gene IDs from TSV File

Reads gene identifiers from a TSV file containing expression data. Typically reads from the first column after skipping header rows. Call the `read_gene_ids_from_tsv_file` subroutine with the following arguments:

Input arguments:

- `filename` `character(len=*)`: Path to the TSV file
- `n_header_rows` `integer(int32)`: Number of header rows to skip (typically 1)
- `gene_col` `integer(int32)`: Column index containing gene IDs (typically 1)

Output arguments:

- `gene_ids` `character(len=*)`: Pre-allocated array for gene identifiers
- `ierr` `integer(int32)`: Error code

Listing 1: Read Gene IDs from TSV File

```
character(len=256) :: gene_ids(n_genes)
! Pre-allocate with expected size

integer(int32) :: ierr

call read_gene_ids_from_tsv_file('expression_data.tsv', &
                                gene_ids, 1, 1, ierr)
```

• Read Expression Vectors

Reads expression data from one or more TSV/CSV files and populates a pre-allocated 2D array (samples x genes). Uses hashmap for efficient gene ID matching and validates for finite values. Call the `read_expression_vectors` subroutine with the following arguments:

Input arguments:

- `file_list` `character(len=*)`: Array of file paths to read from
- `gene_ids` `character(len=*)`: Reference gene IDs for validation and matching
- `expression_vectors` `real(real64)`: Pre-allocated 2D array for expression data
- `n_header_rows` `integer(int32)`: Number of header rows to skip
- `gene_col` `integer(int32)`: Column index containing gene IDs
- `value_cols` `integer(int32)`: Array of column indices containing expression values
- `start_row` `integer(int32)`: Starting row index in `expression_vectors` array
- `delimiter` `character(len=1)`, optional: Column delimiter (default: tab)

Output arguments:

- `expression_vectors` `real(real64)`: Pre-allocated 2D array for expression data populated with data
- `ierr` `integer(int32)`: Error code

Listing 2: Read Expression Vectors

```

! Pre-allocate arrays
real(real64) :: expr_data(n_samples, n_genes)

! Read TSV file with tab delimiter (default)
call read_expression_vectors(files, gene_ids, expr_data, 1, 1, &
                             value_cols, 1, ierr)

! Read CSV file with comma delimiter
call read_expression_vectors(files, gene_ids, expr_data, 1, 1, &
                             value_cols, 1, ierr, ',')

```

• Read OrthoFinder Family Mapping

Reads gene family assignments from OrthoFinder output TSV file and creates mapping arrays between genes and families. The `gene_to_fam` array contains for each gene the index of it's family. Uses hashmap for efficient gene lookup. Call the `read_orthofinder_file` subroutine with the following arguments:

Input arguments:

- `filename` character(len=*): Path to OrthoFinder TSV file
- `gene_ids` character(len=*): Pre-allocated array of gene IDs to match

Output arguments:

- `family_ids` character(len=*): Pre-allocated array for family identifiers
- `gene_to_fam` integer(int32): Pre-allocated array for gene to family mapping
- `ierr` integer(int32): Error code

Listing 3: Read OrthoFinder Family Mapping

```

character(len=256) :: family_ids(n_families)
! Pre-allocate with expected family count

integer(int32) :: gene_to_fam(n_genes)
! Pre-allocate with gene count

integer(int32) :: ierr

call read_orthofinder_file('Orthogroups.tsv', gene_ids, &
                           family_ids, gene_to_fam, ierr)

```

Important Implementation Details:

- All output arrays must be pre-allocated with the correct dimensions and sizes
- Uses hashmap data structure for O(1) gene ID lookups
- Validates for finite values (rejects NaN and Inf) in expression data
- Handles both TSV (tab-separated) and CSV (comma-separated) formats, `gene_to_fam` array uses 1-based indexing for family assignments (0 indicates unassigned)

Typical Data Reading Workflow:

Listing 4: Complete Data Reading Workflow

```

! 1. Pre-allocate arrays

! 2. Read gene IDs from expression file
call read_gene_ids_from_tsv_file('expression.tsv', gene_ids, &
                                 1, 1, ierr)

! 3. Read expression data (columns 2-68 contain sample data)

```



```

character(len=256) :: files(1) = ['expression.tsv']

call read_expression_vectors(files, gene_ids, expr_data, 1, 1, &
                             samples, 1, ierr)

! 4. Read family mapping from OrthoFinder
call read_orthofinder_file('Orthogroups.tsv', gene_ids, &
                           family_ids, gene_to_fam, ierr)

```

2. Data Filtering and Validation

• Get Unassigned Genes Mask

Creates a logical mask identifying genes that have no family assignment (gene_to_fam value = 0). Call the `get_unassigned_mask` subroutine with the following arguments:

Input arguments:

- gene_to_fam integer(int32): Gene to family mapping array

Output arguments:

- mask logical: Logical mask array (same size as gene_to_fam)
- n_genes_kept integer(int32): Number of genes with family assignments
- ierr integer(int32): Error code

Listing 5: Create Unassigned Genes Mask

```

logical :: unassigned_mask(n_genes)
integer(int32) :: n_kept, ierr

call get_unassigned_mask(gene_to_fam, unassigned_mask, n_kept)
! unassigned_mask(i) = .true. if gene i has family assignment
! n_kept = count of genes with family assignments

```

• Apply Unassigned Mask

Filters out genes without family assignments from all data arrays (gene_ids, expression_vectors, gene_to_fam). Arrays are reallocated to the filtered size. Call the `apply_unassigned_mask` subroutine with the following arguments:

Input arguments:

- mask logical: Logical mask from `get_unassigned_mask`
- n_genes_kept integer(int32): Number of genes to keep (from `get_unassigned_mask`)

Input/Output arguments:

- gene_ids character(len=*), allocatable: Gene IDs array (reallocated)
- expression_vectors real(real64), allocatable: Expression data matrix (reallocated)
- gene_to_fam integer(int32), allocatable: Gene to family mapping (reallocated)
- ierr integer(int32): Error code

Listing 6: Filter Unassigned Genes

```

character(len=256), allocatable :: gene_ids(n_genes)
real(real64), allocatable :: expr_data(n_samples, n_genes)
integer(int32), allocatable :: gene_to_fam(n_genes)
logical :: unassigned_mask(n_genes)
integer(int32) :: n_kept, ierr

! First create the mask
call get_unassigned_mask(gene_to_fam, unassigned_mask, n_kept)

```

```

! Then apply it to filter all arrays
call apply_unassigned_mask(gene_ids, expr_data, gene_to_fam, &
                           unassigned_mask, n_kept, ierr)
! Arrays are now reallocated to size n_kept

```

- **Validate Data Structure**

Performs comprehensive validation of data dimensions and consistency across all arrays. Call the `validate_data_structure` subroutine with the following arguments:

Input arguments:

- `n_genes` integer(int32): Number of genes
- `n_families` integer(int32): Number of gene families
- `n_samples` integer(int32): Number of samples
- `gene_ids` character(len=*): Gene identifier array
- `family_ids` character(len=*): Family identifier array
- `gene_to_family` integer(int32): Gene to family mapping
- `expression` real(real64): Expression data matrix
- `centroids` real(real64): Family centroids matrix
- `shift_vectors` real(real64): Shift vectors matrix
- `ierr` integer(int32): Error code

Listing 7: Validate Data Structure

```

call validate_data_structure(n_genes, n_families, n_samples, &
                           gene_ids, family_ids, gene_to_fam, &
                           expr_data, centroids, shift_vectors, &
                           ierr)

```

- **Validate Expression Data**

Validates expression data matrix for NaN, Inf, and optionally negative values. Call the `validate_expression_data` subroutine with the following arguments:

Input arguments:

- `expression` real(real64): Expression data matrix to validate
- `allow_negative` logical: Whether to allow negative expression values
- `ierr` integer(int32): Error code

Listing 8: Validate Expression Data

```

! Validate expression data (disallow negative values)
call validate_expression_data(expr_data, .false., ierr)

! Validate allowing negative values (e.g., for log-fold changes)
call validate_expression_data(expr_data, .true., ierr)

```

- **Validate Empty Strings**

Validates that a string array contains no empty or whitespace-only entries. Call the `validate_empty_strings` subroutine with the following arguments:

Input arguments:

- `string_array` character(len=*): String array to validate
- `array_name` character(len=*): Name for error messages

Output arguments:

- `ierr` integer(int32): Error code

Listing 9: Validate String Array

```
call validate_empty_strings(gene_ids, "Gene IDs", ierr)
call validate_empty_strings(family_ids, "Family IDs", ierr)
```

- **Validate Gene to Family Mapping**

Validates that all gene to family indices are within valid range (1 to n_families). Call the `validate_gene_to_family_mapping` subroutine with the following arguments:

Input arguments:

- `gene_to_family` integer(int32): Gene to family mapping array
- `n_families` integer(int32): Number of valid families

Output arguments:

- `ierr` integer(int32): Error code

Listing 10: Validate Gene-Family Mapping

```
call validate_gene_to_family_mapping(gene_to_fam, n_families, ierr)
```

- **Validate String Array Uniqueness**

Validates that all entries in a string array are unique (no duplicates). Uses a hashtet implementation internally for O(n) runtime. Call the `validate_string_array_uniqueness` subroutine with the following arguments:

Input arguments:

- `string_array` character(len=*): String array to check for duplicates

Output arguments:

- `ierr` integer(int32): Error code

Listing 11: Validate Unique Strings

```
call validate_string_array_uniqueness(gene_ids, ierr)
call validate_string_array_uniqueness(family_ids, ierr)
```

3. Data Accessors

- **Get Gene Index**

Finds the array index for a given gene identifier. Call the `get_gene_index` function with the following arguments:

Input arguments:

- `gene_ids` character(len=*): Array of gene identifiers
- `gene_id` character(len=*): Gene ID to find

Return value:

- `index` integer(int32): Array index (1-based) or 0 if not found

Listing 12: Find Gene Index

```
integer(int32) :: gene_idx

gene_idx = get_gene_index(gene_ids, 'NP_001000001.1')
if (gene_idx > 0) then
    ! Gene found at position gene_idx
else
    ! Gene not found
end if
```

- **Get Family Index**

Finds the array index for a given family identifier. Call the `get_family_index` function with the following arguments:

Input arguments:

- family_ids character(len=*): Array of family identifiers
- family_id character(len=*): Family ID to find

Return value:

- index integer(int32): Array index (1-based) or 0 if not found

Listing 13: Find Family Index

```
integer(int32) :: fam_idx

fam_idx = get_family_index(family_ids, 'OG0000001')
```

- **Get Family for Gene Index**

Retrieves the family index for a given gene index. Call the `get_family_for_gene_index` subroutine with the following arguments:

Input arguments:

- gene_index integer(int32): Gene array index (1-based)
- gene_to_family integer(int32): Gene to family mapping array

Output arguments:

- family_index integer(int32): Family array index (1-based)
- ierr integer(int32): Error code

Listing 14: Get Family for Gene

```
integer(int32) :: gene_idx, fam_idx, ierr

gene_idx = get_gene_index(gene_ids, 'NP_001000001.1')
call get_family_for_gene_index(gene_idx, gene_to_fam, fam_idx, ierr)
```

- **Get Family Centroid**

Retrieves the centroid expression profile for a given family. Call the `get_family_centroid` subroutine with the following arguments:

Input arguments:

- family_index integer(int32): Family array index (1-based)
- family_centroids real(real64): Family centroids matrix

Output arguments:

- ierr integer(int32): Error code
- centroid_vector real(real64): Centroid expression profile

Listing 15: Get Family Centroid

```
real(real64) :: centroid_vector(n_samples)
integer(int32) :: fam_idx, ierr

fam_idx = get_family_index(family_ids, 'OG0000001')
call get_family_centroid(fam_idx, centroids, centroid_vector, ierr)
```

Complete Validation Workflow:

Listing 16: Complete Data Validation Workflow

```
integer(int32) :: ierr

! 1. Filter unassigned genes
logical :: unassigned_mask(n_genes)
```

```

integer(int32) :: n_kept
call get_unassigned_mask(gene_to_fam, unassigned_mask, n_kept)
call apply_unassigned_mask(gene_ids, expr_data, gene_to_fam, &
unassigned_mask, n_kept, ierr)

! 2. Validate individual components
call validate_empty_strings(gene_ids, "Gene IDs", ierr)
call validate_empty_strings(family_ids, "Family IDs", ierr)
call validate_string_array_uniqueness(gene_ids, ierr)
call validate_string_array_uniqueness(family_ids, ierr)
call validate_gene_to_family_mapping(gene_to_fam, n_families, ierr)
call validate_expression_data(expr_data, .false., ierr)

! 3. Comprehensive structure validation
call validate_data_structure(n_genes, n_families, n_samples, gene_ids,
&
family_ids, gene_to_fam, expr_data, &
centroids, shift_vectors, ierr)

! 4. Use accessors for data retrieval
integer(int32) :: gene_idx, fam_idx
real(real64) :: centroid(n_samples)

gene_idx = get_gene_index(gene_ids, 'NP_001000001.1')
call get_family_for_gene_index(gene_idx, gene_to_fam, fam_idx, ierr)
call get_family_centroid(fam_idx, centroids, centroid, ierr)

```

3.1.3 Data Integration

The Fortran module provides comprehensive routines for data integration, based on Jensen-Shannon Divergence. This is very useful for cross-study data.

Key Features:

- **Calculate Jensen-Shannon Compatibility Score:** Score to evaluate how compatible two studies are.
- **Perform JSD Permutation Test:** Random sampling of the input data and recalculation of the JS score to test stability of the compatibility.
- **Family-Wise Score Contribution:** Calculates the impact of selected families on the calculated score.

1. Pool Mean Expression Values (Allocated)

Pools per-gene mean expression values from two studies, sorts them, and computes reference points x_{star} using empirical quantiles.

Arguments:

- `n_genes_S1` `integer(int32)`: number of genes in study S1
- `mean_S1` `real(real64)`, `dimension(n_genes_S1)`: per-gene mean expression values for study S1
- `n_genes_S2` `integer(int32)`: number of genes in study S2
- `mean_S2` `real(real64)`, `dimension(n_genes_S2)`: per-gene mean expression values for study S2
- `n_points` `integer(int32)`: number of reference points to define
- `n_pool` `integer(int32)`: number of included (non-NaN) pooled mean-expression values

- `x_star` `real`(real64), `dimension`(`n_points`): mean-expression reference points
- `ierr` `integer`(int32): error code

Listing 17: Pool Means (Allocated)

```
call pool_means_alloc(n_genes_S1, mean_S1, n_genes_S2, mean_S2, n_points
, n_pool, x_star, ierr)
```

2. Pool Mean Expression Values

Pools per-gene mean expression values across studies, removes trailing NaNs (which are sorted to the end), and computes empirical-quantile reference points `x_star`.

Arguments:

- `pooled_means` `real`(real64), `dimension`(`pool_size`): pooled mean-expression values
- `pooled_means_perm` `integer`(int32), `dimension`(`pool_size`): permutation vector that sorts `pooled_means`
- `pool_size` `integer`(int32): number of pooled means (typically `n_genes_S1` + `n_genes_S2`)
- `n_points` `integer`(int32): number of reference points to compute
- `n_pool` `integer`(int32): number of included (non-NaN) pooled means
- `x_star` `real`(real64), `dimension`(`n_points`): mean-expression reference points
- `ierr` `integer`(int32): error code

Listing 18: Pool Means

```
call pool_means(pooled_means, pooled_means_perm, pool_size, n_points,
n_pool, x_star, ierr)
```

3. Calculate Neighborhood Size

Determines the number of neighbors to use for `construct_neighborhoods`, based on the pooled mean-expression distribution, the number of reference points, and optional user-defined limits.

Arguments:

- `n_pool` `integer`(int32): total number of pooled mean-expression values across both studies
- `n_points` `integer`(int32): number of reference points
- `n_genes_S` `integer`(int32): number of genes in the current study
- `mean_S` `real`(real64), `dimension`(`n_genes_S`): per-gene mean expression values
- `desired_size` `integer`(int32), `optional`: optional upper limit for neighborhood size (default: 1000)
- `n_neighbors` `integer`(int32): calculated neighborhood size

Listing 19: Calculate Neighborhood Size

```
n_neighbors = calc_neighborhood_size(n_pool, n_points, n_genes_S, mean_S
, desired_size)
```

4. Construct Neighborhoods (Allocated)

Constructs neighborhood-based residual sets (kNN) by allocating temporary distance and permutation arrays, then delegating to `construct_neighborhoods`.

Arguments:

- `n_points` `integer`(int32): number of reference points
- `x_star` `real`(real64), `dimension`(`n_points`): mean-expression reference points
- `n_genes_S` `integer`(int32): number of genes in the study
- `mean_S` `real`(real64), `dimension`(`n_genes_S`): per-gene mean expression values
- `n_reps_S` `integer`(int32): number of biological replicates
- `resid_S` `real`(real64), `dimension`(`n_reps_S`, `n_genes_S`): matrix of signed residuals

- `neighborhood_residuals` `real`(real64), `dimension`(`n_reps_S`, `n_neighbors`, `n_points`): residual vectors for each neighborhood
- `neighborhood_indices` `integer`(int32), `dimension`(`n_neighbors`, `n_points`): indices of selected neighborhood genes
- `n_neighbors` `integer`(int32): number of neighbors (compute via `calc_neighborhood_size`)
- `ierr` `integer`(int32): error code

Listing 20: Construct Neighborhoods (Allocated)

```
call construct_neighborhoods_alloc(n_points, x_star, n_genes_S, mean_S,
    n_reps_S, resid_S, &
    neighborhood_residuals, neighborhood_indices, n_neighbors, ierr)
```

5. Construct Neighborhoods

Constructs neighborhood-based residual sets (kNN) for each reference point by computing distances to `x_star`, sorting genes by distance, and selecting the `n_neighbors` closest non-NaN genes.

Arguments:

- `n_points` `integer`(int32): number of reference points
- `x_star` `real`(real64), `dimension`(`n_points`): mean-expression reference points
- `n_genes_S` `integer`(int32): number of genes in the study
- `mean_S` `real`(real64), `dimension`(`n_genes_S`): per-gene mean expression values
- `n_reps_S` `integer`(int32): number of biological replicates
- `resid_S` `real`(real64), `dimension`(`n_reps_S`, `n_genes_S`): matrix of signed residuals
- `tmp_distances` `real`(real64), `dimension`(`n_genes_S`): work array for distances
- `tmp_distances_perm` `integer`(int32), `dimension`(`n_genes_S`): permutation vector for sorting distances
- `neighborhood_residuals` `real`(real64), `dimension`(`n_reps_S`, `n_neighbors`, `n_points`): residual vectors for each neighborhood
- `neighborhood_indices` `integer`(int32), `dimension`(`n_neighbors`, `n_points`): indices of selected neighborhood genes
- `n_neighbors` `integer`(int32): number of neighbors (compute via `calc_neighborhood_size`)
- `ierr` `integer`(int32): error code

Listing 21: Construct Neighborhoods

```
call construct_neighborhoods(n_points, x_star, n_genes_S, mean_S,
    n_reps_S, resid_S, &
    tmp_distances, tmp_distances_perm, neighborhood_residuals,
    neighborhood_indices, &
    n_neighbors, ierr)
```

6. Determine Shared Residual Range

Computes the shared residual range $[-R, R]$ for the absolute residuals pooled from studies S1 and S2.

Arguments:

- `abs_residual_pool` `real`(real64), `dimension`(`pool_size`): absolute residual values of the concatenated S1,S2 residuals
- `abs_residual_pool_perm` `integer`(int32), `dimension`(`pool_size`): permutation vector that sorts `abs_residual_pool`
- `pool_size` `integer`(int32): size of the residual pool
- `shared_residual_range` `real`(real64): computed residual range R

- `ierr` `integer`(int32): error code
- `residual_range_quantile` `real`(real64), `optional`: quantile for determining the residual range (default: 95.0)

Listing 22: Shared Residual Range

```
call determine_shared_residual_range(abs_residual_pool,
    abs_residual_pool_perm, pool_size, shared_residual_range, ierr,
    residual_range_quantile)
```

7. Determine Shared Residual Range (Allocated)

Computes the shared residual range $[-R, R]$ for two studies by allocating and flattening the residual pool from both studies before computing the percentile-based range.

Arguments:

- `neighborhood_residuals_S1` `real`(real64), `dimension`(`n_reps_S1`, `n_neighbors`, `n_points`): neighborhood residuals for study 1, NaN allowed
- `neighborhood_residuals_S2` `real`(real64), `dimension`(`n_reps_S2`, `n_neighbors`, `n_points`): neighborhood residuals for study 2, NaN allowed
- `n_reps_S1` `integer`(int32): number of replicates in study 1
- `n_reps_S2` `integer`(int32): number of replicates in study 2
- `n_neighbors` `integer`(int32): number of neighbors
- `n_points` `integer`(int32): number of reference points
- `shared_residual_range` `real`(real64): computed residual range R
- `ierr` `integer`(int32): error code
- `residual_range_quantile` `real`(real64), `optional`: quantile for determining the residual range (default: 95.0)

Listing 23: Allocated Shared Residual Range

```
call determine_shared_residual_range_alloc(neighborhood_residuals_S1,
    neighborhood_residuals_S2, &
    n_reps_S1, n_reps_S2, n_neighbors, n_points, shared_residual_range,
    ierr, residual_range_quantile)
```

8. Build Residual Histograms

Summarizes neighborhood residuals into absolute histogram counts and probability mass functions `pmf(residual, bin)` for each reference point.

Arguments:

- `neighborhood_residuals` `real`(real64), `dimension`(`n_reps`, `n_neighbors`, `n_points`): neighborhood residuals for a study, NaN allowed
- `n_reps` `integer`(int32): number of replicates
- `n_neighbors` `integer`(int32): number of neighbors
- `n_points` `integer`(int32): number of reference points
- `shared_residual_range` `real`(real64): residual range R from `determine_shared_residual_range_alloc`
- `n_bins` `integer`(int32): number of histogram bins in $[-R, R]$
- `counts` `integer`(int32), `dimension`(`n_points`, `n_bins`): absolute histogram counts
- `pmf` `real`(real64), `dimension`(`n_points`, `n_bins`): normalized histogram counts
- `included_n_reps` `integer`(int32), `dimension`(`n_points`): number of non-NaN residuals per reference point
- `ierr` `integer`(int32): error code

- `neighbor_mask` `logical`, `dimension`(`n_neighbors`, `n_points`), `optional`: mask to exclude specific neighbors

Listing 24: Residual Histograms

```
call build_residual_histograms(neighborhood_residuals, n_reps,
    n_neighbors, n_points, &
    shared_residual_range, n_bins, counts, pmf, included_n_reps, ierr,
    neighbor_mask)
```

9. Compute Divergence per Reference Point

Computes the Jensen–Shannon divergence per reference point using the probability mass functions from two studies.

Arguments:

- `pmf_S1` `real`(`real64`), `dimension`(`n_points`, `n_bins`): normalized histogram counts for study 1
- `pmf_S2` `real`(`real64`), `dimension`(`n_points`, `n_bins`): normalized histogram counts for study 2
- `n_points` `integer`(`int32`): number of reference points
- `n_bins` `integer`(`int32`): number of histogram bins in $[-R, R]$
- `js_divergences` `real`(`real64`), `dimension`(`n_points`): Jensen–Shannon divergence per reference point
- `ierr` `integer`(`int32`): error code

Listing 25: Per-Reference-Point Divergence

```
call compute_divergence_per_reference_point(pmf_S1, pmf_S2, n_points,
    n_bins, js_divergences, ierr)
```

10. Compute Weighted Global Divergence

Computes the global weighted Jensen–Shannon divergence using per-reference-point divergences and the number of included residuals from both studies.

Arguments:

- `js_divergences` `real`(`real64`), `dimension`(`n_points`): Jensen–Shannon divergence per reference point
- `n_points` `integer`(`int32`): number of reference points
- `included_n_reps_S1` `integer`(`int32`), `dimension`(`n_points`): non-NaN residual counts for study 1
- `included_n_reps_S2` `integer`(`int32`), `dimension`(`n_points`): non-NaN residual counts for study 2
- `global_js_divergence` `real`(`real64`): weighted global Jensen–Shannon divergence
- `weights` `real`(`real64`), `dimension`(`n_points`): weights used in the global divergence calculation
- `ierr` `integer`(`int32`): error code

Listing 26: Weighted Global Divergence

```
call compute_weighted_global_divergence(js_divergences, n_points,
    included_n_reps_S1, &
    included_n_reps_S2, global_js_divergence, weights, ierr)
```

11. GJCT Permutation Test (Allocated)

Estimates how likely the observed global Jensen–Shannon divergence is to occur by chance under the null hypothesis that both studies are exchangeable. Allocates all required working arrays and delegates to `gjct_permutation_test`.

Arguments:

- `neighborhood_residuals_S1` `real`(real64), `dimension`(`n_reps_S1`, `n_neighbors`, `n_points`): neighborhood residuals for study 1, NaN allowed
- `neighborhood_residuals_S2` `real`(real64), `dimension`(`n_reps_S2`, `n_neighbors`, `n_points`): neighborhood residuals for study 2, NaN allowed
- `n_reps_S1` `integer`(int32): number of replicates in study 1
- `n_reps_S2` `integer`(int32): number of replicates in study 2
- `n_neighbors` `integer`(int32): number of neighbors
- `n_points` `integer`(int32): number of reference points
- `global_jsd_observed` `real`(real64): observed global JSD value
- `n_bins` `integer`(int32): number of histogram bins
- `shared_residual_range` `real`(real64): shared residual range from `determine_shared_residual_range`
- `n_permutations` `integer`(int32): number of permutations to perform
- `jsd_null` `real`(real64), `dimension`(`n_permutations`): global JSD values under the null hypothesis
- `p_value` `real`(real64): empirical p-value
- `ierr` `integer`(int32): error code
- `random_seed` `integer`(int32), `optional`: seed for shuffling
- `neighbor_mask_S1` `logical`, `dimension`(`n_neighbors`, `n_points`), `optional`: mask to exclude neighbors in study 1
- `neighbor_mask_S2` `logical`, `dimension`(`n_neighbors`, `n_points`), `optional`: mask to exclude neighbors in study 2

Listing 27: GJCT Permutation Test (Allocated)

```
call gjct_permutation_test_alloc(neighborhood_residuals_S1,
    neighborhood_residuals_S2, &
    n_reps_S1, n_reps_S2, n_neighbors, n_points, global_jsd_observed,
    n_bins, &
    shared_residual_range, n_permutations, jsd_null, p_value, ierr,
    random_seed, &
    neighbor_mask_S1, neighbor_mask_S2)
```

12. GJCT Permutation Test

Estimates how likely the observed global Jensen–Shannon divergence is to occur by chance under the null hypothesis that both studies are exchangeable. Reuses preallocated workspace arrays and performs all permutations in-place.

Arguments:

- `neighborhood_residuals_S1_copy` `real`(real64), `dimension`(`n_reps_S1`, `n_neighbors`, `n_points`), `intent`(inout): copy of residuals for study 1, shuffled in-place
- `neighborhood_residuals_S2_copy` `real`(real64), `dimension`(`n_reps_S2`, `n_neighbors`, `n_points`), `intent`(inout): copy of residuals for study 2, shuffled in-place
- `n_reps_S1` `integer`(int32): number of replicates in study 1
- `n_reps_S2` `integer`(int32): number of replicates in study 2
- `n_neighbors` `integer`(int32): number of neighbors
- `n_points` `integer`(int32): number of reference points

- `global_jsd_observed` `real(real64)`: observed global JSD value
- `n_bins` `integer(int32)`: number of histogram bins
- `shared_residual_range` `real(real64)`: shared residual range from `determine_shared_residual_range`
- `n_permutations` `integer(int32)`: number of permutations to perform
- `jsd_null` `real(real64)`, `dimension(n_permutations)`: global JSD values under the null hypothesis
- `p_value` `real(real64)`: empirical p-value
- `tmp_pool` `real(real64)`, `dimension(n_reps_S1 + n_reps_S2, n_neighbors)`: working array for shuffling pooled residuals
- `tmp_pmf_S1` `real(real64)`, `dimension(n_points, n_bins)`: PMF workspace for study 1
- `tmp_pmf_S2` `real(real64)`, `dimension(n_points, n_bins)`: PMF workspace for study 2
- `tmp_counts` `integer(int32)`, `dimension(n_points, n_bins)`: histogram count workspace
- `tmp_included_n_reps_S1` `integer(int32)`, `dimension(n_points)`: workspace for included counts (study 1)
- `tmp_included_n_reps_S2` `integer(int32)`, `dimension(n_points)`: workspace for included counts (study 2)
- `tmp_js_divergences` `real(real64)`, `dimension(n_points)`: workspace for per-point divergences
- `tmp_weights` `real(real64)`, `dimension(n_points)`: workspace for weights
- `ierr` `integer(int32)`: error code
- `random_seed` `integer(int32)`, `optional`: seed for shuffling
- `neighbor_mask_S1` `logical`, `dimension(n_neighbors, n_points)`, `optional`: mask to exclude neighbors in study 1
- `neighbor_mask_S2` `logical`, `dimension(n_neighbors, n_points)`, `optional`: mask to exclude neighbors in study 2

Listing 28: GJCT Permutation Test

```
call gjct_permutation_test(neighborhood_residuals_S1_copy,
    neighborhood_residuals_S2_copy, &
    n_reps_S1, n_reps_S2, n_neighbors, n_points, global_jsd_observed,
    n_bins, &
    shared_residual_range, n_permutations, jsd_null, p_value, tmp_pool,
    tmp_pmf_S1, &
    tmp_pmf_S2, tmp_counts, tmp_included_n_reps_S1,
    tmp_included_n_reps_S2, &
    tmp_js_divergences, tmp_weights, ierr, random_seed, neighbor_mask_S1,
    neighbor_mask_S2)
```

13. FJCT Compute JSD (Allocated)

Computes the family-level compatibility score `global_js_divergence` between two studies for a single gene family by reusing the same conditioning-on-mean-expression pipeline as the global gJCT, but restricting residual samples to genes belonging to the specified family.

Arguments:

- `family_idx` `integer(int32)`: index of the family to analyze
- `gene_to_family_S1` `integer(int32)`, `dimension(n_genes_S1)`: mapping of genes to families in study 1
- `gene_to_family_S2` `integer(int32)`, `dimension(n_genes_S2)`: mapping of genes to families in study 2
- `n_genes_S1` `integer(int32)`: number of genes in study 1

- `n_genes_S2` `integer`(int32): number of genes in study 2
- `n_reps_S1` `integer`(int32): number of replicates in study 1
- `n_reps_S2` `integer`(int32): number of replicates in study 2
- `n_neighbors` `integer`(int32): number of neighbors
- `n_points` `integer`(int32): number of reference points
- `neighborhood_residuals_S1` `real`(real64), `dimension`(`n_reps_S1`, `n_neighbors`, `n_points`): residuals for study 1, NaN allowed
- `neighborhood_residuals_S2` `real`(real64), `dimension`(`n_reps_S2`, `n_neighbors`, `n_points`): residuals for study 2, NaN allowed
- `neighborhood_genes_S1` `integer`(int32), `dimension`(`n_reps_S1`, `n_neighbors`, `n_points`): gene indices for study 1
- `neighborhood_genes_S2` `integer`(int32), `dimension`(`n_reps_S2`, `n_neighbors`, `n_points`): gene indices for study 2
- `n_bins` `integer`(int32): number of histogram bins
- `shared_residual_range` `real`(real64): shared residual range from `determine_shared_residual_range`
- `js_divergences` `real`(real64), `dimension`(`n_points`): Jensen–Shannon divergence per reference point
- `included_n_reps_S1` `integer`(int32), `dimension`(`n_points`): included non-NaN residuals for study 1
- `included_n_reps_S2` `integer`(int32), `dimension`(`n_points`): included non-NaN residuals for study 2
- `total_included_n_reps` `integer`(int32): total included residuals across both studies
- `global_js_divergence` `real`(real64): weighted global Jensen–Shannon divergence
- `weights` `real`(real64), `dimension`(`n_points`): weights used in the global divergence calculation
- `ierr` `integer`(int32): error code

Listing 29: Family-Level JSD (Allocated)

```
call fjct_compute_jsd_alloc(family_idx, gene_to_family_S1,
    gene_to_family_S2, &
    n_genes_S1, n_genes_S2, neighborhood_residuals_S1,
    neighborhood_residuals_S2, &
    neighborhood_genes_S1, neighborhood_genes_S2, n_reps_S1, n_reps_S2,
    n_neighbors, &
    n_points, n_bins, shared_residual_range, js_divergences,
    included_n_reps_S1, &
    included_n_reps_S2, total_included_n_reps, global_js_divergence,
    weights, ierr)
```

14. FJCT Compute JSD

Computes the family-level compatibility score `global_js_divergence` between two studies for a single gene family, reusing preallocated workspace arrays for efficiency.

Arguments:

- `family_idx` `integer`(int32): index of the family to analyze
- `gene_to_family_S1` `integer`(int32), `dimension`(`n_genes_S1`): mapping of genes to families in study 1
- `gene_to_family_S2` `integer`(int32), `dimension`(`n_genes_S2`): mapping of genes to families in study 2

- `n_genes_S1` `integer`(int32): number of genes in study 1
- `n_genes_S2` `integer`(int32): number of genes in study 2
- `n_reps_S1` `integer`(int32): number of replicates in study 1
- `n_reps_S2` `integer`(int32): number of replicates in study 2
- `n_neighbors` `integer`(int32): number of neighbors
- `n_points` `integer`(int32): number of reference points
- `neighborhood_residuals_S1` `real`(real64), `dimension`(`n_reps_S1`, `n_neighbors`, `n_points`): residuals for study 1, NaN allowed
- `neighborhood_residuals_S2` `real`(real64), `dimension`(`n_reps_S2`, `n_neighbors`, `n_points`): residuals for study 2, NaN allowed
- `neighborhood_genes_S1` `integer`(int32), `dimension`(`n_reps_S1`, `n_neighbors`, `n_points`): gene indices for study 1
- `neighborhood_genes_S2` `integer`(int32), `dimension`(`n_reps_S2`, `n_neighbors`, `n_points`): gene indices for study 2
- `n_bins` `integer`(int32): number of histogram bins
- `shared_residual_range` `real`(real64): shared residual range from `determine_shared_residual_range`
- `js_divergences` `real`(real64), `dimension`(`n_points`): Jensen–Shannon divergence per reference point
- `included_n_reps_S1` `integer`(int32), `dimension`(`n_points`): included non-NaN residuals for study 1
- `included_n_reps_S2` `integer`(int32), `dimension`(`n_points`): included non-NaN residuals for study 2
- `total_included_n_reps` `integer`(int32): total included residuals across both studies
- `global_js_divergence` `real`(real64): weighted global Jensen–Shannon divergence
- `weights` `real`(real64), `dimension`(`n_points`): weights used in the global divergence calculation
- `tmp_counts` `integer`(int32), `dimension`(`n_points`, `n_bins`): working array for histogram construction
- `pmf_S1` `real`(real64), `dimension`(`n_points`, `n_bins`): PMF for study 1
- `pmf_S2` `real`(real64), `dimension`(`n_points`, `n_bins`): PMF for study 2
- `ierr` `integer`(int32): error code

Listing 30: Family-Level JSD

```
call fjct_compute_jsd(family_idx, gene_to_family_S1, gene_to_family_S2,
    &
    n_genes_S1, n_genes_S2, neighborhood_residuals_S1,
    neighborhood_residuals_S2, &
    neighborhood_genes_S1, neighborhood_genes_S2, n_reps_S1, n_reps_S2,
    n_neighbors, &
    n_points, n_bins, shared_residual_range, js_divergences,
    included_n_reps_S1, &
    included_n_reps_S2, total_included_n_reps, global_js_divergence,
    weights, &
    tmp_counts, pmf_S1, pmf_S2, ierr)
```

3.1.4 Data Persistence

1. Array Serialization and Deserialization Module

The array serialization and deserialization module provides comprehensive routines for efficiently storing and retrieving multi-dimensional arrays of various data types.

(a) **General Architecture**

The module supports five data types (integer, real, character, logical and complex) across dimensions 1 through 5. All routines follow a consistent naming convention and parameter structure for ease of use.

(b) **Supported Functions**

- `serialize_int_Nd(arr, filename, ierr)` - Serialize N-dimensional integer arrays.
- `deserialize_int_Nd(arr, filename, ierr)` - Deserialize N-dimensional integer arrays.
- `serialize_real_Nd(arr, filename, ierr)` - Serialize N-dimensional real arrays.
- `deserialize_real_Nd(arr, filename, ierr)` - Deserialize N-dimensional real arrays.
- `serialize_char_Nd(arr, filename, ierr)` - Serialize N-dimensional character arrays.
- `deserialize_char_Nd(arr, filename, ierr)` - Deserialize N-dimensional character arrays.
- `serialize_logical_Nd(arr, filename, ierr)` - Serialize N-dimensional logical arrays.
- `deserialize_logical_Nd(arr, filename, ierr)` - Deserialize N-dimensional logical arrays.
- `serialize_complex_Nd(arr, filename, ierr)` - Serialize N-dimensional complex arrays.
- `deserialize_complex_Nd(arr, filename, ierr)` - Deserialize N-dimensional complex arrays.
- `get_array_metadata(filename, dims, max_dims, ndims, ierr, [clen])` - Retrieve array metadata from files

Where N represents the array dimension (1-5).

(c) **Detailed Parameter Specifications**

Serialization Functions Parameters:

- **arr** (input): The array to be serialized.
Type: Allocatable array of type `integer(int32)`, `real(real64)`, `character(len=clen)`, (logical) or `complex`
Dimensions: 1D to 5D
Intent: in
Note: *clen* has to be the size of the arrays elements
- **filename** (input): The name of the binary file to create.
Type: `character(len=*)`
Intent: in
- **ierr** (output): Error code indicating operation status.
Type: `integer(int32)`
Intent: out
Values: 0 indicates success, non-zero values indicate specific error conditions

Listing 31: Example usage of a serialize function

```
call serialize_int_2d(array, "test_file.bin", ierr)
```

(d) **Metadata Retrieval Function Parameters:**

- **filename** (input): The name of the binary file to read metadata from.
Type: `character(len=*)`
Intent: in
- **dims** (output): Array containing the dimensions of the stored array.
Type: `integer(int32)`, `dimension(max_dims)`

- Intent:** out **Note:** Must be preallocated, recommended with size 5
- **max_dims** (input): Size of the dims array (typically 5 for maximum supported dimensions).
Type: integer(int32)
Intent: in
 - **ndims** (output): Number of dimensions actually used by the array.
Type: integer(int32)
Intent: out
 - **ierr** (output): Error code indicating operation status.
Type: integer(int32)
Intent: out
Values: 0 indicates success, non-zero values indicate specific error conditions
 - **clen** (output, optional): For character arrays, the length of character elements.
Type: integer(int32)
Intent: out
Note: This parameter is only used and required for character arrays

Listing 32: Usage of metadata retrieval

```
call get_array_metadata(filename, dims, max_dims, ndims, ierr, clen)
```

Deserialization Functions Parameters:

- **arr** (output): The array to be populated with deserialized data.
Type: Allocatable array of type integer(int32), real(real64), character(len=clen), logical, or complex
Dimensions: 1D to 5D
Intent: out
Note: The array must be allocated with correct dimensions and correct element length before calling deserialization routines
- **filename** (input): The name of the binary file to read.
Type: character(len=*)
Intent: in
- **ierr** (output): Error code indicating operation status.
Type: integer(int32)
Intent: out
Values: 0 indicates success, non-zero values indicate specific error conditions

Listing 33: Example usage of a deserialize function

```
call deserialize_int_2d(array, "test_file.bin", ierr)
```

Listing 34: Full example

```
integer(int32), allocatable :: kallisto_expression(:, :)
integer(int32), allocatable :: kallisto_expression_2(:, :)
integer(int32) :: dims(5)
integer(int32) :: ndims, ierr, max_dims
character(len=64) :: fname

!Assume array is filled with data

call serialize_real_2d(kallisto_expression, &
                      "kallisto_data_v1.bin", ierr)
```



```

if(.not. is_ok(ierr)) error stop ierr

!Reading the data again
call get_array_metadata("kallisto_data_v1.bin", dims, max_dims, &
                        ndims, ierr)
if(.not. is_ok(ierr)) error stop ierr
allocate(kallisto_expression_2(dims(1), dims(2)))

call deserialize_real_2d(kallisto_expression_2, &
                        "kallisto_data_v1.bin", ierr)
if(.not. is_ok(ierr)) error stop ierr

```

(e) **Error Handling**

All functions return an error code through the `ierr` parameter. The error handling utility `tox errors` provides functions to check and interpret these codes.

(f) **Implementation Notes**

- All routines are implemented in standard Fortran for interoperability
- The module uses allocatable arrays for flexible memory management
- File operations use Fortran's native unformatted I/O for efficiency
- Error handling is consistent across all functions for predictable behavior

2. Standard Archive Operations

- **Save Tox Data Archive**

Saves multiple data arrays to a ZIP archive with automatic manifest creation and temporary file management. Handles any combination of available standard data arrays. This function can only be used with standard gene expression data. If other data needs to be saved, see section "Non-standard archive operations". Call the `save_tox_data` subroutine with the following arguments:

Input arguments:

- `zip_filename` character(len=*): Output ZIP file name
- `ierr` integer(int32): Error code

Optional input arguments (provide both array and filename for each data type):

- `gene_ids` character(len=*): Gene IDs array
- `gene_ids_file` character(len=*): Internal filename for gene IDs
- `expression` real(real64): Expression data matrix
- `expression_file` character(len=*): Internal filename for expression data
- `gene_to_family` integer(int32): Gene to family mapping
- `gene_to_family_file` character(len=*): Internal filename for mapping
- `family_ids` character(len=*): Family IDs array
- `family_ids_file` character(len=*): Internal filename for family IDs
- `family_centroids` real(real64): Family centroids matrix
- `family_centroids_file` character(len=*): Internal filename for centroids
- `shift_vectors` real(real64): Shift vectors matrix
- `shift_vectors_file` character(len=*): Internal filename for shift vectors

Listing 35: Save Complete Data Archive

```

integer(int32) :: ierr

! Save all available data arrays
call save_tox_data("complete_data.zip", ierr, &
                  gene_ids=gene_ids, &

```



```

gene_ids_file="genes.bin", &
expression=expr_data, &
expression_file="expression.bin", &
gene_to_family=gene_to_fam, &
gene_to_family_file="mapping.bin", &
family_ids=family_ids, &
family_ids_file="families.bin", &
family_centroids=centroids, &
family_centroids_file="centroids.bin", &
shift_vectors=shift_vecs, &
shift_vectors_file="shifts.bin")

```

Listing 36: Save Partial Data Archive

```

! Save only gene IDs and expression data
call save_tox_data("expression_only.zip", ierr, &
    gene_ids=gene_ids, &
    gene_ids_file="genes.bin", &
    expression=expr_data, &
    expression_file="expression.bin")

! Save only family-related data
call save_tox_data("families_only.zip", ierr, &
    family_ids=family_ids, &
    family_ids_file="families.bin", &
    family_centroids=centroids, &
    family_centroids_file="centroids.bin")

```

• Read Tox Data Archive

Reads data arrays from a ZIP archive created with `save_tox_data`. Automatically handles array allocation based on archive contents. The function will try to read all requested arrays (optional arguments that are provided) from the given archive. If an array can not be found in the archive, the output arrays won't be allocated or returned. Call the `read_tox_data` subroutine with the following arguments:

Input arguments:

- `zip_filename` character(len=*): Input ZIP file name
- `ierr` integer(int32): Error code

Optional output arguments:

- `gene_ids` character(len=:), allocatable: Gene IDs array
- `expression` real(real64), allocatable: Expression data matrix
- `gene_to_family` integer(int32), allocatable: Gene to family mapping
- `family_ids` character(len=:), allocatable: Family IDs array
- `family_centroids` real(real64), allocatable: Family centroids matrix
- `shift_vectors` real(real64), allocatable: Shift vectors matrix

Optional filename output arguments:

- `gene_ids_file` character(len=:), allocatable: Internal filename used
- `expression_file` character(len=:), allocatable: Internal filename used
- `gene_to_family_file` character(len=:), allocatable: Internal filename used
- `family_ids_file` character(len=:), allocatable: Internal filename used
- `family_centroids_file` character(len=:), allocatable: Internal filename used
- `shift_vectors_file` character(len=:), allocatable: Internal filename used

Listing 37: Read Complete Data Archive

```
character(len=:), allocatable :: loaded_gene_ids(:), loaded_family_ids(
    :)
real(real64), allocatable :: loaded_expr(:,:), loaded_centroids(:,:)
real(real64), allocatable :: loaded_shifts(:,:)
integer(int32), allocatable :: loaded_gene_to_fam(:)
integer(int32) :: ierr

! Read all available data from archive
call read_tox_data("complete_data.zip", ierr, &
    gene_ids=loaded_gene_ids, &
    expression=loaded_expr, &
    gene_to_family=loaded_gene_to_fam, &
    family_ids=loaded_family_ids, &
    family_centroids=loaded_centroids, &
    shift_vectors=loaded_shifts)
```

Listing 38: Read Partial Data Archive

```
! Read only specific arrays (others remain unallocated)
call read_tox_data("expression_only.zip", ierr, &
    gene_ids=loaded_gene_ids, &
    expression=loaded_expr)
```

Listing 39: Read with Filename Retrieval

```
character(len=:), allocatable :: actual_gene_file
character(len=:), allocatable :: actual_expr_file
integer(int32) :: ierr

! Get both data and internal filenames
call read_tox_data("complete_data.zip", ierr, &
    gene_ids=loaded_gene_ids, &
    gene_ids_file=actual_gene_file, &
    expression=loaded_expr, &
    expression_file=actual_expr_file)
```

3. Non-Standard Archive Operations

• Create Custom ZIP Archive

Creates a ZIP archive from arbitrary files with key-based manifest for non-standard data. Call the `create_zip_archive` subroutine with the following arguments:

Input arguments:

- `zip_filename` character(len=*): Output ZIP file name
- `keys` character(len=*): Array of keys for manifest entries
- `filenames` character(len=*): Array of file paths to include
- `ierr` integer(int32): Error code

Listing 40: Create Custom Archive

```
character(len=32) :: keys(3) = ['test_data_1', 'test_data_2', &
    'test_data_3']
character(len=256) :: files(3) = ['test_data_1.bin', 'test_2.bin', &
    'test_3.bin']
integer(int32) :: ierr
```

```
call create_zip_archive("custom_data.zip", keys, files, ierr)
```

• Extract ZIP Archive

Extracts all files from a ZIP archive to the disk in the current directory and returns key-value pairs from the manifest. Call the `extract_zip_archive` subroutine with the following arguments:

Input arguments:

- `zip_filename` character(len=*): Input ZIP file name
- `ierr` integer(int32): Error code

Output arguments:

- `keys` character(len=:), allocatable: Array of keys from manifest
- `filenames` character(len=:), allocatable: Array of extracted filenames

Listing 41: Extract Custom Archive

```
character(len=:), allocatable :: archive_keys(:), archive_files(:)
integer(int32) :: ierr

call extract_zip_archive("custom_data.zip", archive_keys, &
                        archive_files, ierr)
```

Complete Archive Workflow:

Listing 42: Complete Archive Management Workflow

```
integer(int32) :: ierr

! 1. Save processed data
call save_tox_data("analysis_results.zip", ierr, &
gene_ids=gene_ids, gene_ids_file="genes.bin", &
expression=expr_data, expression_file="expression.bin", &
family_centroids=centroids, family_centroids_file="centroids.bin")

! 2. Later, read the data back
character(len=:), allocatable :: loaded_genes(:)
real(real64), allocatable :: loaded_expr(:, :), loaded_centroids(:, :)
call read_tox_data("analysis_results.zip", ierr, &
gene_ids=loaded_genes, &
expression=loaded_expr, &
family_centroids=loaded_centroids)

! 3. Create custom archive with additional results
character(len=32) :: custom_keys(2) = ['data_1', 'data_2']
character(len=256) :: custom_files(2) = ['data_1_v2.bin', &
'data_2_v1.bin']
call create_zip_archive("full_analysis.zip", custom_keys, &
custom_files, ierr)

! 4. Extract and verify custom archive
character(len=:), allocatable :: extracted_keys(:), extracted_files(:)
call extract_zip_archive("full_analysis.zip", extracted_keys, &
extracted_files, ierr)
```

Key Features of Archive Operations:

- **Automatic Memory Management:** Arrays automatically allocated to correct sizes
- **Flexible Data Selection:** Save/read any combination of data arrays
- **Cross-Platform Compatibility:** Archives work across R, Python, and Fortran implementations
- **Manifest-Based:** Automatic tracking of file contents and relationships
- **Temporary File Handling:** Automatic cleanup of intermediate files
- **Error Recovery:** Comprehensive error checking and reporting

Archive Structure:

```
archive.zip
|- manifest.txt          (key=filename mappings)
|- genes.bin             (gene IDs, if provided)
|- expression.bin        (expression data, if provided)
|- mapping.bin           (gene to family, if provided)
|- families.bin          (family IDs, if provided)
|- centroids.bin         (family centroids, if provided)
|- shifts.bin            (shift vectors, if provided)
```

4. Individual Array Save/Load Operations

• Save Gene IDs

Saves a gene IDs array to a binary file with automatic character length preservation. Call the `save_gene_ids` subroutine with the following arguments:

Input arguments:

- `gene_ids` `character(len=*)`: Array of gene identifiers
- `filename` `character(len=*)`: Output binary file name
- `ierr` `integer(int32)`: Error code

Listing 43: Save Gene IDs Array

```
character(len=256) :: gene_ids(10000)
integer(int32) :: ierr

call save_gene_ids(gene_ids, 'gene_ids.bin', ierr)
```

• Load Gene IDs

Loads a gene IDs array from a binary file with automatic allocation and character length detection. Call the `load_gene_ids` subroutine with the following arguments:

Input arguments:

- `filename` `character(len=*)`: Input binary file name
- `ierr` `integer(int32)`: Error code

Output arguments:

- `gene_ids` `character(len=:)`, `allocatable`: Loaded gene IDs array

Listing 44: Load Gene IDs Array

```
character(len=:), allocatable :: loaded_gene_ids(:)
integer(int32) :: ierr

call load_gene_ids(loaded_gene_ids, 'gene_ids.bin', ierr)
! loaded_gene_ids automatically allocated with correct size
! and character length
```

- **Save Expression Vectors**

Saves an expression data matrix to a binary file. Call the `save_expression_vectors` subroutine with the following arguments:

Input arguments:

- `expression` `real(real64)`: 2D expression data matrix
- `filename` `character(len=*)`: Output binary file name
- `ierr` `integer(int32)`: Error code

Listing 45: Save Expression Data

```
real(real64) :: expr_data(n_samples, n_genes)
integer(int32) :: ierr

call save_expression_vectors(expr_data, 'expression.bin', ierr)
```

- **Load Expression Vectors**

Loads an expression data matrix from a binary file with automatic allocation. Call the `load_expression_vectors` subroutine with the following arguments:

Input arguments:

- `filename` `character(len=*)`: Input binary file name
- `ierr` `integer(int32)`: Error code

Output arguments:

- `expression` `real(real64)`, `allocatable`: Loaded expression data matrix

Listing 46: Load Expression Data

```
real(real64), allocatable :: loaded_expr(:, :)
integer(int32) :: ierr

call load_expression_vectors(loaded_expr, 'expression.bin', ierr)
! loaded_expr automatically allocated as (samples x genes)
```

- **Save Gene to Family Mapping**

Saves a gene to family mapping array to a binary file. Call the `save_gene_to_family` subroutine with the following arguments:

Input arguments:

- `gene_to_family` `integer(int32)`: Gene to family mapping array
- `filename` `character(len=*)`: Output binary file name
- `ierr` `integer(int32)`: Error code

Listing 47: Save Gene-Family Mapping

```
integer(int32) :: gene_to_fam(n_genes)
integer(int32) :: ierr

call save_gene_to_family(gene_to_fam, 'gene_to_fam.bin', ierr)
```

- **Load Gene to Family Mapping**

Loads a gene to family mapping array from a binary file with automatic allocation. Call the `load_gene_to_family` subroutine with the following arguments:

Input arguments:

- `filename` `character(len=*)`: Input binary file name
- `ierr` `integer(int32)`: Error code

Output arguments:

– gene_to_family integer(int32), allocatable: Loaded gene to family mapping

Listing 48: Load Gene-Family Mapping

```
integer(int32), allocatable :: loaded_gene_to_fam(:)
integer(int32) :: ierr

call load_gene_to_family(loaded_gene_to_fam, 'gene_to_fam.bin', ierr)
```

- **Save Family IDs**

Saves a family IDs array to a binary file with automatic character length preservation. Call the `save_family_ids` subroutine with the following arguments:

Input arguments:

- family_ids character(len=*): Array of family identifiers
- filename character(len=*): Output binary file name
- ierr integer(int32): Error code

Listing 49: Save Family IDs

```
character(len=256) :: family_ids(n_families)
integer(int32) :: ierr

call save_family_ids(family_ids, 'family_ids.bin', ierr)
```

- **Load Family IDs**

Loads a family IDs array from a binary file with automatic allocation and character length detection. Call the `load_family_ids` subroutine with the following arguments:

Input arguments:

- filename character(len=*): Input binary file name
- ierr integer(int32): Error code

Output arguments:

- family_ids character(len=:), allocatable: Loaded family IDs array

Listing 50: Load Family IDs

```
character(len=:), allocatable :: loaded_family_ids(:)
integer(int32) :: ierr

call load_family_ids(loaded_family_ids, 'family_ids.bin', ierr)
```

- **Save Family Centroids**

Saves a family centroids matrix to a binary file. Call the `save_family_centroids` subroutine with the following arguments:

Input arguments:

- family_centroids real(real64): 2D family centroids matrix
- filename character(len=*): Output binary file name
- ierr integer(int32): Error code

Listing 51: Save Family Centroids

```
real(real64) :: centroids(67, 15512)
integer(int32) :: ierr

call save_family_centroids(centroids, 'centroids.bin', ierr)
```

- **Load Family Centroids**

Loads a family centroids matrix from a binary file with automatic allocation. Call the `load_family_centroids` subroutine with the following arguments:

Input arguments:

- `filename` character(len=*): Input binary file name
- `ierr` integer(int32): Error code

Output arguments:

- `family_centroids` real(real64), allocatable: Loaded family centroids matrix

Listing 52: Load Family Centroids

```
real(real64), allocatable :: loaded_centroids(:, :)
integer(int32) :: ierr

call load_family_centroids(loaded_centroids, 'centroids.bin', ierr)
```

- **Save Shift Vectors**

Saves a shift vectors matrix to a binary file. Call the `save_shift_vectors` subroutine with the following arguments:

Input arguments:

- `shift_vectors` real(real64): 2D shift vectors matrix
- `filename` character(len=*): Output binary file name
- `ierr` integer(int32): Error code

Listing 53: Save Shift Vectors

```
real(real64) :: shift_vecs(134, 88327) ! 2xsamples x genes
integer(int32) :: ierr

call save_shift_vectors(shift_vecs, 'shift_vectors.bin', ierr)
```

- **Load Shift Vectors**

Loads a shift vectors matrix from a binary file with automatic allocation. Call the `load_shift_vectors` subroutine with the following arguments:

Input arguments:

- `filename` character(len=*): Input binary file name
- `ierr` integer(int32): Error code

Output arguments:

- `shift_vectors` real(real64), allocatable: Loaded shift vectors matrix

Listing 54: Load Shift Vectors

```
real(real64), allocatable :: loaded_shifts(:, :)
integer(int32) :: ierr

call load_shift_vectors(loaded_shifts, 'shift_vectors.bin', ierr)
```

- **Get Array Metadata**

Retrieves metadata (dimensions, character length) from a binary file without loading the entire array. Call the `get_array_metadata` subroutine with the following arguments:

Input arguments:

- `filename` character(len=*): Binary file name to inspect
- `max_dims` integer(int32): Maximum number of dimensions to return

– ierr integer(int32): Error code

Output arguments:

– dims integer(int32): Array dimensions

– ndims integer(int32): Number of dimensions

– char_len integer(int32), optional: Character length for string arrays

Listing 55: Inspect Array File Metadata

```
integer(int32) :: dims(5), ndims, char_len, ierr

! Get metadata for gene IDs file
call get_array_metadata('gene_ids.bin', dims, 5, ndims, &
                        ierr, char_len)

if (is_ok(ierr)) then
    print *, "Dimensions:", dims(1:ndims)
    print *, "Character length:", char_len
end if

! Get metadata for expression data file
call get_array_metadata('expression.bin', dims, 5, ndims, ierr)
if (is_ok(ierr)) then
    print *, "Expression data shape:", dims(1), "x", dims(2)
end if
```

Complete Individual Array Workflow:

Listing 56: Individual Array Save/Load Workflow

```
integer(int32) :: ierr

! 1. Save individual arrays
call save_gene_ids(gene_ids, 'gene_ids.bin', ierr)
call save_expression_vectors(expr_data, 'expression.bin', ierr)
call save_gene_to_family(gene_to_fam, 'mapping.bin', ierr)
call save_family_ids(family_ids, 'families.bin', ierr)
call save_family_centroids(centroids, 'centroids.bin', ierr)
call save_shift_vectors(shift_vecs, 'shifts.bin', ierr)

! 2. Check file metadata before loading
integer(int32) :: dims(5), ndims, char_len
call get_array_metadata('expression.bin', dims, 5, ndims, ierr)

! 3. Load arrays (automatic allocation)
character(len=:), allocatable :: loaded_genes(:)
real(real64), allocatable :: loaded_expr(:, :)
integer(int32), allocatable :: loaded_mapping(:)
character(len=:), allocatable :: loaded_families(:)
real(real64), allocatable :: loaded_centroids(:, :)
real(real64), allocatable :: loaded_shifts(:, :)

call load_gene_ids(loaded_genes, 'gene_ids.bin', ierr)
call load_expression_vectors(loaded_expr, 'expression.bin', ierr)
call load_gene_to_family(loaded_mapping, 'mapping.bin', ierr)
call load_family_ids(loaded_families, 'families.bin', ierr)
```



```
call load_family_centroids(loaded_centroids, 'centroids.bin', ierr)
call load_shift_vectors(loaded_shifts, 'shifts.bin', ierr)
```

Manual Archive Creation with Individual Arrays:

Listing 57: Manual Archive Creation

```
integer(int32) :: ierr

! 1. Save arrays to individual files
call save_gene_ids(gene_ids, 'temp_genes.bin', ierr)
call save_expression_vectors(expr_data, 'temp_expr.bin', ierr)

! 2. Create custom archive with specific keys
character(len=32) :: keys(2) = ['gene_data', 'expression_data']
character(len=256) :: files(2) = ['temp_genes.bin', 'temp_expr.bin']
call create_zip_archive('manual_archive.zip', keys, files, ierr)

! 3. Clean up temporary files (optional)
call delete_file('temp_genes.bin', ierr)
call delete_file('temp_expr.bin', ierr)
```

Key Features of Individual Array Operations:

- **Automatic Allocation:** Output arrays automatically allocated to correct sizes
- **Character Length Preservation:** String arrays maintain original character lengths
- **Metadata Inspection:** Check array properties without loading data
- **Binary Format:** Efficient storage for large datasets
- **Error Handling:** Comprehensive error checking for file operations
- **Memory Efficient:** Can process arrays larger than available memory

Performance Considerations:

- Binary format provides fast I/O for large arrays
- Automatic dimension detection eliminates manual size tracking
- Character length preservation avoids fixed-width string limitations
- Suitable for datasets with millions of genes and thousands of samples

3.1.5 Data Processing

1. Normalization of Expression Values

The subroutine runs a complete normalization pipeline on gene expression data. It first applies standard deviation normalization, then quantile normalization, followed by replicate averaging, and finally a log2 transformation. The result is returned in the `buf_log` array, which contains the fully normalized expression values. Additionally intermediate results of the performed calculations are available in output buffer arrays. Call the `normalization_pipeline` subroutine with the following arguments:

Input arguments:

- `n_genes` integer(int32): Total number of genes
- `n_tissues` integer(int32): Number of axes (tissues / dimensions)
- `input_matrix` real(real64): Expression vector matrix with size `n_genes × n_tissues`
- `max_stack` integer(int32): Stack size for quicksort
- `group_s` integer(int32): Start column index for each replicate group (`n_grps`)
- `group_c` integer(int32): Number of columns per replicate group (`n_grps`)
- `n_grps` integer(int32): Number of replicate groups

Output arguments:

- `buf_stddev` `real(real64)`: Buffer for std dev normalization (`n_genes * n_tissues`)
- `buf_quant` `real(real64)`: Buffer for quantile normalization (`n_genes * n_tissues`)
- `buf_avg` `real(real64)`: Buffer for replicate averaging (`n_genes * n_grps`)
- `buf_log` `real(real64)`: Buffer for $\log_2(x+1)$ transformation (`n_genes * n_grps`)
- `temp_col` `real(real64)`: Temporary column vector for sorting (`n_genes`)
- `rank_means` `real(real64)`: Buffer for rank means (`n_genes`)
- `perm` `integer(int32)`: Permutation vector for sorting (`n_genes`)
- `stack_left` `integer(int32)`: Left stack for quicksort (`max_stack`)
- `stack_right` `integer(int32)`: Right stack for quicksort (`max_stack`)
- `ierr` `integer(int32)`: Error code (for error details please check [error handling section](#))

Listing 58: Normalization Pipeline

```
call normalization_pipeline(n_genes, n_tissues, input_matrix, &
                           buf_stddev, buf_quant, buf_avg, buf_log, &
                           temp_col, rank_means, perm, stack_left, &
                           stack_right, max_stack, group_s, &
                           group_c, n_grps, ierr)
```

Alternatively you can calculate each step of the normalization pipeline manually by calling the following subroutines in sequence:

(a) **Normalize by Standard Deviation**

The subroutine normalizes each gene's expression vector using `sqrt(mean(x^2))` across tissues (not classical standard deviation). Call the `normalize_by_std_dev` subroutine with the following arguments:

Input arguments:

- `n_genes` `integer(int32)`: Number of genes (rows)
- `n_tissues` `integer(int32)`: Number of tissues (columns)
- `input_matrix` `real(real64)`: Flattened input matrix of gene expression values (column-major)

Output arguments:

- `output_matrix` `real(real64)`: Output normalized matrix (same shape as input)
- `ierr` `integer(int32)`: Error code (for error details please check [error handling section](#))

Listing 59: Standard Deviation Normalization

```
call normalize_by_std_dev(n_genes, n_tissues, input_matrix,
                          output_matrix, ierr)
```

(b) **Quantile Normalization**

The subroutine performs quantile normalization of a gene expression matrix (F42-compliant) and computes average expression per rank across tissues. Call the `quantile_normalization` subroutine with the following arguments:

Input arguments:

- `n_genes` `integer(int32)`: Number of genes (rows)
- `n_tissues` `integer(int32)`: Number of tissues (columns)
- `input_matrix` `real(real64)`: Flattened input matrix (column-major)
- `max_stack` `integer(int32)`: Stack size passed from R

Output arguments:

- `output_matrix` `real(real64)`: Output normalized matrix (same shape as input)
- `temp_col` `real(real64)`: Temporary vector for column sorting (size `n_genes`)

- `rank_means` `real(real64)`: Preallocated vector to store rank means (size `n_genes`)
- `perm` `integer(int32)`: Permutation vector (size `n_genes`)
- `stack_left` `integer(int32)`: Manual quicksort stack ($\geq \log_2(n_genes) + 10$)
- `stack_right` `integer(int32)`: Manual quicksort stack (same size as `stack_left`)
- `ierr` `integer(int32)`: Error code (for error details please check [error handling section](#))

Listing 60: Quantile Normalization

```
call quantile_normalization(n_genes, n_tissues, input_matrix,
    output_matrix, temp_col, rank_means, perm, stack_left, stack_right
    , max_stack, ierr)
```

(c) Calculate Tissue Averages

The subroutine calculates tissue averages by averaging replicates within each group. For each group of tissue replicates, it computes the average expression per gene. The input matrix is column-major, flattened as a 1D array. Call the `calc_tiss_avg` subroutine with the following arguments:

Input arguments:

- `n_gene` `integer(int32)`: Number of genes (rows)
- `n_grps` `integer(int32)`: Number of tissue groups
- `group_s` `integer(int32)`: Start column index for each group (length: `n_grps`)
- `group_c` `integer(int32)`: Number of columns per group (length: `n_grps`)
- `input_matrix` `real(real64)`: Flattened input matrix (length: `n_gene * sum(group_c)`)

Output arguments:

- `output_matrix` `real(real64)`: Flattened output matrix (length: `n_gene * n_grps`)
- `ierr` `integer(int32)`: Error code (for error details please check [error handling section](#))

Listing 61: Tissue Averages Calculation

```
call calc_tiss_avg(n_gene, n_grps, group_s, group_c, input_matrix, &
    output_matrix, ierr)
```

(d) Log2 Transformation

The subroutine applies $\log_2(x + 1)$ transformation to each element of the input matrix. This is computed via $\log(x + 1) / \log(2)$ for compatibility with WebAssembly (WASM). Call the `log2_transformation` subroutine with the following arguments:

Input arguments:

- `n_genes` `integer(int32)`: Number of genes (rows)
- `n_tissues` `integer(int32)`: Number of tissues (columns)
- `input_matrix` `real(real64)`: Flattened input matrix (size: `n_genes * n_tissues`)

Output arguments:

- `output_matrix` `real(real64)`: Output matrix (same size as input)
- `ierr` `integer(int32)`: Error code (for error details please check [error handling section](#))

Listing 62: Log2 Transformation

```
call log2_transformation(n_genes, n_tissues, input_matrix, &
    output_matrix, ierr)
```

2. Calculate Fold Change

If fold change calculation is needed, this subroutine can be used after normalization. The subroutine calculates $\log_2$ fold changes between condition and control columns. For each control-condition pair, this subroutine computes the $\log_2$ fold change by subtracting the expression

value in the control column from the corresponding value in the condition column, for all genes. Call the `calc_fchange` subroutine with the following arguments:

Input arguments:

- `n_genes` integer(int32): Total number of genes
- `n_cols` integer(int32): Number of columns in the input matrix
- `n_pairs` integer(int32): Number of control-condition pairs
- `control_cols` integer(int32): Control column indices (length `n_pairs`)
- `cond_cols` integer(int32): Condition column indices (length `n_pairs`)
- `i_matrix` real(real64): Input matrix, flattened (length: `n_genes × n_cols`)

Output arguments:

- `o_matrix` real(real64): Output matrix for fold changes (length: `n_genes × n_pairs`)
- `ierr` integer(int32): Error code (for error details please check [error handling section](#))

Listing 63: Fold Change Calculation

```
call calc_fchange(n_genes, n_cols, n_pairs, control_cols, cond_cols, &
                 i_matrix, o_matrix, ierr)
```

3.1.6 Family-Based Analysis

1. Calculation of family Centroids

The subroutine computes the centroid (mean expression vector) for each gene family in a gene expression dataset. It iterates over all families, selects the relevant genes for each family and calculates the mean vector for each family. There are two possible modes to use. To use all genes for calculation, select `all` mode. If you want to specify relevant genes for calculation, select `orthologs` mode. It is important that you provide a logical `ortholog_set` array when using `orthologs` mode. Call the `group_centroid` subroutine with the following arguments:

Input arguments:

- `expression_vectors` real(real64): Expression vector matrix with size `n_axes × n_genes`
- `n_axes` integer(int32): Number of axes (tissues / dimensions)
- `n_genes` integer(int32): Total number of genes
- `gene_to_family` integer(int32): Gene-to-family mapping array with the length of `n_genes` (1-based indexing)
- `n_families` integer(int32): Total number of gene families
- `mode` integer(int32): Calculation mode ("GROUP_ALL" or "GROUP_ORTHOLOGS") where "GROUP_ALL" = 1 and "GROUP_ORTHOLOGS" = 0 are parameters
- (Optional) `ortholog_set` logical: Logical array indicating if a gene is part of the calculation subset (only used in `GROUP_ORTHOLOGS` mode)

Output arguments:

- `centroid_matrix` real(real64): Matrix of calculated family centroid vectors with size `n_axes × n_families`
- `selected_indices` integer(int32): Array of gene indices used for the calculation with length of `n_genes`
- `ierr` integer(int32): Error code (for error details please check [error handling section](#))

Listing 64: Calculation of Family Centroids

```
call group_centroid(expression_vectors, n_axes, n_genes, &
                   gene_to_family, n_families, centroid_matrix, &
                   mode, selected_indices, ierr, ortholog_set)
```

2. Compute Shift Vector Field

Computes shift vectors by subtracting corresponding family centroids from gene expression vectors. The output stores both centroids and shift vectors in a single matrix.

Mathematical Operation:

$$\text{shift_vectors}(d + i, j) = \text{expression_vectors}(i, j) - \text{family_centroids}(i, \text{gene_to_centroid}(j))$$

$$\text{shift_vectors}(1 : d, j) = \text{family_centroids}(:, \text{gene_to_centroid}(j))$$

Arguments:

- `d integer(int32)`: Expression vector dimension
- `n_genes integer(int32)`: Total number of genes
- `n_families integer(int32)`: Total number of families
- `expression_vectors real(real64)`: Gene expression matrix [$d \times n_genes$]
- `family_centroids real(real64)`: Family centroid matrix [$d \times n_families$]
- `gene_to_centroid integer(int32)`: Mapping from genes to family centroids [n_genes]
- `shift_vectors real(real64)`: Output matrix [$2*d \times n_genes$] storing centroids (rows 1-d) and shift vectors (rows d+1-2d)
- `ierr integer(int32)`: Error code

Listing 65: Shift Vector Field Computation

```
call compute_shift_vector_field(d, n_genes, n_families, &
                               expression_vectors, family_centroids, &
                               gene_to_centroid, shift_vectors, ierr)
```

Output Structure:

- `shift_vectors(1:d, :)`: Contains the family centroids for each gene
- `shift_vectors(d+1:2*d, :)`: Contains the shift vectors (expression vector - family centroid)

3. Calculate Mean Vector

Alternatively you can calculate a single mean vector for a given set of vectors by calling the `mean_vector` subroutine directly with the following arguments:

Input arguments:

- `expression_vectors real(real64)`: Expression vector matrix with size `n_axes` $\times$ `n_genes`
- `n_axes integer(int32)`: Number of axes (tissues / dimensions)
- `n_genes integer(int32)`: Total number of genes
- `gene_indices integer(int32)`: An array containing the column indices of the selected genes in `expression_vectors`
- `n_selected_genes integer(int32)`: Total number of genes in the current family to be calculated

Output arguments:

- `centroid real(real64)`: Calculated mean vector (centroid) with length of `n_axes`
- `ierr integer(int32)`: Error code (for error details please check [error handling section](#))

Listing 66: Mean Vector Calculation

```
call mean_vector(expression_vectors, n_axes, n_genes, gene_indices, &
                 n_selected_genes, centroid, ierr)
```

4. Calculate Distance to Centroid

For each gene in the expression vector matrix, the subroutine computes the Euclidean distance to its corresponding family centroid and returns a distances array containing the distance value

for each expression vector to its family centroid. Call the `distance_to_centroid` subroutine with the following arguments:

Input arguments:

- `n_genes` integer(int32): Total number of genes
- `n_families` integer(int32): Total number of gene families
- `genes` real(real64): Gene expression matrix ($d \times n_genes$)
- `centroids` real(real64): Family centroid matrix ($d \times n_genes$)
- `gene_to_fam` integer(int32): Gene-to-family mapping array with the length of `n_genes` (1-based indexing)
- `d` integer(int32): Expression vector dimension

Output arguments:

- `distances` real(real64): Array of distances for each gene with length of `n_genes`

Listing 67: Distance to Centroid Calculation

```
call distance_to_centroid(n_genes, n_families, genes, centroids, &
                        gene_to_fam, distances, d)
```

(a) **Calculate Euclidean Distance**

Alternatively you can calculate a single Euclidean distance between two vectors manually by calling `euclidean_distance` subroutine with the following arguments:

Input arguments:

- `vec1` real(real64): First expression vector as array with length of dimension `d`
- `vec2` real(real64): Second expression vector as array with length of dimension `d`
- `d` integer(int32): Dimension value

Output arguments:

- `result` real(real64): Euclidean distance value between `vec1` and `vec2`

Listing 68: Euclidean distance calculation

```
call euclidean_distance(vec1, vec2, d, result)
```

5. Detect Gene Outliers

The subroutine first computes family scaling, then calculates relative distance indices (RDI) and identifies outliers based on the specified percentile threshold. It returns a boolean array indicating whether each gene is an outlier. Call the `detect_outliers` subroutine with the following arguments:

Input arguments:

- `n_genes` integer(int32): Total number of genes
- `n_families` integer(int32): Total number of gene families
- `distances` real(real64): Array of distances for each gene to its centroid with length of `n_genes`
- `gene_to_fam` integer(int32): Gene-to-family mapping array with the length of `n_genes` (1-based indexing)
- **(Optional)** `percentile` real(real64): Percentile threshold value for outlier detection (default is 95)

Input / Output arguments:

- `work_array` real(real64): Work array for sorting with length of `n_genes`
- `perm` integer(int32): Permutation array for sorting with length of `n_genes`
- `stack_left` integer(int32): Stack array for left indices during sorting with length of `n_genes`

- `stack_right` integer(int32): Stack array for right indices during sorting with length of `n_genes`
- `loess_x` real(real64): Reference x-coordinates for loess smoothing with length of `n_families`
- `loess_y` real(real64): Reference y-coordinates for loess smoothing with length of `n_families`
- `loess_n` integer(int32): Indices of reference points used for smoothing with length of `n_families`

Output arguments:

- `is_outlier` logical: Boolean array with length of `n_genes` indicating whether each gene is an outlier
- `ierr` integer(int32): Error code (for error details please check [error handling section](#))

Listing 69: Gene Outlier Calculation

```
call detect_outliers(n_genes, n_families, distances, gene_to_fam, &
                    work_array, perm, stack_left, stack_right, &
                    is_outlier, loess_x, loess_y, loess_n, ierr, &
                    percentile)
```

Alternatively you can calculate each step of `detect_outliers` manually with the following subroutine:

(a) **Compute Family Scaling**

The subroutine calculates a scaling factor for each gene family by computing the median and standard deviation of gene-to-centroid distances within each family, then applies LOESS smoothing to these values. Call the `compute_family_scaling` subroutine with the following arguments:

Input arguments:

- `n_genes` integer(int32): Total number of genes
- `n_families` integer(int32): Total number of gene families
- `distances` real(real64): Array of distances for each gene to its centroid with length of `n_genes`
- `gene_to_fam` integer(int32): Gene-to-family mapping array with the length of `n_genes` (1-based indexing)

Input / Output arguments:

- `loess_x` real(real64): Reference x-coordinates for loess smoothing with length of `n_families`
- `loess_y` real(real64): Reference y-coordinates for loess smoothing with length of `n_families`
- `indices_used` integer(int32): Indices of reference points used for smoothing with length of `n_families`
- `perm_tmp` integer(int32): Permutation array for sorting with length of `n_genes`

Output arguments:

- `dscale` real(real64): Array of scaling factors per family with length of `n_families`
- `family_distances` real(real64): Temporary work array with length of `n_genes` to store distances of genes within each family during calculation
- `ierr` integer(int32): Error code (for error details please check [error handling section](#))

Listing 70: Compute Family Scaling

```
call compute_family_scaling(n_genes, n_families, distances, &
                           gene_to_fam, dscale, loess_x, loess_y, &
                           indices_used, perm_tmp, stack_left_tmp, &
                           stack_right_tmp, family_distances, ierr)
```


(b) **Compute Family Scaling Helper**

This helper subroutine allocates internal arrays for easier usage and calls `compute_family_scaling`. Call the `compute_family_scaling_alloc` subroutine with the following arguments:

Input arguments:

- `n_genes` integer(int32): Total number of genes
- `n_families` integer(int32): Total number of gene families
- `distances` real(real64): Array of distances for each gene to its centroid with length of `n_genes`
- `gene_to_fam` integer(int32): Gene-to-family mapping array with the length of `n_genes` (1-based indexing)

Input / Output arguments:

- `loess_x` real(real64): Reference x-coordinates for loess smoothing with length of `n_families`
- `loess_y` real(real64): Reference y-coordinates for loess smoothing with length of `n_families`
- `indices_used` integer(int32): Indices of reference points used for smoothing with length of `n_families`

Output arguments:

- `dscale` real(real64): Array of scaling factors per family with length of `n_families`
- `ierr` integer(int32): Error code (for error details please check [error handling section](#))

Listing 71: Compute Family Scaling

```
call compute_family_scaling_alloc(n_genes, n_families, distances, &
                                gene_to_fam, dscale, loess_x, &
                                loess_y, indices_used, ierr)
```

(c) **Compute Relative Distance Index (RDI)**

The subroutine calculates the Relative Distance Index (RDI) for each gene by dividing its Ecudclidean distance to the family centroid by the scaling factor of its family. Call the `compute_rdi` subroutine with the following arguments:

Input arguments:

- `n_genes` integer(int32): Total number of genes
- `distances` real(real64): Array of distances for each gene to its centroid with length of `n_genes`
- `gene_to_fam` integer(int32): Gene-to-family mapping array with the length of `n_genes` (1-based indexing)
- `dscale` real(real64): Array of scaling factors per family with length of `n_families`

Input / Output arguments:

- `sorted_rdi` real(real64): Work array for sorting with length of `n_genes`
- `perm` integer(int32): Pre-initialized (1: `n_genes`) permutation array for sorting with length of `n_genes`
- `stack_left` integer(int32): Stack array for sorting with length of `n_genes`
- `stack_right` integer(int32): Stack array for sorting with length of `n_genes`

Output arguments:

- `rdi` real(real64): Array of RDI values for each gene with length of `n_genes`

Listing 72: Compute Relative Distance Index (RDI)

```
call compute_rdi(n_genes, distances, gene_to_fam, dscale, rdi, &
                sorted_rdi, perm, stack_left, stack_right)
```

(d) **Identify Outliers**

The subroutine identifies outliers based on the top percentile of RDI values. Call the `identify_outliers` subroutine with the following arguments:

Input arguments:

- `n_genes` integer(int32): Total number of genes
- `rdi` real(real64): Array of RDI values for each gene with length of `n_genes`
- `sorted_rdi` real(real64): Array of sorted RDI values for each gene with length of `n_genes` - **this array must be filtered to remove negatives and be sorted in ascending order!**
- **(Optional)** `percentile` real(real64): Percentile threshold value for outlier detection (default is 95)

Output arguments:

- `is_outlier` logical: Boolean array with length of `n_genes` indicating whether each gene is an outlier
- `threshold`: Actual threshold value used for outlier detection

Listing 73: Identify Outliers

```
call identify_outliers(n_genes, rdi, sorted_rdi, is_outlier, &
                      threshold, percentile)
```

6. Compute Normalized Tissue Versatility

The subroutine calculates a normalized tissue versatility score for selected gene expression vectors. It is based on the angle between each gene expression vector and the space diagonal. Versatility is normalized to $[0, 1]$, where 0 means uniform expression 1 means expression in only one axis. Call the `compute_tissue_versatility` subroutine with the following parameters:

Input arguments:

- `n_genes` integer(int32): Total number of genes
- `n_vectors` integer(int32): Total number of expression vectors
- `n_selected_axes` integer(int32): Total number of selected axes
- `n_selected_vectors` integer(int32): Total number of selected vectors
- `expression_vectors` real(real64): Expression vector matrix with size `n_axes` $\times$ `n_vectors`
- `exp_vecs_selection_index` logical: Logical array with length of `n_vectors` to specify which vectors should be included in the calculation
- `axes_selection` logical: Logical array with length of `n_axes` to specify which axes should be included in the calculation

Output arguments:

- `tissue_versatilities` real(real64): Array with the length of `n_selected_vectors` containing the calculated tissue versatilities
- `tissue_angles_deg` real(real64): Array with the length of `n_selected_vectors` containing the calculated angles in degrees
- `ierr` integer(int32): Error code (for error details please check [error handling section](#))

Listing 74: Tissue Versatility Calculation

```
call compute_tissue_versatility(n_axes, n_vectors, &
                               expression_vectors, exp_vecs_selection_index, &
                               n_selected_vectors, axes_selection, &
                               n_selected_axes, tissue_versatilities, &
                               tissue_angles_deg, ierr)
```

3.1.7 RAP Projection and Analysis Functions

The Fortran module provides comprehensive routines for RAP (Relative Axes Projection) analysis, including vector projection, angle calculation, and axis contribution analysis.

Key Features:

- **RAP Projection:** Removes diagonal component to focus on relative differences
- **Signed Angles:** Calculates both magnitude and directionality of angular separation
- **Dimensional Flexibility:** Handles 2D, 3D, and higher-dimensional cases
- **Axis Contribution Analysis:** Quantifies relative importance of each axis

1. Omics Vector RAP Projection

Projects selected expression vectors onto the RAP constructed from a selected set of axes. This performs both vector selection and RAP projection in a single operation.

Arguments:

- `vecs` `real(real64)`: Matrix with expression vectors [$n_axes \times n_vecs$]
- `n_axes` `integer(int32)`: Number of axes
- `n_vecs` `integer(int32)`: Number of vectors per axis
- `vecs_selection_mask` `logical`: Mask for vectors where projection is computed [n_vecs]
- `n_selected_vecs` `integer(int32)`: Count of selected vectors
- `axes_selection_mask` `logical`: Mask for axes included in RAP [n_axes]
- `n_selected_axes` `integer(int32)`: Count of selected axes
- `projections` `real(real64)`: Output projected vectors [$n_selected_axes \times n_selected_vecs$]
- `ierr` `integer(int32)`: Error code

Listing 75: Vector RAP Projection

```
call omics_vector_RAP_projection(vecs, n_axes, n_vecs, &
                                vecs_selection_mask, n_selected_vecs, &
                                axes_selection_mask, n_selected_axes, &
                                projections, ierr)
```

2. Omics Field RAP Projection

Projects selected vector fields (e.g., shift vectors) onto the RAP constructed from a selected set of axes. Computes shift vectors as differences between origins and targets before projection.

Arguments:

- `vecs` `real(real64)`: Matrix with vector fields [$2 \times n_axes \times n_vecs$] (first n_axes rows: origins, last n_axes rows: targets)
- `n_axes` `integer(int32)`: Number of axes
- `n_vecs` `integer(int32)`: Number of vectors per axis
- `vecs_selection_mask` `logical`: Mask for vectors where projection is computed [n_vecs]
- `n_selected_vecs` `integer(int32)`: Count of selected vectors
- `axes_selection_mask` `logical`: Mask for axes included in RAP [n_axes]
- `n_selected_axes` `integer(int32)`: Count of selected axes
- `projections` `real(real64)`: Output projected vectors [$n_selected_axes \times n_selected_vecs$]
- `ierr` `integer(int32)`: Error code

Listing 76: Vector Field RAP Projection

```
call omics_field_RAP_projection(vecs, n_axes, n_vecs, &
                                vecs_selection_mask, &
                                n_selected_vecs, axes_selection_mask, &
                                n_selected_axes, projections, ierr)
```

3. Project Selected Vectors onto RAP

Core RAP projection routine that subtracts the diagonal component from selected vectors.

Mathematical Operation:

$$\text{projection} = \text{vector} - \frac{\sum_{i=1}^n \text{vector}_i}{n}$$

Arguments:

- `selected_vecs` `real(real64)`: Matrix with vectors for selected axes [`n_selected_axes` × `n_selected_vecs`] (input/output)
- `n_selected_axes` `integer(int32)`: Number of selected axes
- `n_selected_vecs` `integer(int32)`: Number of selected vectors

Listing 77: Core RAP Projection

```
call project_selected_vecs_onto_rap(selected_vecs, n_selected_axes, &
                                   n_selected_vecs)
```

4. Clock Hand Angle Between Vectors

Computes the signed rotation angle between two RAP-projected and normalized vectors. Handles 2D, 3D, and higher-dimensional cases with automatic directionality calculation.

Arguments:

- `v1` `real(real64)`: First normalized vector in RAP space [`n_dims`]
- `v2` `real(real64)`: Second normalized vector in RAP space [`n_dims`]
- `n_dims` `integer(int32)`: Dimension of both vectors
- `signed_angle` `real(real64)`: Signed angle between vectors in radians [$[-\pi, \pi]$]
- `selected_axes_for_signed` `integer(int32)`: Indices of 3 axes for directionality calculation [3] (ignored if `n_dims` ≤ 3)
- `ierr` `integer(int32)`: Error code

Listing 78: Signed Angle Calculation

```
call clock_hand_angle_between_vectors(v1, v2, n_dims, signed_angle, &
                                     selected_axes_for_signed, ierr)
```

5. Clock Hand Angles for Shift Vectors

Computes signed rotation angles between corresponding pairs of RAP-projected and normalized vectors (e.g., expression centroids and paralogs).

Arguments:

- `origins` `real(real64)`: First set of RAP-projected vectors (e.g., centroids) [`n_dims` × `n_vecs`]
- `targets` `real(real64)`: Second set of RAP-projected vectors (e.g., paralogs) [`n_dims` × `n_vecs`]
- `n_dims` `integer(int32)`: Dimension of each vector in RAP space
- `n_vecs` `integer(int32)`: Number of vector pairs
- `vecs_selection_mask` `logical`: Mask for vector pairs where angle should be computed [`n_vecs`]
- `n_selected_vecs` `integer(int32)`: Count of selected vector pairs
- `selected_axes_for_signed` `integer(int32)`: Indices of 3 axes for directionality [3]
- `signed_angles` `real(real64)`: Output signed angles in radians [$[-\pi, \pi]$] [`n_selected_vecs`]
- `ierr` `integer(int32)`: Error code

Listing 79: Batch Angle Calculation

```
call clock_hand_angles_for_shift_vectors(origins, targets, n_dims, &
                                         n_vecs, vecs_selection_mask, &
                                         n_selected_vecs, &
                                         selected_axes_for_signed, &
                                         signed_angles, ierr)
```

6. Compute Relative Axis Contributions

Computes fractional contribution of each axis to a RAP-projected and normalized vector. Values range [0,1] and sum to 1.

Arguments:

- `vec` `real(real64)`: RAP-projected and normalized vector [`n_axes`]
- `n_axes` `integer(int32)`: Number of axes
- `contributions` `real(real64)`: Fractional contribution of each axis [`n_axes`]
- `ierr` `integer(int32)`: Error code

Listing 80: Axis Contribution Analysis

```
call compute_relative_axis_contributions(vec, n_axes, &
                                         contributions, ierr)
```

7. Relative Axes Changes from Shift Vector

Wrapper for computing axis contributions from shift vectors.

Arguments:

- `vec` `real(real64)`: RAP-projected and normalized shift vector [`n_axes`]
- `n_axes` `integer(int32)`: Number of axes
- `contributions` `real(real64)`: Fractional contribution of each axis [`n_axes`]
- `ierr` `integer(int32)`: Error code

Listing 81: Shift Vector Axis Contributions

```
call relative_axes_changes_from_shift_vector(vec, n_axes, &
                                              contributions, ierr)
```

8. Relative Axes Expression from Expression Vector

Wrapper for computing axis contributions from expression vectors.

Arguments:

- `vec` `real(real64)`: RAP-projected and normalized expression vector [`n_axes`]
- `n_axes` `integer(int32)`: Number of axes
- `contributions` `real(real64)`: Fractional contribution of each axis [`n_axes`]
- `ierr` `integer(int32)`: Error code

Listing 82: Expression Vector Axis Contributions

```
call relative_axes_expression_from_expression_vector(vec, n_axes, &
                                                    contributions, ierr)
```

3.1.8 Paralog Analysis

The Fortran module provides comprehensive routines for paralog analysis.

Key Features:

- **Detect Neofunctionalization:** Identifies neofunctionalization for genes by checking whether the difference of expression to its ancestor exceeds the threshold for the respective axis.

- **Detect Dosage Effect:** Identifies subsets of paralogs with small angle to the ancestor (`max_angle`) and sum to a magnitude significantly exceeding `norm(ancestor)` (`gain`).
- **Detect Subfunctionalization:** Identifies subsets of paralogs exhibiting significant angles to the ancestor.

1. Detect Neofunctionalization

Identifies neofunctionalization for genes by checking whether the difference of expression to its ancestor exceeds the threshold for the respective axis.

Arguments:

- `ancestors` `real`(real64), `dimension`(n_axes, n_families): RAP projected unit length expression vector of ancestral ortholog
- `n_families` `integer`(int32): number of vectors in `ancestors`
- `genes` `real`(real64), `dimension`(n_axes, n_genes): RAP projected unit length expression vectors of genes
- `n_axes` `integer`(int32): size of `ancestors` vector and vectors in `genes`
- `gene_to_fam` `integer`(int32), `dimension`(n_genes): mapping of gene index to family index
- `n_genes` `integer`(int32): number of vectors in `genes`
- `thresholds` `real`(real64), `dimension`(n_axes): threshold per axis that defines significant change in expression, may be a percentile of all genes' changes per axis
- `neofunc` `logical`, `dimension`(n_genes, n_axes): `.true.` if neofunctionalization has been detected for the respective axes
- `ierr` `integer`(int32): error code

Listing 83: Vector RAP Projection

```
call detect_neofunctionalization(ancestors, n_families, genes, n_axes,
    gene_to_fam, n_genes, thresholds, neofunc, ierr)
```

2. Detect Dosage Effect

Identifies subsets of paralogs with small angle to the ancestor (`max_angle`) and sum to a magnitude significantly exceeding `norm(ancestor)` (`gain`).

The size of the `work_arr_paralog_subsets` has binomial complexity. This is why this subroutine only handles one single family, the size varies too much across them. If it is wanted to have all results in one array, consider an array of pointers, using a derived type.

Arguments:

- `ancestor` `real`(real64), `dimension`(n_dims): expression vector of ancestral ortholog
- `genes` `real`(real64), `dimension`(n_dims, n_genes): expression vectors of genes
- `n_genes` `integer`(int32): number of vectors in `genes`
- `n_dims` `integer`(int32): size of `ancestor` vector and vectors in `genes`
- `filtered_paralogs_mask` `integer`(int32), `dimension`(n_mask_chunks): bit mask with genes' indices kept by pattern set to 1, else 0. Use `filter_paralogs_by_pattern` for its calculation
- `n_mask_chunks` `integer`(int32): number of 32 bit chunks a mask needs to encode `n_genes` genes. Use subroutine `mask_chunk_count` for calculation
- `n_results` `integer`(int32): number of resulting subsets. They are stored as the first `n_results` elements of `work_arr_paralog_subsets`
- `max_subset_size` `integer`(int32): maximum subset size of checked gene subsets.
USE `calc_work_arr_paralog_subsets_size` **TO DETERMINE THIS NUMBER**
- `work_arr_paralog_subsets` `integer`(int32), `dimension`(n_mask_chunks, n_paralog_subsets): working array to hold bitmask encoded subsets for detection. Each bitmask is built of 32 bit chunks. $\frac{n_genes+31}{32}$ is equivalent to $ceil(\frac{n_genes}{32})$ and represents the number of chunks.

- `n_paralog_subsets` `integer(int32)`: number of gene subsets that can be stored in `work_arr_paralog_subsets`.
USE `calc_work_arr_paralog_subsets_size` TO DETERMINE THIS NUMBER
- `active_mask` `integer(int32)`, `dimension(n_mask_chunks)`: working array to hold the extended subsets
- `temp_paralog_vector` `real(real64)`, `dimension(n_dims)`: vector used for pruning subsets
- `ierr` `integer(int32)`: error code
- `max_angle` `real(real64)`, `optional`: in dosage mode maximum angle in radians $0 \leq \text{angle} \leq \pi$ that a subset candidate must not exceed, otherwise pruned, default is π
- `gain_gamma` `real(real64)`, `optional`: positive magnitude gain for dosage effect, default 0.1

Listing 84: Dosage Effect Detection

```
call detect_dosage_effect(ancestor, genes, n_genes, n_dims,
    filtered_paralogs_mask, n_mask_chunks, &
    n_results, max_subset_size,
    work_arr_paralog_subsets, n_paralog_subsets
    , &
    active_mask, temp_paralog_vector, ierr,
    max_angle, gain_gamma)
```

3. Detect Subfunctionalization

Identifies subsets of paralogs exhibiting significant angles to the ancestor.

The size of the `work_arr_paralog_subsets` has binomial complexity. This is why this subroutine only handles one single family, the size varies too much across them. If it is wanted to have all results in one array, consider an array of pointers, using a derived type.

Arguments:

- `ancestor` `real(real64)`, `dimension(n_dims)`: expression vector of ancestral ortholog
- `genes` `real(real64)`, `dimension(n_dims, n_genes)`: expression vectors of genes
- `n_genes` `integer(int32)`: number of vectors in genes
- `n_dims` `integer(int32)`: size of ancestor vector and vectors in genes
- `rdi_threshold` `real(real64)`: max allowed residual distance from ancestor
- `filtered_paralogs_mask` `integer(int32)`, `dimension(n_mask_chunks)`: bit mask with genes' indices kept by pattern set to 1, else 0. Use `filter_paralogs_by_pattern` for its calculation
- `n_mask_chunks` `integer(int32)`: number of 32 bit chunks a mask needs to encode `n_genes` genes. Use subroutine `mask_chunk_count` for calculation
- `n_results` `integer(int32)`: number of resulting subsets. They are stored as the first `n_results` elements of `work_arr_paralog_subsets`
- `max_subset_size` `integer(int32)`: maximum subset size of checked gene subsets.
USE `calc_work_arr_paralog_subsets_size` TO DETERMINE THIS NUMBER
- `work_arr_paralog_subsets` `integer(int32)`, `dimension(n_mask_chunks, n_paralog_subsets)`: working array to hold bitmask encoded subsets for detection. Each bitmask is built of 32 bit chunks. $\frac{n_genes+31}{32}$ is equivalent to $\text{ceil}(\frac{n_genes}{32})$ and represents the number of chunks.
- `n_paralog_subsets` `integer(int32)`: number of gene subsets that can be stored in `work_arr_paralog_subsets`.
USE `calc_work_arr_paralog_subsets_size` TO DETERMINE THIS NUMBER
- `active_mask` `integer(int32)`, `dimension(n_mask_chunks)`: working array to hold the extended subsets
- `temp_paralog_vector` `real(real64)`, `dimension(n_dims)`: vector used for pruning subsets
- `paralog_norms` `real(real64)`, `dimension(n_genes)`: needed for subset pruning, holds the eu-

clidean norms of genes (you can use the `norm` function from `f42_utils` for this)

- `sorted_paralog_norms_perm integer(int32), dimension(n_genes)`: needed for subset pruning, as the minimum norm of the genes that could extend a subset should not be lower than the subset angle to the ancestor
- `temp_work_array real(real64), dimension(n_genes)`: needed for efficient check of minimum value after a certain index
- `ierr integer(int32)`: error code

Listing 85: Subfunctionalization Detection

```
call detect_subfunctionalization(ancestor, genes, n_genes, n_dims,
    rdi_threshold, filtered_paralogs_mask, &
                                n_mask_chunks, n_results,
                                max_subset_size,
                                work_arr_paralog_subsets, &
                                n_paralog_subsets, active_mask,
                                temp_paralog_vector, paralog_norms,
                                &
                                sorted_paralog_norms_perm,
                                temp_work_array, ierr)
```

4. Mask Chunk Count

This subroutine easily determines the needed chunk count for subset bit masks, as an integer has only 32 bits.

Arguments:

- `n_genes integer(int32)`: number of genes
- `count integer(int32)`: number of 32 bit chunks a mask needs to encode `n_genes` genes

Listing 86: Mask Chunk Count

```
call mask_chunk_count(n_genes, count)
```

5. Filter Paralogs By Pattern Subfunctionalization

This subroutine prefilters the genes for subfunctionalization, as genes that are already too close in angle to the ancestor don't match the pattern and don't need to be tried as subset extensions.

Arguments:

- `gene_angles real(real64), dimension(n_genes)`: vector holding the angles between ancestor and genes ($0 \leq \text{angle} \leq \pi$)
- `threshold real(real64)`: filter threshold
- `n_genes integer(int32)`: number of genes
- `n_families integer(int32)`: number of families
- `gene_to_fam integer(int32), dimension(n_genes)`: mapping of gene index to family index, so gene `i` is related to `gene_angles(i)` and part of family `j=gene_to_fam(i)`
- `masks integer(int32), dimension(n_mask_chunks, n_families)`: bit mask that will have indices of genes kept by pattern set to 1, else 0
- `n_mask_chunks integer(int32)`: number of 32 bit chunks a mask needs to encode `n_genes` genes
- `ierr integer(int32)`: error code

Listing 87: Prefiltering Paralogs for Subfunctionalization

```
call filter_paralogs_by_pattern_subfunctionalization(gene_angles,
    threshold, n_genes, n_families, &
```



```
gene_to_fam, masks,
n_mask_chunks,
ierr)
```

6. Filter Paralogs By Pattern Dosage Effect

This subroutine prefilters the genes for dosage effect, as genes that are already too distant in angle to the ancestor don't match the pattern and don't need to be tried as subset extensions.

Arguments:

- `gene_angles` `real(real64)`, `dimension(n_genes)`: vector holding the angles between ancestor and genes ($0 \leq \text{angle} \leq \pi$)
- `threshold` `real(real64)`: filter threshold
- `n_genes` `integer(int32)`: number of genes
- `n_families` `integer(int32)`: number of families
- `gene_to_fam` `integer(int32)`, `dimension(n_genes)`: mapping of gene index to family index, so gene *i* is related to `gene_angles(i)` and part of family `j=gene_to_fam(i)`
- `masks` `integer(int32)`, `dimension(n_mask_chunks, n_families)`: bit mask that will have indices of genes kept by pattern set to 1, else 0
- `n_mask_chunks` `integer(int32)`: number of 32 bit chunks a mask needs to encode `n_genes` genes
- `ierr` `integer(int32)`: error code

Listing 88: Prefiltering Paralogs for Dosage Effect

```
call filter_paralogs_by_pattern_dosage_effect(gene_angles, threshold,
n_genes, n_families, &
gene_to_fam, masks,
n_mask_chunks, ierr)
```

7. Calculate Work Array Paralog Subsets Size

The `detect_*` subroutines need a work array for the to-be-tested subsets. In worst case, all need to be tried and subsets that cannot be extended will be kept as results. This is the reason why the work array holds the results as well, as all subsets that are stored in the array can be results as well.

This subroutine calculates the needed size for the work array.

Arguments:

- `max_subset_size` `integer(int32)`: maximum size that a subset must not exceed. If the desired size is too large and leads to an integer overflow, `max_subset_size` will be set to the maximum valid size. Also, size will be set to number of genes in `filtered_paralogs_mask` if larger.
- `n_genes` `integer(int32)`: number of genes
- `work_array_size` `integer(int32)`: the calculated needed work array size in absolute worst case scenario. Look into source for details.
- `filtered_paralogs_mask` `integer(int32)`, `dimension(n_mask_chunks)`: output mask with all genes disabled that did not pass the filter
- `n_mask_chunks` `integer(int32)`: number of 32 bit chunks a mask needs to encode `n_genes` genes
- `ierr` `integer(int32)`: error code

Listing 89: Work Array Size Calculation

```
call calc_work_arr_paralog_subsets_size(max_subset_size, n_genes,
work_array_size, &
```



```
filtered_paralogs_mask,
n_mask_chunks, ierr)
```

8. Mask Set State

Sets the state of a bit/gene in `bit_mask`.

Arguments:

- `bit_mask` `integer(int32)`, `dimension(:)`: chunked mask to mark active paralogs
- `i_gene` `integer(int32)`: index of paralog to be marked active
- `state` `logical`: state the bit should be set to
- `ierr` `integer(int32)`: error code

Listing 90: Set Bit State in Mask

```
call mask_set_state(bit_mask, i_gene, state, ierr)
```

9. Mask Check State

Checks the state of a bit/paralog in `bit_mask` $\Rightarrow$ `.true.` if 1 else `false`.

Arguments:

- `bit_mask` `integer(int32)`, `dimension(:)`: chunked mask to mark active paralogs
- `i_gene` `integer(int32)`: index of paralog to be marked active

Result:

- `state` `logical`: check result

Listing 91: Check Bit State in Mask

```
state = mask_check_state(bit_mask, i_gene)
```

3.1.9 Trajectory Contribution Analysis

1. Fortran Core Contribution Analysis Functions

The Fortran module provides core routines for trajectory contribution analysis using cosine similarity and spike detection.

(a) Trajectory Contribution Calculation

Calculates integrated trajectory contribution using cosine similarity between factor and dependent vectors.

Mathematical Formula:

$$\text{contribution} = \frac{\text{factor} \cdot \text{dependent}}{\|\text{factor}\| \cdot \|\text{dependent}\|}$$

Arguments:

- `n_timepoints` `integer(int32)`: Number of timepoints
- `factor` `real(real64)`: Vector of independent variable [`n_timepoints`]
- `dependent` `real(real64)`: Vector of dependent variable [`n_timepoints`]
- `mode` `integer(int32)`: Processing mode (1 = Normal, 2 = RAP)
- `contribution` `real(real64)`: Output contribution value
- `ierr` `integer(int32)`: Error code

Listing 92: Trajectory Contribution Calculation

```
call trajectory_contribution(factor, dependent, n_timepoints, mode, &
                             contribution, ierr)
```

(b) **Spike Contribution Calculation**

Calculates timepoint-wise spike contributions using cosine similarity for each individual timepoint.

Mathematical Formula:

$$\text{contribution}(i) = \frac{\text{factor}(i) \cdot \text{dependent}(i)}{\|\text{factor}\| \cdot \|\text{dependent}\|} \quad \text{for } 1 \leq i \leq n_timepoints$$

Arguments:

- `n_timepoints` integer(int32): Number of timepoints
- `factor` real(real64): Vector of independent variable [n_timepoints]
- `dependent` real(real64): Vector of dependent variable [n_timepoints]
- `mode` integer(int32): Processing mode (1 = Normal, 2 = RAP)
- `contribution` real(real64): Output spike contributions [n_timepoints]
- `ierr` integer(int32): Error code

Listing 93: Spike Contribution Calculation

```
call spike_contribution(factor, dependent, n_timepoints, mode, &
                        contribution, ierr)
```

(c) **Calculate Contributions (Expert Version)**

Calculates both spike and integrated contributions for a specific factor with pre-allocated work arrays.

Arguments:

- `n_factors` integer(int32): Number of factors in trajectories
- `n_samples` integer(int32): Number of samples in trajectories
- `n_timepoints` integer(int32): Number of timepoints in trajectories
- `trajectories` real(real64): 3D trajectories array [n_factors, n_samples, n_timepoints]
- `i_factor` integer(int32): Factor index to calculate contributions for
- `dependent_idx` integer(int32): Dependent variable index
- `mode` integer(int32): Processing mode (1 = Normal, 2 = RAP)
- `integrated_contri` real(real64): Output integrated contributions [n_samples]
- `spike_contri` real(real64): Output spike contributions [n_timepoints, n_samples]
- `temp_factor_vector` real(real64): Work array [n_timepoints]
- `temp_dependent_vector` real(real64): Work array [n_timepoints]
- `ierr` integer(int32): Error code

Listing 94: Expert Contributions Calculation

```
call calc_contributions(trajectories, n_factors, n_samples, &
                        n_timepoints, i_factor, dependent_idx, mode, &
                        spike_contri, integrated_contri, &
                        temp_factor_vector, temp_dependent_vector, ierr)
```

(d) **Calculate Contributions (Allocation Version)**

Calculates both spike and integrated contributions with internal memory allocation.

Arguments:

- `n_factors` integer(int32): Number of factors in trajectories
- `n_samples` integer(int32): Number of samples in trajectories
- `n_timepoints` integer(int32): Number of timepoints in trajectories
- `trajectories` real(real64): 3D trajectories array [n_factors, n_samples, n_timepoints]

- `i_factor` integer(int32): Factor index to calculate contributions for
- `dependent_idx` integer(int32): Dependent variable index
- `mode` integer(int32): Processing mode (1 = Normal, 2 = RAP)
- `integrated_contribs` real(real64): Output integrated contributions [n_samples]
- `spike_contribs` real(real64): Output spike contributions [n_timepoints, n_samples]
- `ierr` integer(int32): Error code

Listing 95: Allocation-Based Contributions Calculation

```
call calc_contributions_alloc(trajectories, n_factors, n_samples, &
                             n_timepoints, i_factor, dependent_idx, mode, &
                             spike_contribs, integrated_contribs, ierr)
```

(e) Vector Accessor Routines

Extract specific slices from the 3D trajectories array.

Arguments for Sample Extraction:

- `n_factors` integer(int32): Number of factors
- `n_samples` integer(int32): Number of samples
- `n_timepoints` integer(int32): Number of timepoints
- `trajectories` real(real64): 3D trajectories array
- `i_factor` integer(int32): Factor index
- `i_timepoint` integer(int32): Timepoint index
- `result_vector` real(real64): Output vector [n_samples]
- `ierr` integer(int32): Error code

Listing 96: Extract Samples Vector

```
call get_vec_across_samples(trajectories, n_factors, n_samples, &
                           n_timepoints, i_factor, i_timepoint, &
                           result_vector, ierr)
```

Arguments for Timepoint Extraction:

- `n_factors` integer(int32): Number of factors
- `n_samples` integer(int32): Number of samples
- `n_timepoints` integer(int32): Number of timepoints
- `trajectories` real(real64): 3D trajectories array
- `i_factor` integer(int32): Factor index
- `i_sample` integer(int32): Sample index
- `result_vector` real(real64): Output vector [n_timepoints]
- `ierr` integer(int32): Error code

Listing 97: Extract Timepoints Vector

```
call get_vec_across_timepoints(trajectories, n_factors, n_samples, &
                              n_timepoints, i_factor, i_sample, &
                              result_vector, ierr)
```

(f) Outlier detection

The outlier detection workflow consists of two main pathways: integrated (trajectory-level) analysis and spike (timepoint-level) analysis. You can perform outlier detection by calling the following subroutines in sequence:

i. Calculate Integrated Threshold

The subroutine calculates an empirical threshold for integrated (trajectory-level) contributions using pre-computed permutations. This determines which samples are significant

overall across all genes. Call the `calc_integrated_threshold` subroutine with the following arguments:

Input arguments:

- `n_samples` integer(int32): Number of samples
- `contributions` real(real64): 1D array of integrated contributions (size: `n_samples`)
- `percentile_val` real(real64): Percentile value for threshold (0.0–100.0)
- `permutation` integer(int32): Pre-computed permutation indices for sorted contributions (size: `n_samples`)

Output arguments:

- `threshold` real(real64): Scalar threshold value
- `ierr` integer(int32): Error code (for error details please check [error handling section](#))

Listing 98: Integrated Threshold Calculation

```
call calc_integrated_threshold(contributions, n_samples, &
                              percentile_val, threshold, &
                              permutation, ierr)
```

ii. **Detect Integrated Outliers**

The subroutine identifies outlier samples where the integrated contribution exceeds the empirical threshold. Used to find trajectories with significant overall explanatory contributions. Call the `detect_outliers_integrated` subroutine with the following arguments:

Input arguments:

- `n_samples` integer(int32): Number of samples
- `contributions` real(real64): Array of integrated contributions (size: `n_samples`)
- `threshold` real(real64): Scalar threshold value for outlier detection

Output arguments:

- `outlier_mask` logical: 1D logical array indicating outliers (size: `n_samples`)
- `ierr` integer(int32): Error code (for error details please check [error handling section](#))

Listing 99: Integrated Outlier Detection

```
call detect_outliers_integrated(contributions, n_samples, &
                               threshold, outlier_mask, ierr)
```

iii. **Calculate Spike Thresholds**

The subroutine calculates empirical thresholds for spike contributions using pre-sorted permutations. Computes percentile-based thresholds for each timepoint in spike contribution data. Call the `calc_spike_thresholds` subroutine with the following arguments:

Input arguments:

- `n_timepoints` integer(int32): Number of timepoints
- `n_samples` integer(int32): Number of samples
- `spike_contribs` real(real64): 2D array of spike contributions (size: `n_timepoints` × `n_samples`)
- `percentile_val` real(real64): Percentile value for threshold (0.0–100.0)
- `permutation` integer(int32): Pre-computed permutation indices (size: `n_samples` × `n_timepoints`).

Output arguments:

- `thresholds` real(real64): 1D array of thresholds for each timepoint (size: `n_timepoints`)

- `ierr` integer(int32): Error code (for error details please check [error handling section](#))

Listing 100: Spike Thresholds Calculation

```
call calc_spike_thresholds(spike_contribs, n_timepoints, &
                           n_samples, percentile_val, thresholds, &
                           permutation, ierr)
```

iv. Detect Spike Outliers

The subroutine identifies outliers in spike contribution data by comparing against timepoint-specific thresholds. Performs element-wise comparison across all timepoints and samples. Call the `detect_outliers_spike` subroutine with the following arguments:

Input arguments:

- `n_timepoints` integer(int32): Number of timepoints
- `n_samples` integer(int32): Number of samples
- `spike_contribs` real(real64): 2D array of spike contributions (size: `n_timepoints` × `n_samples`)
- `thresholds` real(real64): Array of thresholds for each timepoint (size: `n_timepoints`)

Output arguments:

- `outlier_mask` logical: 2D logical array indicating outliers (size: `n_timepoints` × `n_samples`)
- `ierr` integer(int32): Error code (for error details please check [error handling section](#))

Listing 101: Spike Outlier Detection

```
call detect_outliers_spike(spike_contribs, n_timepoints, &
                           n_samples, thresholds, outlier_mask, ierr)
```

For convenience, allocation-friendly versions are also available:

- `calc_integrated_threshold_alloc`: Handles internal allocation and sorting
- `calc_spike_thresholds_alloc`: Handles internal allocation and sorting

(g) Process Trajectories with Per-Timepoint Percentiles

Complete pipeline processing with timepoint-specific spike thresholds.

Arguments:

- `n_factors` integer(int32): Number of factors
- `n_samples` integer(int32): Number of samples
- `n_timepoints` integer(int32): Number of timepoints
- `factor_mask_count` integer(int32): Number of true values in factor mask
- `trajectories` real(real64): 3D trajectories array
- `factor_mask` logical: Mask for included factors [`n_factors`]
- `dependent_idx` integer(int32): Dependent variable index
- `mode` integer(int32): Processing mode (1 = Normal, 2 = RAP)
- `percentile` real(real64): Percentile value (0.0–100.0)
- `integrated_contribs` real(real64): Output integrated contributions [`n_samples`, `factor_mask_count`]
- `spike_contribs` real(real64): Output spike contributions [`n_timepoints`, `n_samples`, `factor_mask_count`]
- `thresholds_integrated_contrib` real(real64): Integrated thresholds [`factor_mask_count`]
- `thresholds_spike_contrib` real(real64): Spike thresholds [`n_timepoints`, `factor_mask_count`]

- `outliers_integrated_contrib` logical: Integrated outliers [n_samples, factor_mask_count]
- `outliers_spike_contrib` logical: Spike outliers [n_timepoints, n_samples, factor_mask_count]
- `ierr` integer(int32): Error code

Listing 102: Complete Pipeline with Per-Timepoint Thresholds

```
call process_trajectories_alloc(trajectories, n_factors, n_samples, &
                               n_timepoints, factor_mask, factor_mask_count, &
                               dependent_idx, mode, percentile, &
                               integrated_contribs, spike_contribs, &
                               thresholds_integrated_contrib, &
                               outliers_integrated_contrib, &
                               thresholds_spike_contrib, &
                               outliers_spike_contrib, ierr)
```

(h) Process Trajectories with Global Percentile

Complete pipeline processing with global spike thresholds (flattened approach).

Arguments:

- `n_factors` integer(int32): Number of factors
- `n_samples` integer(int32): Number of samples
- `n_timepoints` integer(int32): Number of timepoints
- `factor_mask_count` integer(int32): Number of true values in factor mask
- `trajectories` real(real64): 3D trajectories array
- `factor_mask` logical: Mask for included factors [n_factors]
- `dependent_idx` integer(int32): Dependent variable index
- `mode` integer(int32): Processing mode (1 = Normal, 2 = RAP)
- `percentile` real(real64): Percentile value (0.0–100.0)
- `integrated_contribs` real(real64): Output integrated contributions [n_samples, factor_mask_count]
- `spike_contribs` real(real64): Output spike contributions [n_timepoints, n_samples, factor_mask_count]
- `thresholds_integrated_contrib` real(real64): Integrated thresholds [factor_mask_count]
- `thresholds_spike_contrib` real(real64): Spike thresholds [factor_mask_count] (1D)
- `outliers_integrated_contrib` logical: Integrated outliers [n_samples, factor_mask_count]
- `outliers_spike_contrib` logical: Spike outliers [n_timepoints, n_samples, factor_mask_count]
- `ierr` integer(int32): Error code

Listing 103: Complete Pipeline with Global Threshold

```
call process_trajectories_flat_alloc(trajectories, n_factors, &
                                     n_samples, n_timepoints, factor_mask, &
                                     factor_mask_count, dependent_idx, mode, &
                                     percentile, integrated_contribs, &
                                     spike_contribs, thresholds_integrated_contrib, &
                                     outliers_integrated_contrib, &
                                     thresholds_spike_contrib, &
                                     outliers_spike_contrib, ierr)
```

Processing Modes:

- `MODE.NORMAL` = 1: Returns cosine similarity for normal vectors
- `MODE.RAP` = 2: Returns arccos of cosine similarity for RAP projected vectors

Key Features:

- **Pure Subroutines:** Most routines are pure with no side effects
- **Robust Error Handling:** Comprehensive error checking and validation
- **Memory Efficient:** Expert versions allow pre-allocated work arrays
- **Mathematically Sound:** Uses cosine similarity with proper normalization
- **Flexible Processing:** Supports both per-timepoint and global thresholding

3.1.10 Fortran Utils

1. Binary Search Trees for 1D Range Queries

Binary Search Trees provide efficient indexing and range query capabilities for one-dimensional data using flat array implementations for optimal memory performance.

Fortran BST Implementation

(a) Build BST Index

Constructs a Binary Search Tree index by sorting the input values and storing permutation indices. Uses an efficient stack-based sorting algorithm.

Input:

- `values` `real(real64)`: 1D array of values to index
- `num_values` `integer(int32)`: Number of elements

Output:

- `sorted_indices` `integer(int32)`: Permutation indices after sorting (1-based)
- `left_stack`, `right_stack` `integer(int32)`: Sorting workspace
- `ierr` `integer(int32)`: Error code

```
call build_bst_index(values, num_values, sorted_indices, left_stack, &
                    right_stack, ierr)
```

(b) Get Sorted Value

Retrieves the value at a specific position in the sorted order defined by the BST index.

Input:

- `values` `real(real64)`: Original values array
- `sorted_indices` `integer(int32)`: BST index array (1-based)
- `position` `integer(int32)`: Position in sorted order (1-based)

Output:

- `sorted_value` `real(real64)`: Value at specified position
- `ierr` `integer(int32)`: Error code

```
sorted_value = get_sorted_value(values, sorted_indices, &
                               position, ierr)
```

(c) BST Range Query

Efficiently finds all values within a specified range using the precomputed BST index.

Input:

- `values` `real(real64)`: Original values array
- `sorted_indices` `integer(int32)`: BST index array (1-based)
- `num_values` `integer(int32)`: Number of elements
- `lower_bound`, `upper_bound` `real(real64)`: Range boundaries

Output:

- `output_indices` `integer(int32)`: Indices of matching values (1-based)
- `num_matches` `integer(int32)`: Number of matches found
- `ierr` `integer(int32)`: Error code


```
call bst_range_query(values, sorted_indices, num_values, &
                    lower_bound, upper_bound, output_indices, &
                    num_matches, ierr)
```

2. k-d Trees for Multi-Dimensional Indexing

k-d trees provide efficient space partitioning for multi-dimensional data with support for both Euclidean and spherical geometries.

Fortran k-d Tree Implementation

(a) Build k-d Tree Index

Constructs a k-d tree index using stack-based non-recursive partitioning along specified dimensions.

Input:

- `points` `real(real64)`: Data points matrix (dimensions $\times$ points)
- `num_dimensions` `integer(int32)`: Number of dimensions
- `num_points` `integer(int32)`: Number of points
- `dimension_order` `integer(int32)`: Dimension ordering for splitting (1-based)

Output:

- `kd_indices` `integer(int32)`: Output index array (1-based)
- `workspace` `integer(int32)`: Workspace array
- `value_buffer` `real(real64)`: Value buffer for sorting
- `permutation` `integer(int32)`: Permutation array
- `left_stack`, `right_stack` `integer(int32)`: Sorting stacks
- `recursion_stack` `integer(int32)`: Recursion stack
- `ierr` `integer(int32)`: Error code

```
call build_kd_index(points, num_dimensions, num_points, kd_indices, &
                  dimension_order, workspace, value_buffer, &
                  permutation, left_stack, right_stack, &
                  recursion_stack, ierr)
```

(b) Build Spherical k-d Tree

Specialized k-d tree construction optimized for unit vectors on hyperspheres.

Input:

- `vectors` `real(real64)`: Unit vectors matrix (dimensions $\times$ vectors)
- `num_dimensions` `integer(int32)`: Number of dimensions
- `num_vectors` `integer(int32)`: Number of vectors
- `dimension_order` `integer(int32)`: Dimension ordering for splitting (1-based)

Output:

- `sphere_indices` `integer(int32)`: Output index array (1-based)
- Various workspace arrays (same as `build_kd_index`)
- `ierr` `integer(int32)`: Error code

```
call build_spherical_kd(vectors, num_dimensions, num_vectors, &
                      sphere_indices, dimension_order, workspace, &
                      value_buffer, permutation, left_stack, &
                      right_stack, recursion_stack, ierr)
```

(c) Get Point from k-d Index

Retrieves point coordinates from specified position in k-d tree ordering.

Input:

- `points` `real(real64)`: Original points matrix
- `kd_indices` `integer(int32)`: k-d tree index array (1-based)
- `position` `integer(int32)`: Position in index (1-based)

Output:

- `point_values` `real(real64)`: Retrieved point values
- `ierr` `integer(int32)`: Error code

```
call get_kd_point(points, kd_indices, position, point_values, ierr)
```

3. Hashing

The TensorOmics package provides high-performance hashing utilities using the XXH3 algorithm for both hash maps and hash sets. These utilities are implemented in Fortran with separate chaining collision resolution and automatic resizing.

(a) Hashmap

The hashmap implementation provides key-value storage with string keys and integer values, using the XXH3 hashing algorithm for optimal performance.

- **Type Definition:**

```
type(hashmap_type) :: map
```

- **Create Hashmap:** Initialize a new hashmap with optional initial size. The Hashmap size will be set to the next greater power of two.

```
call hashmap_create(map, initial_size)
```

- `map`: Hashmap object to create
- **(Optional)** `initial_size`: Initial size of the hashmap

- **Insert Key-Value Pair:** Insert or update a key-value pair in the hashmap.

```
call hashmap_put(map, key, value)
```

- `map`: Hashmap to insert into
- `key`: String key to store
- `value`: Integer value to store

- **Lookup Key:** Retrieve the value associated with a key.

```
value = hashmap_get(map, key)
```

- `map`: Hashmap to search
- `key`: String key to look for
- **Returns:** Integer value, or -1 if key not found

- **Destroy Hashmap:** Clean up and deallocate the hashmap. It is important to always call this function when the hashmap is no longer in use to avoid memory leaks.

```
call hashmap_destroy(map)
```

- `map`: Hashmap to destroy

Hashmap Usage Example:

```
use xxh3_hashmap_module

type(hashmap_type) :: gene_map
integer :: gene_index
```

```

! Create hashmap for gene IDs
call hashmap_create(gene_map, initial_size=1000)

! Store gene indices
call hashmap_put(gene_map, "Gene_001", 1)
call hashmap_put(gene_map, "Gene_002", 2)
call hashmap_put(gene_map, "Gene_003", 3)

! Lookup gene index
gene_index = hashmap_get(gene_map, "Gene_002")
! gene_index = 2

! Clean up
call hashmap_destroy(gene_map)

```

(b) Hashset

The hashset implementation provides unique string storage with fast membership testing, using the same XXH3 hashing algorithm as the hashmap.

- **Type Definition:**

```
type(hashset_type) :: set
```

- **Create Hashset:** Initialize a new hashset with optional initial size. The hashset size will be increased to the next greatest power of two.

```
call hashset_create(set, initial_size)
```

- `set`: Hashset object to create
- **(Optional)** `initial_size`: Initial size of the hashset

- **Insert Key:** Insert a unique key into the hashset.

```
call hashset_put(set, key, ierr)
```

- `set`: Hashset to insert into
- `key`: String key to store
- `ierr`: Error code (0 if successful, error if duplicate)

- **Check Membership:** Test if a key exists in the hashset.

```
exists = is_in_hashset(set, key)
```

- `set`: Hashset to search
- `key`: String key to check
- **Returns:** Logical indicating membership

- **Destroy Hashset:** Clean up and deallocate the hashset. It is important that this function is always called when the hashset is no longer in use to avoid memory leaks.

```
call hashset_destroy(set)
```

- `set`: Hashset to destroy

Hashset Usage Example:

```

use xxh3_hashmap_module

type(hashset_type) :: unique_genes
integer :: ierr

```

```

logical :: exists

! Create hashset for unique gene validation
call hashset_create(unique_genes, initial_size=5000)

! Add genes to set
call hashset_put(unique_genes, "Gene_001", ierr)
call hashset_put(unique_genes, "Gene_002", ierr)
call hashset_put(unique_genes, "Gene_003", ierr)

! Check for duplicates (will return error)
call hashset_put(unique_genes, "Gene_001", ierr)
! ierr != 0 indicates duplicate key

! Test membership
exists = is_in_hashset(unique_genes, "Gene_002")
! exists = .true.

! Clean up
call hashset_destroy(unique_genes)

```

Technical Features:

- Uses XXH3 non-cryptographic hash algorithm for optimal performance
- Automatic resizing when load factor exceeds 0.75
- Separate chaining collision resolution
- Table sizes are powers of two for efficient modulo operations
- Keys are automatically trimmed and normalized
- Memory efficient with proper cleanup procedures
- Thread-safe for read operations (concurrent writes not supported)

Performance Characteristics:

- Average case $O(1)$ for insertions and lookups
- Memory usage proportional to number of elements + overhead
- Resizing operations are $O(n)$ but amortized over many insertions

4. Sorting Algorithms

The TensorOmics package provides high-performance quicksort implementations for various data types using iterative algorithms with manual stack management. These sorting routines operate indirectly through permutation vectors to preserve the original data order.

(a) Generic Sorting Interface

A unified interface for sorting different data types through the `sort_array` generic procedure.

• Generic Procedure:

```
call sort_array(array, perm, stack_left, stack_right)
```

- `array`: Input array to sort (real, integer, or character)
- `perm`: Permutation vector that will be reordered
- `stack_left`: Work array for left indices (size = array size)
- `stack_right`: Work array for right indices (size = array size)

Supported Types:

```

! Real arrays
real(real64) :: data_real(:)
! Integer arrays

```

```
integer(int32) :: data_int(:)
! Character arrays
character(len=*) :: data_char(:)
```

(b) Type-Specific Sorting Procedures

Direct access to type-specific sorting routines for explicit control.

• Sort Real Arrays:

```
call sort_real(array, perm, stack_left, stack_right)
```

- array: Real(real64) input array to sort
- perm: Permutation vector that will be sorted
- stack_left: Manual stack of left indices
- stack_right: Manual stack of right indices

• Sort Integer Arrays:

```
call sort_integer(array, perm, stack_left, stack_right)
```

- array: Integer(int32) input array to sort
- perm: Permutation vector that will be sorted
- stack_left: Manual stack of left indices
- stack_right: Manual stack of right indices

• Sort Character Arrays:

```
call sort_character(array, perm, stack_left, stack_right)
```

- array: Character input array to sort (lexicographic order)
- perm: Permutation vector that will be sorted
- stack_left: Manual stack of left indices
- stack_right: Manual stack of right indices

Sorting Usage Example:

```
use f42_utils
use iso_fortran_env, only: real64, int32

real(real64) :: expression_data(1000)
integer(int32) :: perm(1000), stack_left(1000), stack_right(1000)
integer :: i

! Initialize permutation
do i = 1, 1000
  perm(i) = i
end do

! Sort expression data indirectly
call sort_real(expression_data, perm, stack_left, stack_right)

! Access sorted data through permutation
! expression_data(perm(1)) is the smallest value
! expression_data(perm(1000)) is the largest value
```

(c) Technical Implementation Details

- **Algorithm:** Iterative quicksort with manual stack management
- **Sorting Method:** Indirect sorting via permutation vectors

- **Pivot Selection:** Median-of-three (center element)
- **Memory:** Requires two work arrays of same size as input
- **Performance:** Average case $O(n \log n)$, worst case $O(n^2)$
- **Stability:** Not stable (order of equal elements not preserved)

(d) **Work Array Requirements**

For sorting an array of size n , you must provide:

- `perm(n)`: Permutation vector (initialize with 1, 2, ..., n)
- `stack_left(n)`: Work array for stack operations
- `stack_right(n)`: Work array for stack operations

Initialization Pattern:

```
do i = 1, n
    perm(i) = i          ! Identity permutation
end do
```

(e) **Advanced Usage: Direct Algorithm Access**

For maximum performance in tight loops, you can call the internal quicksort procedures directly:

```
! Internal quicksort for real arrays
call quicksort_real(array, perm, n, stack_left, stack_right)

! Internal quicksort for integer arrays
call quicksort_int(array, perm, n, stack_left, stack_right)

! Internal quicksort for character arrays
call quicksort_char(array, perm, n, stack_left, stack_right)
```

- These procedures require explicit array size parameter n
- Work arrays must be properly initialized
- Provides slightly better performance by skipping interface overhead

Performance Considerations:

- Manual stack management avoids recursion limits for large arrays
- Indirect sorting preserves original data and allows multiple sort orders
- Character sorting uses Fortran's intrinsic lexicographic comparison
- Memory usage is $4n$ for all data including input, work array, and both stacks.
- In case of memory considerations, heapsort might be used.

Error Handling:

- All sorting procedures are **pure** (no side effects)
- No explicit error codes - assumes valid input dimensions
- User must ensure work arrays are properly sized
- Invalid permutations may result in incorrect sorting

3.2 Python Pipeline

3.2.1 Overview

The Python workflow is exposed through the `tensoromics_functions.py` module. The functions provide a high-level interface that mirrors the Fortran pipeline.

3.2.2 Data Management

1. Data Reading and Processing

- **Read Gene IDs from TSV File**

The function reads gene identifiers from a tab-separated value (TSV) file, extracting specific gene IDs from a designated column while skipping header rows. This function is typically used as the first step in processing gene expression data to obtain the complete list of gene identifiers. Call the `read_gene_ids_from_tsv_file` function with the following arguments:

- `filename`: Path to the TSV file containing gene IDs (string or list with single file)
- `n_genes`: Total number of genes in the file
- `gene_ids_len`: Maximum character length for each gene ID
- `n_header_rows`: Number of header rows to skip at the beginning of the file
- `gene_col`: Column index (1-based) where gene IDs are located

Returns:

- `gene_ids`: Numpy array of shape (`n_genes`) containing gene ID strings

Listing 104: Python Read Gene IDs from TSV File

```
gene_ids = read_gene_ids_from_tsv_file(filename, n_genes,
                                       gene_ids_len, n_header_rows, gene_col)
```

- **Read Expression Vectors**

The function reads gene expression data from multiple tabular files (CSV/TSV format), extracting expression values for specified genes across multiple samples. This function consolidates expression data from multiple files into a single 2D array. Call the `read_expression_vectors` function with the following arguments:

- `file_list`: List of file paths containing expression data
- `gene_ids`: Array of gene IDs to extract expression values for
- `n_samples`: Number of samples
- `n_header_rows`: Number of header rows to skip in each file
- `gene_col`: Column index (1-based) containing gene identifiers
- `value_cols`: List of column indices (1-based) containing expression values
- **(Optional)** `delimiter`: Delimiter used in the files (default: tab)

Returns:

- `expression_vectors`: 2D numpy array of shape (`n_samples`, `n_genes`) containing expression values as 64-bit floats

Listing 105: Python Read Expression Vectors

```
expression_vectors = read_expression_vectors(file_list, gene_ids,
                                             n_samples, n_header_rows, gene_col,
                                             value_cols, delimiter='\t')
```

- **Read OrthoFinder File**

The function processes an OrthoFinder output file to map genes to their respective orthologous families. This function establishes the gene-family relationships essential for downstream orthology-based analyses. Call the `read_orthofinder_file` function with the following arguments:

- `filename`: Path to the OrthoFinder TSV file (string or list with single file)
- `gene_ids`: Complete list of gene identifiers to map
- `family_ids_len`: Maximum character length for family IDs

- **n_families**: Total number of gene families expected

Return dictionary:

- **family_ids**: Numpy array of shape (**n_families**,) containing family ID strings
- **gene_to_fam**: Numpy array of shape (**n_genes**,) containing family indices for each gene (0 = unassigned)

Listing 106: Python Read OrthoFinder File

```
family_result = read_orthofinder_file(filename, gene_ids,
                                     family_ids_len, n_families)
```

Complete Data Loading Workflow Example:

Listing 107: Python Complete Data Loading Workflow

```
# Step 1: Load gene identifiers
gene_ids = read_gene_ids_from_tsv_file(
    filename="data/expression_data.tsv",
    n_genes=50000,
    gene_ids_len=32,
    n_header_rows=1,
    gene_col=1
)

# Step 2: Load expression data from multiple samples
samples = ["sample1.tsv", "sample2.tsv", "sample3.tsv", "sample4.tsv"]
expression_vectors = read_expression_vectors(
    file_list=samples,
    gene_ids=gene_ids,
    n_samples=12, # assuming 3 samples per file
    n_header_rows=1,
    gene_col=1,
    value_cols=[2, 3]
)

# Step 3: Load orthology information
orthology_data = read_orthofinder_file(
    filename="data/Orthogroups.tsv",
    gene_ids=gene_ids,
    family_ids_len=32,
    n_families=15000
)
```

2. Data Filtering and Validation

• Filter Unassigned Genes

The function filters out genes that are not assigned to any family (where `gene_to_fam = 0`). This is essential for cleaning data before orthology-based analyses, ensuring only genes with valid family assignments are processed. Call the `filter_unassigned_genes` function with the following arguments:

- **gene_to_fam**: Family assignment array where 0 indicates unassigned genes

Return dictionary:

- **mask**: Boolean array indicating which genes to keep (1 = assigned to family, 0 = unassigned)
- **n_genes_kept**: Number of genes kept after filtering

Listing 108: Python Filter Unassigned Genes

```
filtered_genes = filter_unassigned_genes(gene_to_fam)
```

- **Validate Gene to Family Mapping**

The function validates the gene-to-family mapping array, ensuring all family indices are within valid range and properly assigned. This prevents downstream errors from invalid family indices. Call the `validate_gene_to_family_mapping` function with the following arguments:

- `gene_to_fam`: Array mapping each gene to a family index (0 = unassigned)
- `n_families`: Total number of families

Listing 109: Python Validate Gene to Family Mapping

```
validate_gene_to_family_mapping(gene_to_fam, n_families)
```

- **Validate Expression Data**

The function validates expression data, checking for NaN/Inf values and optionally verifying non-negative values. This ensures data quality before computational analyses. Call the `validate_expression_data` function with the following arguments:

- `expression_vectors`: 2D array of expression data (n_samples x n_genes)
- **(Optional)** `check_non_negative`: Whether to check for non-negative values (default: True)

Listing 110: Python Validate Expression Data

```
validate_expression_data(expression_vectors, check_non_negative=True)
```

- **Validate Family Centroids**

The function validates family centroids data, checking for NaN/Inf values and proper data structure. Centroids represent the central tendency of each gene family's expression profile. Call the `validate_family_centroids` function with the following arguments:

- `family_centroids`: 2D array of family centroids (samples x families)

Listing 111: Python Validate Family Centroids

```
validate_family_centroids(family_centroids)
```

- **Validate Shift Vectors**

The function validates shift vectors, ensuring data types are correct and structure matches expected patterns. Shift vectors represent the deviation of each gene from its family centroid. Call the `validate_shift_vectors` function with the following arguments:

- `shift_vectors`: 2D array of shift vectors (samples × genes)
- `expression_vectors`: 2D array of expression data (samples × genes)
- `family_centroids`: 2D array of family centroids (samples × families)
- `gene_to_fam`: Array mapping each gene to a family index
- `n_genes`: Number of genes
- `n_samples`: Number of samples
- `n_families`: Number of families

Listing 112: Python Validate Shift Vectors

```
validate_shift_vectors(shift_vectors, expression_vectors,
                      family_centroids, gene_to_fam, n_genes,
                      n_samples, n_families)
```


- **Validate String Array Uniqueness**

The function validates that all strings in an array are unique, using a hashset internally. Note:

This may increase memory usage temporarily for large datasets. Call the `validate_string_array_uniqueness` function with the following arguments:

- `strings`: Array of strings to validate (typically gene IDs or family IDs)

Listing 113: Python Validate String Array Uniqueness

```
validate_string_array_uniqueness(strings)
```

- **Validate Data Structure**

The function performs comprehensive validation of the overall data structure consistency, confirming sizes and dependencies between different data components as far as possible. Call the `validate_data_structure` function with the following arguments:

- `n_genes`: Number of genes
- `n_families`: Number of families
- `n_samples`: Number of samples
- `gene_ids`: Array of gene IDs
- `gene_family_ids`: Array of family IDs
- `gene_to_fam`: Array mapping each gene to a family index
- `expression_vectors`: 2D array of expression data (samples $\times$ genes)
- `family_centroids`: 2D array of family centroids (samples $\times$ families)
- `shift_vectors`: 2D array of shift vectors (genes $\times$ samples)

Listing 114: Python Validate Data Structure

```
validate_data_structure(n_genes, n_families, n_samples, gene_ids,
                        gene_family_ids, gene_to_fam, expression_vectors,
                        family_centroids, shift_vectors)
```

- **Validate All Data**

The function performs comprehensive validation of all data components in one call. This function executes all individual validations sequentially for complete data quality assurance. Call the `validate_all_data` function with the following arguments:

- `n_genes`: Number of genes
- `n_families`: Number of families
- `n_samples`: Number of samples
- `gene_ids`: Array of gene IDs
- `gene_family_ids`: Array of family IDs
- `gene_to_fam`: Array mapping each gene to a family index
- `expression_vectors`: 2D array of expression data (samples $\times$ genes)
- `family_centroids`: 2D array of family centroids (samples $\times$ families)
- `shift_vectors`: 2D array of shift vectors (samples $\times$ genes)

Listing 115: Python Validate All Data

```
validate_all_data(n_genes, n_families, n_samples, gene_ids,
                  gene_family_ids, gene_to_fam, expression_vectors,
                  family_centroids, shift_vectors)
```

Data Filtering and Validation Workflow Example:

Listing 116: Python Data Filtering and Validation Workflow

```

# After loading data, filter unassigned genes
filter_result = filter_unassigned_genes(gene_to_fam)
mask = np.array(filter_result['mask'], dtype=bool)
n_genes_kept = filter_result['n_genes_kept']

# Apply filter to all gene-based arrays
filtered_gene_ids = gene_ids[mask]
filtered_expression = expression_vectors[:, mask]
filtered_gene_to_fam = gene_to_fam[mask]

# Run comprehensive validations
validate_string_array_uniqueness(filtered_gene_ids)
validate_string_array_uniqueness(family_ids)
validate_expression_data(filtered_expression, check_non_negative=True)
validate_gene_to_family_mapping(filtered_gene_to_fam, n_families)

# For complete analysis with centroids and shift vectors
validate_data_structure(
    n_genes_kept, n_families, n_samples,
    filtered_gene_ids, family_ids, filtered_gene_to_fam,
    filtered_expression, family_centroids, shift_vectors
)

# Or use the comprehensive validation
validate_all_data(
    n_genes_kept, n_families, n_samples,
    filtered_gene_ids, family_ids, filtered_gene_to_fam,
    filtered_expression, family_centroids, shift_vectors
)

```

3.2.3 Data Integration

The Python module provides comprehensive routines for data integration, based on Jensen-Shannon Divergence. This is very useful for cross-study data.

Key Features:

- **Calculate Jensen-Shannon Compatibility Score:** Score to evaluate how compatible two studies are.
- **Perform JSD Permutation Test:** Random sampling of the input data and recalculation of the JS score to test stability of the compatibility.
- **Family-Wise Score Contribution:** Calculates the impact of selected families on the calculated score.

1. Compute Gene Means

Compute per-gene mean expression values across replicates, ignoring any NaN entries. The input expression matrix is converted to Fortran order for efficient passing to the underlying compiled routine.

Arguments:

- **expr** (array-like): array of shape (n_reps, n_genes) containing expression values.

Returns:

- **np.ndarray:** array of shape (n_genes) containing per-gene mean expression values.

Listing 117: Per-Gene Mean Expression

```
means = tox_compute_gene_means(expr)
```

2. Compute Residuals

Compute signed residuals for each gene by subtracting the per-gene mean expression from each replicate. The expression matrix is converted to Fortran order for efficient interaction with the compiled routine.

Arguments:

- `expr` (array-like): array of shape `(n_reps, n_genes)` containing expression values.
- `means` (array-like): array of shape `(n_genes)` containing per-gene mean expression values.

Returns:

- `np.ndarray`: array of shape `(n_reps, n_genes)` containing signed residuals.

Listing 118: Signed Residual Computation

```
resid = tox_compute_residuals(expr, means)
```

3. Pool Mean Expression Values

Pool per-gene mean expression values across two studies and compute empirical reference points `x_star` using the underlying compiled routine. Both input vectors are flattened, concatenated internally, and NaN values are handled by the Fortran implementation.

Arguments:

- `mean_S1` (array-like): array of shape `(n_genes_S1)` containing per-gene mean expression values for study 1.
- `mean_S2` (array-like): array of shape `(n_genes_S2)` containing per-gene mean expression values for study 2.
- `n_points` (`int`): number of reference points to compute.

Returns:

- `dict`: { `'n_pool'`: `int`, `'x_star'`: `np.ndarray (n_points,)` }

Listing 119: Pooling Mean Expression Values

```
out = tox_pool_means(mean_S1, mean_S2, n_points)

n_pool = out["n_pool"]
x_star = out["x_star"]
```

4. Pool Mean Expression Values (Expert)

Pool mean-expression values using pre-pooled and pre-sorted arrays. This expert interface assumes that pooling and sorting have already been performed externally, allowing direct computation of reference points `x_star` without re-allocating or re-sorting data.

Arguments:

- `pooled_means` (array-like): 1D array of shape `(pool_size)`, containing pre-pooled mean expression values.
- `pooled_perm` (array-like): permutation vector of shape `(pool_size)` that sorts `pooled_means`.
- `n_points` (`int`): number of reference points to compute.

Returns:

- `dict`: { `'n_pool'`: `int`, `'x_star'`: `np.ndarray (n_points,)` }

Listing 120: Pooling Means (Expert Interface)

```
out = tox_pool_means_expert(
    pooled_means,
```

```

        pooled_perm,
        n_points
    )

    n_pool = out["n_pool"]
    x_star = out["x_star"]

```

5. Compute Neighborhood Size

Compute the number of neighbors to use for neighborhood-based residual analysis. The calculation mirrors the Fortran routine: it enforces minimum and maximum limits, accounts for the number of reference points, and excludes genes with NaN mean expression values.

Arguments:

- `n_pool` (`int`): total number of pooled mean-expression values across both studies.
- `n_points` (`int`): number of reference points.
- `n_genes_S` (`int`): number of genes in the current study.
- `mean_S` (array-like): array of shape (`n_genes_S`) containing per-gene mean expression values.
- `desired_size` (`int`): optional desired neighborhood size (default 0, meaning “use Fortran default logic”).

Returns:

- `int`: computed neighborhood size.

Listing 121: Neighborhood Size Computation

```

n_neighbors = tox_calc_neighborhood_size(
    n_pool,
    n_points,
    n_genes_S,
    mean_S,
    desired_size=0
)

```

6. Construct Neighborhoods

Construct neighborhood-based residual sets by selecting, for each reference point, the closest genes in mean-expression space. The neighborhood size is computed automatically unless a desired size is provided. Residuals and indices are returned in arrays sized to the actual number of neighbors.

Arguments:

- `x_star` (array-like): array of shape (`n_points`) containing reference mean-expression values.
- `mean_S` (array-like): array of shape (`n_genes_S`) with per-gene mean expression values.
- `resid_S` (array-like): array of shape (`n_reps_S`, `n_genes_S`) containing signed residuals.
- `n_pool` (`int`): total number of pooled mean-expression values across studies.
- `desired_n_neighbors` (`int`): optional desired neighborhood size (default 0, meaning “compute automatically”).

Returns:

- `dict`: { `'neighborhood_residuals'`: `np.ndarray` (`n_reps_S`, `actual_n_neighbors`, `n_points`), `'neighborhood_indices'`: `np.ndarray` (`actual_n_neighbors`, `n_points`) }

Listing 122: Neighborhood Construction

```

out = tox_construct_neighborhoods(
    x_star,

```

```

    mean_S,
    resid_S,
    n_pool,
    desired_n_neighbors=0
)

neigh_res = out["neighborhood_residuals"]
neigh_idx = out["neighborhood_indices"]

```

7. Determine Shared Residual Range (Expert)

Compute the shared residual range R from a pre-pooled and pre-sorted residual vector using a percentile threshold.

Arguments:

- `residual_pool` (array-like): 1D array of shape `(pool_size)`, typically `(n_reps_S1 + n_reps_S2) * n_neighbors * n_points`.
- `residual_pool_perm` (array-like): permutation vector of shape `(pool_size)` that sorts `residual_pool`.
- `residual_range_quantile` (`float`): percentile used to determine the residual range (default 95.0).

Returns:

- `float`: shared residual range R

Listing 123: Shared Residual Range (Expert)

```

R = tox_determine_shared_residual_range_expert(
    residual_pool,
    residual_pool_perm,
    residual_range_quantile=95.0
)

```

8. Determine Shared Residual Range

Compute the shared residual range R from two residual matrices. Both residual tensors are flattened (absolute values), pooled, and the percentile threshold is applied to determine the symmetric residual range $[-R, R]$.

Arguments:

- `neighborhood_residuals_S1` (array-like): array of shape `(n_reps_S1, n_neighbors, n_points)`.
- `neighborhood_residuals_S2` (array-like): array of shape `(n_reps_S2, n_neighbors, n_points)`.
- `residual_range_quantile` (`float`): percentile used to determine the residual range (default 95.0).

Returns:

- `float`: shared residual range R

Listing 124: Shared Residual Range

```

R = tox_determine_shared_residual_range(
    neighborhood_residuals_S1,
    neighborhood_residuals_S2,
    residual_range_quantile=95.0
)

```

9. Build Residual Histograms

Build histogram counts and probability mass functions (PMFs) for one study, using a shared residual range and a fixed number of bins. Residuals are grouped per reference point, NaNs are ignored, and both absolute counts and normalized PMFs are returned.

Arguments:

- `neighborhood_residuals` (array-like): array of shape $(n_reps, n_neighbors, n_points)$ containing residuals.
- `shared_residual_range` (float): symmetric residual range $[-R, R]$ used for histogram binning.
- `n_bins` (int): number of equally sized histogram bins.

Returns:

- `dict`: { `'counts'`: `np.ndarray` (n_points, n_bins), `'pmf'`: `np.ndarray` (n_points, n_bins), `'included_n_residuals'`: `np.ndarray` (n_points ,) }

Listing 125: Residual Histogram Construction

```
out = tox_build_residual_histograms(  
    neighborhood_residuals,  
    shared_residual_range,  
    n_bins  
)  
  
counts = out["counts"]  
pmf = out["pmf"]  
included = out["included_n_residuals"]
```

10. Build Residual Histograms (Filtered)

Build histogram counts and probability mass functions (PMFs) for one study, using a shared residual range and a fixed number of bins. A boolean neighbor mask selects which neighbors contribute to each reference point's histogram.

Arguments:

- `neighborhood_residuals` (array-like): array of shape $(n_reps, n_neighbors, n_points)$ containing residuals.
- `shared_residual_range` (float): symmetric residual range $[-R, R]$ used for histogram binning.
- `n_bins` (int): number of histogram bins.
- `neighbor_mask` (array-like): boolean mask of shape $(n_neighbors, n_points)$ selecting neighbors to include.

Returns:

- `dict`: { `'counts'`: `np.ndarray` (n_points, n_bins), `'pmf'`: `np.ndarray` (n_points, n_bins), `'included_n_residuals'`: `np.ndarray` (n_points ,) }

Listing 126: Filtered Residual Histogram Construction

```
out = tox_build_residual_histograms_filtered(  
    neighborhood_residuals,  
    shared_residual_range,  
    n_bins,  
    neighbor_mask  
)  
  
counts = out["counts"]
```

```
pmf = out["pmf"]
included = out["included_n_residuals"]
```

11. Compute Divergence per Reference Point

Compute the Jensen–Shannon divergence for each reference point using two probability mass function (PMF) matrices. Each row corresponds to a reference point, and each column to a histogram bin.

Arguments:

- `pmf_S1` (array-like): array of shape `(n_points, n_bins)` containing PMFs for study 1.
- `pmf_S2` (array-like): array of shape `(n_points, n_bins)` containing PMFs for study 2.

Returns:

- `np.ndarray`: array of shape `(n_points,)` containing per-point Jensen–Shannon divergences.

Listing 127: Per-Point JSD Computation

```
jsd = tox_compute_divergence_per_reference_point(
    pmf_S1,
    pmf_S2
)
```

12. Compute Weighted Global Divergence

Compute the weighted global Jensen–Shannon divergence using per-point JSD values and the number of included residuals from both studies. Weights are proportional to the total number of included residuals per reference point.

Arguments:

- `js_divergences` (array-like): array of shape `(n_points)` containing per-point JSD values.
- `included_S1` (array-like): array of shape `(n_points)` with included residual counts for study 1.
- `included_S2` (array-like): array of shape `(n_points)` with included residual counts for study 2.

Returns:

- `dict`: { `'global_js_divergence'`: `float`, `'weights'`: `np.ndarray (n_points,)` }

Listing 128: Weighted Global JSD

```
out = tox_compute_weighted_global_divergence(
    js_divergences,
    included_S1,
    included_S2
)

global_jsd = out["global_js_divergence"]
weights = out["weights"]
```

13. Permutation Test for Global JSD

Estimate how likely the observed global Jensen–Shannon divergence is to occur by chance under the null hypothesis that both studies are exchangeable. Residuals from both studies are repeatedly pooled and shuffled, histograms are rebuilt, and global JSD values are recomputed for each permutation.

Arguments:

- `neighborhood_residuals_S1` (array-like): array of shape `(n_reps, n_neighbors, n_points)` for study 1.

- `neighborhood_residuals_S2` (array-like): array of shape $(n_reps, n_neighbors, n_points)$ for study 2.
- `global_jsd_observed` (float): observed global Jensen–Shannon divergence.
- `n_bins` (int): number of histogram bins.
- `shared_residual_range` (float): shared residual range $[-R, R]$ used for histogram binning.
- `n_permutations` (int): number of permutations to perform.
- `random_seed` (int): seed for reproducible shuffling (default 42).

Returns:

- `dict`: { `'jsd_null'`: np.ndarray $(n_permutations,)$, `'p_value'`: float }

Listing 129: Permutation Test for Global JSD

```
out = gjct_permutation_test(
    neighborhood_residuals_S1,
    neighborhood_residuals_S2,
    global_jsd_observed,
    n_bins,
    shared_residual_range,
    n_permutations,
    random_seed=42
)

jsd_null = out["jsd_null"]
p_value = out["p_value"]
```

14. Permutation Test for Global JSD (Filtered)

Estimate how likely the observed global Jensen–Shannon divergence is to occur by chance under the null hypothesis that both studies are exchangeable. Neighbor masks allow excluding specific neighbors for each reference point in each study during histogram construction.

Arguments:

- `neighborhood_residuals_S1` (array-like): array of shape $(n_reps, n_neighbors, n_points)$ for study 1.
- `neighborhood_residuals_S2` (array-like): array of shape $(n_reps, n_neighbors, n_points)$ for study 2.
- `global_jsd_observed` (float): observed global Jensen–Shannon divergence.
- `n_bins` (int): number of histogram bins.
- `shared_residual_range` (float): shared residual range $[-R, R]$ used for histogram binning.
- `n_permutations` (int): number of permutations to perform.
- `neighbor_mask_S1` (array-like): boolean mask $(n_neighbors, n_points)$ selecting neighbors for study 1.
- `neighbor_mask_S2` (array-like): boolean mask $(n_neighbors, n_points)$ selecting neighbors for study 2.
- `random_seed` (int): seed for reproducible shuffling (default 42).

Returns:

- `dict`: { `'jsd_null'`: np.ndarray $(n_permutations,)$, `'p_value'`: float }

Listing 130: Filtered Permutation Test for Global JSD

```
out = gjct_permutation_test_filtered(
    neighborhood_residuals_S1,
    neighborhood_residuals_S2,
    global_jsd_observed,
```



```

        n_bins ,
        shared_residual_range ,
        n_permutations ,
        neighbor_mask_S1 ,
        neighbor_mask_S2 ,
        random_seed=42
    )

jsd_null = out["jsd_null"]
p_value = out["p_value"]

```

15. Compute Family-Level JSD

Compute the family-level Jensen–Shannon divergence for a given paralog family. This routine builds family-specific neighbor masks, constructs masked residual histograms, computes per-point JSD values, and aggregates them into a weighted global divergence.

Arguments:

- `family_idx` (`int`): index of the gene family to analyze.
- `gene_to_family_S1` (array-like): (`n_genes_S1`) `int32` vector mapping each gene in study 1 to a family.
- `gene_to_family_S2` (array-like): (`n_genes_S2`) `int32` vector mapping each gene in study 2 to a family.
- `neighborhood_residuals_S1` (array-like): array of shape (`n_reps_S1`, `n_neighbors`, `n_points`).
- `neighborhood_residuals_S2` (array-like): array of shape (`n_reps_S2`, `n_neighbors`, `n_points`).
- `neighborhood_genes_S1` (array-like): array of shape (`n_neighbors`, `n_points`) with gene indices for study 1.
- `neighborhood_genes_S2` (array-like): array of shape (`n_neighbors`, `n_points`) with gene indices for study 2.
- `n_bins` (`int`): number of histogram bins.
- `shared_residual_range` (`float`): shared residual range $[-R, R]$ used for histogram construction.

Returns:

- `dict`: { `'js_divergences'`: `np.ndarray` (`n_points`,),
`'included_n_reps_S1'`: `np.ndarray` (`n_points`,),
`'included_n_reps_S2'`: `np.ndarray` (`n_points`,),
`'total_included_n_reps'`: `int`,
`'global_js_divergence'`: `float`,
`'weights'`: `np.ndarray` (`n_points`,),
`'ierr'`: `int` }

Listing 131: Family-Level JSD Computation

```

out = fjct_compute_jsd(
    family_idx ,
    gene_to_family_S1 ,
    gene_to_family_S2 ,
    neighborhood_residuals_S1 ,
    neighborhood_residuals_S2 ,
    neighborhood_genes_S1 ,
    neighborhood_genes_S2 ,

```

```

        n_bins,
        shared_residual_range
    )

    jsd = out["js_divergences"]
    global_jsd = out["global_js_divergence"]
    weights = out["weights"]

```

3.2.4 Data Persistence

1. Python Interface for Array Serialization and Deserialization

The Python interface provides bindings to the Fortran serialization and deserialization routines, enabling efficient handling of multi-dimensional arrays. This interface supports NumPy arrays with native data type compatibility and Fortran-style column-major memory layout.

(a) General Architecture

The Python module `tensoromics_functions` provides ten main functions for array handling:

- `tox_serialize_int_nd(array, filename)` - Serialize N-dimensional integer arrays.
- `tox_deserialize_int_nd(filename)` - Deserialize N-dimensional integer arrays.
- `tox_serialize_real_nd(array, filename)` - Serialize N-dimensional real arrays.
- `tox_deserialize_real_nd(filename)` - Deserialize N-dimensional real arrays.
- `tox_serialize_char_nd(array, filename)` - Serialize N-dimensional character arrays.
- `tox_deserialize_char_nd(filename)` - Deserialize N-dimensional character arrays.
- `tox_serialize_logical_nd(array, filename)` - Serialize N-dimensional logical arrays.
- `tox_deserialize_logical_nd(filename)` - Deserialize N-dimensional logical arrays.
- `tox_serialize_complex_nd(array, filename)` - Serialize N-dimensional complex arrays.
- `tox_deserialize_complex_nd(filename)` - Deserialize N-dimensional complex arrays.
- `get_array_metadata(filename)` - Retrieve array dimensions and metadata

(b) Serialization Functions:

- `array` (input): NumPy array to be serialized

Type: `numpy.ndarray` with dtype:

- `np.int32` for integer
- `np.float64` for float values
- `U5` for characters
- `boolean` for logical
- `np.complex128` for complex numbers

Layout: Column-major (Fortran) order (`order='F'`)

Dimensions: Since the array is being passed flat, theoretically all dimensions are supported, but practical use cases are typically 1D to 5D

- `filename` (input): Name of the binary file to create

Type: `str`

Listing 132: Example usage of serialization function

```

tox_serialize_int_nd(int_array, filename)
tox_serialize_real_nd(real_array, filename)

```

(c) Deserialization Functions:

- `filename` (input): Name of the binary file to read

Type: `str`

- **Return Value:** Deserialized NumPy array
- Type:** `numpy.ndarray` with original dtype and dimensions
- Layout:** Column-major (Fortran) order

Listing 133: Example usage of deserialization function

```
restored_int = tox_deserialize_int_nd(filename)
restored_real = tox_deserialize_real_nd(filename)
```

(d) **Implementation Details**

- The Python interface uses Fortran-style column-major array ordering for optimal compatibility
- All arrays are converted to the appropriate Fortran data types before serialization
- The interface automatically handles metadata including dimensions and data types
- Error handling is implemented through Python exceptions

(e) **Usage Example**

Listing 134: Full example of serialization and deserialization

```
# Real array example
real_array = np.array([1.5, 2.3, 3.2, 4.0, 5.0], dtype=np.float64,
    order='F')
tox_serialize_real_nd(real_array, "test_real_1d.bin")
restored_real = tox_deserialize_real_nd("test_real_1d.bin")

# Character array example with metadata
char_array = np.asfortranarray(["hello", "world"], dtype='U5')
tox_serialize_char_nd(char_array, "test_char_1d.bin")

restored_char = tox_deserialize_char_nd("test_char_1d.bin")

# Multi-dimensional examples
int_2d = np.array([[1, 2, 3], [4, 5, 6]], dtype=np.int32, order='F')
tox_serialize_int_nd(int_2d, "test_int_2d.bin")

restored_2d = tox_deserialize_int_nd("test_int_2d.bin")
```

(f) **Compatibility Notes**

- Requires NumPy installation
- Arrays must use Fortran-style column-major ordering
- Character arrays must use fixed-length string data types
- The interface supports N dimensions, but practical use cases are typically 1D to 5D
- Data types are restricted to int32, float64, and U5 for characters

2. Standard Archive Operations

• Create Zip Archive (Low-Level)

The low-level function creates a zip archive from keys and filenames, directly calling the Fortran backend. Use this function for custom data. Note that the identifiers at position i belongs to position i of the filenames. All files named in the filenames have to be saved to the disk before the function is called. For more details on how to save arrays, see section array serialization and deserialization. Call the `create_zip_archive` function with the following arguments:

- `zip_filename`: Name of the zip file to create
- `keys`: List of keys for the manifest (identifiers for each file)

- `filenames`: List of filenames to include in archive

Listing 135: Python Create Zip Archive (Low-Level)

```
create_zip_archive(zip_filename, keys, filenames)
```

- **Extract Zip Archive**

The function extracts a zip archive created by `create_zip_archive` and returns a dictionary mapping data keys to extracted filenames. All extracted files will be written to the disk. For more details on how to continue working with the files, see section array serialization and deserialization. Call the `extract_zip_archive` function with the following arguments:

- `zip_filename`: Path to the zip file to extract

Returns:

- Dictionary mapping data keys to extracted temporary filenames

Listing 136: Python Extract Zip Archive

```
extract_zip_archive(zip_filename)
```

- **Save TOX Data (High-Level)**

The high-level function saves TOX data to a zip archive, handling serialization, and archive creation for standard-conform tox gene data. This function automatically manages temporary files and cleanup. Call the `save_tox_data` function with the following arguments:

- `zip_filename`: Name of the zip file to create
- **(Optional)** `gene_ids`: Array of gene IDs
- **(Optional)** `expression_vectors`: 2D array of expression data
- **(Optional)** `gene_to_fam`: Array mapping genes to families
- **(Optional)** `family_ids`: Array of family IDs
- **(Optional)** `family_centroids`: 2D array of family centroids
- **(Optional)** `shift_vectors`: 2D array of shift vectors
- Corresponding `*_name` parameters for temporary filenames (e.g., `gene_ids_name`, `expression_vectors_name`, etc.)

Listing 137: Python Save TOX Data

```
save_tox_data(zip_filename, gene_ids=None, expression_vectors=None,
              gene_to_fam=None, family_ids=None, family_centroids=None,
              shift_vectors=None, gene_ids_name=None,
              expression_vectors_name=None, gene_to_fam_name=None,
              family_ids_name=None, family_centroids_name=None,
              shift_vectors_name=None)
```

- **Read TOX Data**

The function reads data from a zip archive created by `save_tox_data`, extracting and deserializing the requested data components for standard-conform tox gene data. This function automatically handles file extraction, cleanup and array allocation. Call the `read_tox_data` function with the following arguments:

- `zip_filename`: Name of the zip file to read
- **(Optional)** `load_gene_ids`: Whether to load gene IDs (default: False)
- **(Optional)** `load_expression_vectors`: Whether to load expression vectors (default: False)
- **(Optional)** `load_gene_to_fam`: Whether to load gene-to-family mapping (default: False)
- **(Optional)** `load_family_ids`: Whether to load family IDs (default: False)

- (Optional) `load_family_centroids`: Whether to load family centroids (default: False)
- (Optional) `load_shift_vectors`: Whether to load shift vectors (default: False)

Return dictionary:

- `gene_ids`: Loaded gene IDs array (if requested)
- `expression_vectors`: Loaded expression vectors array (if requested)
- `gene_to_fam`: Loaded gene-to-family mapping array (if requested)
- `family_ids`: Loaded family IDs array (if requested)
- `family_centroids`: Loaded family centroids array (if requested)
- `shift_vectors`: Loaded shift vectors array (if requested)

Listing 138: Python Read TOX Data

```
read_tox_data(zip_filename, load_gene_ids=False,
              load_expression_vectors=False,
              load_gene_to_fam=False, load_family_ids=False,
              load_family_centroids=False, load_shift_vectors=False)
```

Data Archiving Workflow Examples:

Listing 139: Python Standard TOX Data Archiving

```
# Save complete TOX dataset
save_tox_data(
    zip_filename="complete_tox_dataset.zip",
    gene_ids=filtered_gene_ids,
    gene_ids_name="gene_ids.bin",
    expression_vectors=filtered_expression,
    expression_vectors_name="expression.bin",
    gene_to_fam=filtered_gene_to_fam,
    gene_to_fam_name="gene_to_fam.bin",
    family_ids=family_ids,
    family_ids_name="family_ids.bin",
    family_centroids=family_centroids,
    family_centroids_name="centroids.bin",
    shift_vectors=shift_vectors,
    shift_vectors_name="shift_vectors.bin"
)

# Save partial dataset
save_tox_data(
    zip_filename="basic_data.zip",
    gene_ids=filtered_gene_ids,
    gene_ids_name="gene_ids.bin",
    expression_vectors=filtered_expression,
    expression_vectors_name="expression.bin"
)

# Read data back with selective loading
loaded_data = read_tox_data(
    zip_filename="complete_tox_dataset.zip",
    load_gene_ids=True,
    load_expression_vectors=True,
    load_gene_to_fam=True
)
```

Listing 140: Python Custom Archive Creation and Extraction

```
# Create custom archive with non-standard data
create_zip_archive(
    zip_filename="custom_data.zip",
    keys=["3d_tensor", "results"],
    filenames=["tensor_data.bin", "analysis_results.bin"]
)

# Extract and access custom archive
file_mapping = extract_zip_archive("custom_data.zip")
print("Files in archive:", file_mapping)
# Output: {'3d_tensor': 'tensor_data.bin', 'results': '
analysis_results.bin'}
```

Listing 141: Python Mixed Standard and Custom Data Archiving

```
# Serialize custom arrays
tox_serialize_real_nd(custom_array, "custom_data.bin")
tox_serialize_int_nd(another_array, "another_data.bin")

# Create mixed archive with both standard and custom data
create_zip_archive(
    zip_filename="mixed_archive.zip",
    keys=["standard_expression", "custom_feature",
          "analysis_results"],
    filenames=["expression.bin", "custom_data.bin",
              "another_data.bin"]
)

# Later, extract and process
mapping = extract_zip_archive("mixed_archive.zip")
expression_data = tox_deserialize_real_nd(
    mapping["standard_expression"])

custom_features = tox_deserialize_real_nd(
    mapping["custom_feature"])

results = tox_deserialize_int_nd(mapping["analysis_results"])
```

3.2.5 Data Processing

1. Normalization of Expression Values

The function runs a complete normalization pipeline on gene expression data. It first applies standard deviation normalization, then quantile normalization, followed by replicate averaging, and finally a log2 transformation. The result is returned in an array, which contains the fully normalized expression values. Call the `tox_normalization_pipeline` function with the following arguments:

- **input_matrix**: Numeric expression vector matrix with size `genes × tissues`
- **group_starts**: Integer array, start column index for each replicate group (1-based)
- **group_counts**: Integer array, number of columns per replicate group

Return:

- Numpy array with $\log_2(x+1)$ normalized expression with size (genes $\times$ groups)

Listing 142: Python Normalization Pipeline

```
tox_normalization_pipeline(input_matrix, group_starts, group_counts)
```

Alternatively you can calculate each step of the normalization pipeline manually by calling the following functions in sequence:

(a) **Normalize by Standard Deviation**

The function normalizes each gene's expression vector using `sqrt(mean(x2))` across tissues (not classical standard deviation). Call the `tox_normalize_by_std_dev` function with the following arguments:

- **input_matrix:** A numeric matrix with genes as rows and tissues as columns

Returns:

- Numpy array: Contains a normalized matrix with same dimensions as input

Listing 143: Python Standard Deviation Normalization

```
tox_normalize_by_std_dev(input_matrix)
```

(b) **Quantile Normalization**

The function performs quantile normalization of a gene expression matrix (F42-compliant) and computes average expression per rank across tissues. Call the `tox_quantile_normalization` function with the following arguments:

- **input_matrix:** A numeric matrix with genes as rows and tissues as columns

Returns:

- Numpy array: Quantile-normalized matrix with same dimensions as input

Listing 144: Python Quantile Normalization

```
tox_quantile_normalization(input_matrix)
```

(c) **Calculate Tissue Averages**

The function calculates tissue averages by averaging replicates within each group. For each group of tissue replicates, it computes the average expression per gene. The input matrix is column-major, flattened as a 1D array. Call the `tox_calculate_tissue_averages` function with the following arguments:

- **input_matrix:** A numeric matrix with genes as rows and tissue replicates as columns
- **group_starts:** Array of starting column indices for each group (1-based for Fortran)
- **group_counts:** Array of counts for each group

Returns:

- Numpy array: Matrix with genes as rows and averaged tissues as columns

Listing 145: Python Tissue Averages Calculation

```
tox_calculate_tissue_averages(input_matrix, group_starts,
                               group_counts)
```

(d) **Log2 Transformation**

The function applies $\log_2(x + 1)$ transformation to each element of the input matrix. This is computed via $\log(x + 1) / \log(2)$ for compatibility with WebAssembly (WASM). Call the `tox_log2_transformation` function with the following arguments:

- **input_matrix:** A numeric matrix with genes as rows and tissues as columns

Returns:

- Numpy array: Log2-transformed matrix with same dimensions as input

Listing 146: Python Log2 Transformation

```
tox_log2_transformation(input_matrix)
```

2. Calculate Fold Change

If fold change calculation is needed, this function can be used after normalization. The function calculates log2 fold changes between condition and control columns. For each control-condition pair, this function computes the log2 fold change by subtracting the expression value in the control column from the corresponding value in the condition column, for all genes. Call the `tox_calculate_fold_changes` function with the following arguments:

- **input_matrix**: A numeric matrix with genes as rows and tissues/conditions as columns
- **control_cols**: Array of control column indices (1-based for Fortran)
- **condition_cols**: Array of condition column indices (1-based for Fortran)

Returns:

- Numpy array: Matrix with genes as rows and fold change values as columns

Listing 147: Python Fold Change Calculation

```
tox_calculate_fold_changes(input_matrix, control_cols, condition_cols)
```

3.2.6 Family-Based Analysis

1. Calculation of family Centroids

The function computes the centroid (mean expression vector) for each gene family in a gene expression dataset. It iterates over all families, selects the relevant genes for each family and calculates the mean vector for each family. There are two possible modes to use. To use all genes for calculation, select `all` mode. If you want to specify relevant genes for calculation, select `orthologs` mode. It is important that you provide a logical `ortholog_set` array when using `orthologs` mode. Call the `tox_group_centroid` function with the following arguments:

- **expression_vectors**: Expression vector matrix with size `n_axes × n_genes`
- **gene_to_family**: Gene-to-family mapping array with the length of `n_genes` (1-based indexing)
- **n_families**: Total number of gene families
- **mode**: Calculation mode ("all" or "orthologs")
- (Optional) **ortholog_set**: Logical array indicating if a gene is part of the calculation subset (only used in `orthologs` mode)

Return:

- **centroids_out**: Matrix of calculated family centroid vectors with size `n_axes × n_families` (read-only)

Listing 148: Python Calculation of Family Centroids

```
tox_group_centroid(expression_vectors, gene_to_family, n_families,
                    mode, ortholog_set)
```

(a) Calculate Mean Vector

Alternatively you can calculate a single mean vector for a given set of vectors by calling the `mean_vector` function directly with the following arguments:

- **expression_vectors real(real64)**: Expression vector matrix with size `n_axes × n_genes`

- **gene_indices** integer(int32): An array containing the column indices of the selected genes in **expression_vectors**

Return:

- **centroid**: Calculated mean vector (centroid) with length of **n_axes** (read-only)

Listing 149: Python Mean Vector Calculation

```
tox_mean_vector(expression_vectors , gene_indices)
```

2. Compute Shift Vector Field

Calculates the shift vector field for each gene expression vector based on its family centroid. The shift vector represents the difference between the gene expression vector and its corresponding family centroid, starting at the expression vector and pointing to its family centroid.

Mathematical Definition:

$$\text{shift_vector} = \text{expression_vector} - \text{family_centroid}$$

Arguments:

- **expression_vectors**: 2D numpy array of shape (**n_axes**, **n_vectors**) where each column is a gene expression vector
- **family_centroids**: 2D numpy array of shape (**n_axes**, **n_families**) where each column is a family centroid vector
- **gene_to_centroid**: 1D numpy array of length **n_vectors** containing 1-based family indices mapping each gene to its centroid

Returns:

- **shift_vectors**: 2D numpy array of shape (**2*n_axes**, **n_vectors**) containing centroids (rows 0:**n_axes**-1) and shift vectors (rows **n_axes**:**2*n_axes**-1) (read-only)

Listing 150: Python Shift Vector Field Computation

```
shift_vectors = tox_compute_shift_vector_field(expression_vectors ,
                                              family_centroids , gene_to_centroid)
```

3. Calculate Distance to Centroid

For each gene in the gene expression matrix, the function computes the Euclidean distance to its corresponding family centroid and returns a read-only distances array containing the distance value for each expression vector to its family centroid. Call the **tox.distance_to_centroid** function with the following arguments:

- **genes**: Gene expression matrix array with size of (**n_genes** × **d**)
- **centroids**: Family centroid matrix array with size of (**n_families** × **d**)
- **gene_to_fam**: Gene-to-family mapping array with the length of **n_genes** (0 = no family, >0 = family index)
- **d**: Gene expression matrix dimension as number

Listing 151: Python Distance to Centroid Calculation

```
tox_distance_to_centroid(genes , centroids , gene_to_fam , d)
```

(a) Calculate Euclidean Distance

Alternatively you can calculate a single Euclidean distance between two vectors manually by calling **tox.euclidean_distance** function with the following arguments:

- **vec1**: First vector as array

- **vec2**: Second vector as array

Both vectors must have same length!

Listing 152: Python Euclidean distance calculation

```
tox_euclidean_distance(vec1, vec2)
```

4. Detect Gene Outliers

The function first computes family scaling, then calculates relative distance indices (RDI) and identifies outliers based on the specified percentile threshold. It returns a read-only dictionary containing the outlier boolean values and intermediate results. Call the `tox_detect_outliers` function with the following arguments:

- **distances**: Array of distances for each gene to its centroid
- **gene_to_fam**: Gene-to-family mapping array with the same length as the distances array (0 = no family, >0 = family index)
- **(Optional) percentile**: Percentile threshold value for outlier detection (default is 95)

Return dictionary:

- **outliers**: Boolean array indicating whether each gene is an outlier
- **loess_x**: Array of x coordinates used as reference points for LOESS smoothing (median distance)
- **loess_y**: Array of y coordinates used as reference points for LOESS smoothing (standard deviation)
- **loess_n**: Array with numbers of genes in each family used to compute the corresponding median and standard deviation

Listing 153: Python Gene Outlier Calculation

```
tox_detect_outliers(distances, gene_to_fam, percentile=95.0)
```

Alternatively you can calculate each step of `tox_detect_outliers` manually with the following functions:

(a) Compute Family Scaling

The function calculates a scaling factor for each gene family by computing the median and standard deviation of gene-to-centroid distances within each family, then applies LOESS smoothing to these values. Call the `tox_compute_family_scaling` function with the following arguments:

- **distances**: Array of distances for each gene to its centroid
- **gene_to_fam**: Gene-to-family mapping array with the same length as the distances array (0 = no family, >0 = family index)

Return dictionary:

- **dscale**: Array of scaling factors per family
- **loess_x**: Array of x coordinates used as reference points for LOESS smoothing (median distance)
- **loess_y**: Array of y coordinates used as reference points for LOESS smoothing (standard deviation)
- **indices_used**: Array of indices of families that have been included in the calculation

Listing 154: Python Compute Family Scaling

```
tox_compute_family_scaling(distances, gene_to_fam)
```

(b) **Compute Family Scaling (Expert Version)**

This function is the expert version of `tox_detect_outliers` and requires pre-allocated work arrays for maximum performance and control. It can be used when you need fine-grained control over memory allocation or you are calling this function many times in a tight loop. Call the `tox_compute_family_scaling_expert` function with the following arguments:

- **distances**: Array of distances for each gene to its centroid
- **gene_to_fam**: Gene-to-family mapping array with the same length as the distances array (0 = no family, >0 = family index)
- **perm_tmp**: Permutation array for sorting with the same length as the distances array
- **stack_left_tmp**: Stack array for sorting with the same length as the distances array
- **stack_right_tmp**: Stack array for sorting with the same length as the distances array
- **family_distances**: Work array for sorting with the same length as the distances array

Return dictionary:

- **dscale**: Array of scaling factors per family
- **loess_x**: Array of x coordinates used as reference points for LOESS smoothing (median distance)
- **loess_y**: Array of y coordinates used as reference points for LOESS smoothing (standard deviation)
- **indices_used**: Array of indices of families that have been included in the calculation
- **perm_tmp**: Permutation array for sorting
- **stack_left_tmp**: Stack array for sorting
- **stack_right_tmp**: Stack array for sorting
- **family_distances**: Work array for sorting

Listing 155: Python Compute Family Scaling (Expert Version)

```
tox_compute_family_scaling_expert(distances, gene_to_fam, perm_tmp,
                                  stack_left_tmp, stack_right_tmp, family_distances)
```

(c) **Compute Relative Distance Index (RDI)**

The function calculates the Relative Distance Index (RDI) for each gene by dividing its Euclidean distance to the family centroid by the scaling factor of its family. The function returns a numpy array with RDI values for each gene. Call the `tox_compute_rdi` function with the following arguments:

- **distances**: Array of distances for each gene to its centroid
- **gene_to_fam**: Gene-to-family mapping array with the same length as the distances array (0 = no family, >0 = family index)
- **dscale**: Array of scaling factors per family

Listing 156: Python Compute Relative Distance Index (RDI)

```
tox_compute_rdi(distances, gene_to_fam, dscale)
```

(d) **Identify Outliers**

The function identifies outliers based on the top percentile of RDI values. Call the `tox_identify_outliers` function with the following arguments:

- **rdi**: Array of RDI values for each gene
- **(Optional) threshold**: Fixed RDI threshold number (if None, percentile value is used)
- **(Optional) percentile**: Percentile threshold number for outlier detection (default is 95)

Return dictionary:

- **outliers**: Boolean array indicating whether each gene is an outlier

- **threshold:** Actual threshold value used for outlier detection

Listing 157: Python Identify Outliers

```
tox_identify_outliers(rdi, threshold=None, percentile=95.0)
```

5. Compute Normalized Tissue Versatility:

The function calculates a normalized tissue versatility score for selected gene expression vectors. It is based on the angle between each gene expression vector and the space diagonal. Versatility is normalized to $[0, 1]$, where 0 means uniform expression 1 means expression in only one axis. Call the `tox_compute_tissue_versatility` function with the following parameters:

- **expression_vectors:** Expression vector matrix array with size `n_axes × n_vectors`
- **vector_selection:** Logical array with length of `n_vectors` to specify which vectors should be included in the calculation
- **axes_selection:** Logical array with length of `n_axes` to specify which axes should be included in the calculation

Return dictionary:

- **tissue_versatilities:** Array containing the calculated tissue versatilities for selected vectors
- **tissue_angles_deg:** Array containing the calculated angles in degrees for selected vectors
- **n_selected_vectors:** Number of vectors processed
- **n_selected_axes:** Number of axes used in calculation

Listing 158: Python Tissue Versatility Calculation

```
tox_calculate_tissue_versatility(expression_vectors, vector_selection,
                                axis_selection)
```

3.2.7 Python RAP Projection and Analysis Functions

The Python interface provides comprehensive wrappers for RAP (Relative Axes Projection) analysis with automatic memory management, input validation, and error handling.

Python Implementation Features:

- **Automatic Memory Layout:** Uses Fortran order for matrices and C order for vectors as required by underlying Fortran code
- **Comprehensive Input Validation:** Validates array dimensions, types, and consistency before Fortran calls
- **Flexible Mask Input:** Accepts both boolean and integer arrays for selection masks
- **Automatic Dimension Handling:** Infers dimensions from input arrays and handles 2D/3D edge cases
- **Read-only Outputs:** All return arrays are marked as read-only to prevent accidental modification
- **Error Conversion:** Converts Fortran error codes to informative Python exceptions

Validation Checks:

- Vector dimensions must match between inputs
- Selection mask lengths must match corresponding array dimensions
- For vector field projection, input must have exactly $2 \times n_axes$ rows
- For dimensions ≥ 3 , selected axes must be valid indices and have exactly 3 elements
- All arrays are converted to appropriate data types (float64 for vectors, int32 for masks)

Usage Example:

Listing 159: Complete Python RAP Analysis Workflow

```
import numpy as np

# Example: 4D vectors, 8 vector pairs
n_dims, n_vecs = 4, 8

# Create sample data
origins = np.random.rand(n_dims, n_vecs).astype(np.float64)
targets = np.random.rand(n_dims, n_vecs).astype(np.float64)

# Single vector angle calculation
v1 = origins[:, 0]
v2 = targets[:, 0]
selected_axes = np.array([1, 2, 3], dtype=np.int32)
angle = tox_clock_hand_angle_between_vectors(v1, v2, selected_axes)

# Batch angle calculation
vecs_mask = np.array([True, True, False, True, False, True, True, False])
angles = tox_clock_hand_angles_for_shift_vectors(
    origins,
    targets,
    vecs_mask,
    selected_axes
)

# Axis contribution analysis
shift_vector = v2 - v1
axis_contributions = relative_axes_changes_from_shift_vector(shift_vector)

# RAP projection
expression_matrix = np.random.rand(6, 10).astype(np.float64)
vecs_mask = np.ones(10, dtype=bool) # Select all vectors
axes_mask = np.array([1, 1, 1, 1, 0, 0], dtype=bool) # Select first 4
axes
projections = tox_vector_RAP_projection(expression_matrix, vecs_mask,
    axes_mask)
```

1. Vector RAP Projection

Projects selected expression vectors onto the RAP constructed from a selected set of axes.

Arguments:

- **vecs**: 2D numpy array of expression vectors [$n_{\text{axes}} \times n_{\text{vecs}}$]
- **vecs_selection_mask**: Boolean or integer array of length n_{vecs} indicating which vectors to project
- **axes_selection_mask**: Boolean or integer array of length n_{axes} indicating which axes to include in RAP

Returns:

- **projections**: 2D numpy array of projected vectors [$n_{\text{selected_axes}} \times n_{\text{selected_vecs}}$] (read-only)

Listing 160: Vector RAP Projection

```
projections = tox_vector_RAP_projection(vecs, vecs_selection_mask,
    axes_selection_mask)
```

2. Field RAP Projection

Projects selected vector fields (e.g., shift vectors) onto the RAP constructed from a selected set of axes. Automatically computes shift vectors as differences between origins and targets.

Arguments:

- **vecs**: 2D numpy array of vector fields [$2 \times n_axes \times n_vecs$] (first n_axes rows: origins, last n_axes rows: targets)
- **vecs_selection_mask**: Boolean or integer array of length n_vecs indicating which vectors to project
- **axes_selection_mask**: Boolean or integer array of length n_axes indicating which axes to include in RAP

Returns:

- **projections**: 2D numpy array of projected vectors [$n_selected_axes \times n_selected_vecs$] (read-only)

Listing 161: Vector Field RAP Projection

```
projections = tox_field_RAP_projection(vecs, vecs_selection_mask,
                                       axes_selection_mask)
```

3. Clock Hand Angle Between Vectors

Calculates the signed rotation angle between two RAP-projected and normalized vectors. Handles 2D, 3D, and higher-dimensional cases with automatic directionality calculation.

Arguments:

- **v1**: First normalized vector in RAP space (1D numpy array)
- **v2**: Second normalized vector in RAP space (1D numpy array)
- **selected_axes_for_signed**: Integer array of 3 axes to use for directionality calculation

Returns:

- **signed_angle**: Signed angle between vectors in radians $[-\pi, \pi]$ (read-only float)

Listing 162: Signed Angle Between Vectors

```
angle = tox_clock_hand_angle_between_vectors(v1, v2,
                                             selected_axes_for_signed)
```

4. Clock Hand Angles for Shift Vectors

Computes signed rotation angles between corresponding pairs of RAP-projected and normalized vectors (e.g., expression centroids and paralogs).

Arguments:

- **origins**: 2D numpy array of origin vectors [$n_dims \times n_vecs$] (e.g., centroids)
- **targets**: 2D numpy array of target vectors [$n_dims \times n_vecs$] (e.g., paralogs)
- **vecs_selection_mask**: Boolean array of length n_vecs indicating which vector pairs to compute
- **selected_axes_for_signed**: Integer array of 3 axes to use for directionality calculation

Returns:

- **signed_angles**: 1D numpy array of signed angles in radians $[-\pi, \pi]$ (read-only)

Listing 163: Batch Angle Calculation for Vector Pairs

```
angles = tox_clock_hand_angles_for_shift_vectors(origins, targets,
                                                  vecs_selection_mask,
                                                  selected_axes_for_signed)
```

5. Relative Axis Contributions from Shift Vector

Computes fractional contribution of each axis to a RAP-projected and normalized shift vector. Values range [0,1] and sum to 1.

Arguments:

- **shift_vector**: RAP-projected and normalized shift vector (1D numpy array)

Returns:

- **contrib**: 1D numpy array of relative axis contributions (sums to 1, read-only)

Listing 164: Shift Vector Axis Contributions

```
contributions = relative_axes_changes_from_shift_vector(shift_vector)
```

6. Relative Axis Contributions from Expression Vector

Computes fractional contribution of each axis to a RAP-projected and normalized expression vector. Values range [0,1] and sum to 1.

Arguments:

- **expression_vector**: RAP-projected and normalized expression vector (1D numpy array)

Returns:

- **contrib**: 1D numpy array of relative axis contributions (sums to 1, read-only)

Listing 165: Expression Vector Axis Contributions

```
contributions = relative_axes_expression_from_expression_vector(  
    expression_vector)
```

3.2.8 Paralog Analysis

The Python module provides comprehensive wrappers for paralog analysis.

Key Features:

- **Detect Neofunctionalization**: Identifies neofunctionalization for genes by checking whether the difference of expression to its ancestor exceeds the threshold for the respective axis.
- **Detect Dosage Effect**: Identifies subsets of paralogs with small angle to the ancestor (**max_angle**) and sum to a magnitude significantly exceeding **norm(ancestor)** (**gain**).
- **Detect Subfunctionalization**: Identifies subsets of paralogs exhibiting significant angles to the ancestor.

1. Detect Neofunctionalization

Identify neofunctionalization for genes by checking whether the difference of expression to its ancestor exceeds the threshold for each axis.

Arguments:

- **ancestors** (np.ndarray): 2D array of shape (n_axes, n_families).
- **genes** (np.ndarray): 2D array of shape (n_axes, n_genes).
- **gene_to_fam** (np.ndarray): 1D integer array of length n_genes mapping gene index to family index.
- **thresholds** (np.ndarray): 1D array of shape (n_axes,) with per-axis thresholds.

Returns:

- **np.ndarray**: Integer array of shape (n_genes, n_axes) with 0/1 values (non-zero interpreted as True).

Listing 166: Neofunctionalization Detection

```
neofunc_mask = tox_detect_neofunctionalization(ancestors, genes,  
    gene_to_fam, thresholds)
```

2. Detect Dosage Effect

Detect dosage effect among paralogs using gain and angle thresholds.

This function has binomial memory complexity, depending on included genes. This is why this subroutine only handles one single family, the size varies too much across them. If it is wanted to have all results in one array, just use a python list.

Arguments:

- `ancestor` (array-like): vector of shape `(n_dims)`.
- `genes` (array-like): 2D array of shape `(n_dims, n_genes)` containing all genes.
- `filtered_paralogs_mask` (array-like): int32 mask (chunks of 32 bits) for `n_genes`.
- `max_subset_size` (`int`): desired maximum subset size (may be adjusted).
- `gain_gamma` (`float`): required magnitude gain (default 0.1).
- `max_angle` (`float`): maximum allowed angle in radians (default π).

Returns:

- `dict`: { `'n_results'`: `int`, `'results'`: `np.ndarray` }

Listing 167: Dosage Effect Detection

```
n_results, results = tox_detect_dosage_effect(
    ancestor, genes, filtered_paralogs_mask, max_subset_size,
    gain_gamma=0.1, max_angle=np.pi
).values()
```

3. Detect Subfunctionalization

Detect subfunctionalization among paralogs based on residual distance and pruning.

This function has binomial memory complexity, depending on included genes. This is why this subroutine only handles one single family, the size varies too much across them. If it is wanted to have all results in one array, just use a python list.

Arguments:

- `ancestor` (array-like): vector of shape `(n_dims)`.
- `genes` (array-like): 2D array of shape `(n_dims, n_genes)` containing all genes.
- `rdi_threshold` (`float`): maximum allowed residual distance to `ancestor`.
- `filtered_paralogs_mask` (array-like): int32 mask (chunks of 32 bits) for `n_genes`.
- `max_subset_size` (`int`): desired maximum subset size (may be adjusted).
- `paralog_norms` (array-like): float64 norms for all genes (`n_genes`).
- `sorted_paralog_norms_perm` (array-like): int32 permutation of norms (`n_genes`).

Returns:

- `dict`: { `'n_results'`: `int`, `'results'`: `np.ndarray` }

Listing 168: Subfunctionalization Detection

```
n_results, results = tox_detect_subfunctionalization(
    ancestor, genes, rdi_threshold,
    filtered_paralogs_mask, max_subset_size,
    paralog_norms, sorted_paralog_norms_perm
).values()
```

4. Mask Chunk Count

Compute the number of 32-bit chunks needed to encode a given number of genes.

Arguments:

- `n_genes` (`int`): number of genes to encode.

Returns:

- `int`: number of 32-bit chunks required.

Listing 169: Mask Chunk Count Calculation

```
n_chunks = tox_mask_chunk_count(n_genes)
```

5. Filter Paralogs By Pattern Subfunctionalization

Filter paralogs by subfunctionalization pattern using angle threshold, within the grouped slice.

Arguments:

- `gene_angles` (array-like): angles for all genes (length = `n_genes`).
- `threshold` (`float`): filtering threshold.
- `gene_to_fam` (array-like): gene-to-family mapping (`n_genes`).
- `n_families` (`int`): number of families.

Returns:

- `np.ndarray`: int32 mask (chunks of 32 bits) with 1 for kept, 0 otherwise.

Listing 170: Subfunctionalization Pattern Filtering

```
mask = tox_filter_paralogs_by_pattern_subfunctionalization(
    gene_angles, threshold, gene_to_fam, n_families
)
```

6. Filter Paralogs By Pattern Dosage Effect

Filter paralogs by dosage effect using angle threshold, within the grouped slice.

Arguments:

- `gene_angles` (array-like): angles for all genes (length = `n_genes`).
- `threshold` (`float`): filtering threshold.
- `gene_to_fam` (array-like): gene-to-family mapping (`n_genes`).
- `n_families` (`int`): number of families.

Returns:

- `np.ndarray`: int32 mask (chunks of 32 bits) with 1 for kept, 0 otherwise.

Listing 171: Dosage Effect Pattern Filtering

```
mask = tox_filter_paralogs_by_pattern_dosage_effect(
    gene_angles, threshold, gene_to_fam, n_families
)
```

7. Calculate Work Array Paralog Subsets Size

Calculate the required work array size for paralog subset analysis.

Arguments:

- `max_subset_size` (`int`): maximum subset size (may be adjusted by the routine).
- `n_genes` (`int`): total number of genes.
- `filtered_paralogs_mask` (array-like): integer bit mask (chunks of 32 bits) marking filtered paralogs.

Returns:

- `dict`: { `'actual_max_subset_size'`: `int`, `'work_array_size'`: `int` }

Listing 172: Work Array Size Calculation

```
actual_max_subset_size, work_array_size =
    tox_calc_work_arr_paralog_subsets_size(
        max_subset_size, n_genes, filtered_paralogs_mask
    ).values()
```

8. Mask Check State

Compute the number of 32-bit chunks needed to encode a given number of genes.

Arguments:

- `n_genes` (`int`): number of genes to encode.

Returns:

- `int`: number of 32-bit chunks required.

Listing 173: Mask Chunk Count Calculation

```
n_chunks = tox_mask_chunk_count(n_genes)
```

3.2.9 Trajectory Contribution Analysis

1. Fortran Core Contribution Analysis Functions

2. Trajectory Contribution Calculation

Computes integrated trajectory-level contribution between two vectors using cosine similarity.

Arguments:

- `factor`: 1D numpy array of shape `(n_timepoints,)` — independent variable
- `dependent`: 1D numpy array of shape `(n_timepoints,)` — dependent variable
- `mode`: Processing mode (1 = cosine similarity, 2 = angle/acos)

Returns:

- `contribution`: Scalar contribution value (float)

Listing 174: Trajectory-Level Contribution Calculation

```
contribution = tox_trajectory_contribution(factor, dependent, mode)
```

3. Spike Contribution Calculation

Computes timepoint-wise spike contributions between two vectors.

Arguments:

- `factor`: 1D numpy array of shape `(n_timepoints,)` — independent variable
- `dependent`: 1D numpy array of shape `(n_timepoints,)` — dependent variable
- `mode`: Processing mode (1 = Normal, 2 = RAP)

Returns:

- `contribution`: 1D numpy array of shape `(n_timepoints,)` with contribution values per timepoint (read-only)

Listing 175: Spike-Wise Contribution Calculation

```
spike_contributions = tox_spike_contribution(factor, dependent, mode)
```

4. Calculate Contributions (Allocation Version)

Calculates both spike and integrated contributions for a specific factor with internal memory allocation.

Arguments:

- `trajectories`: 3D numpy array of shape `(n_factors, n_samples, n_timepoints)` in Fortran order
- `i_factor`: 0-based index of the independent variable
- `dependent_idx`: 0-based index of the dependent variable
- `mode`: Processing mode (1 = Normal, 2 = RAP)

Returns:

- Dictionary containing:

- spikes: Spike contributions [n_timepoints, n_samples] (read-only)
- trajectory: Integrated contributions [n_samples] (read-only)

Listing 176: Contributions Calculation with Internal Allocation

```
result = tox_calc_contributions(trajectories, i_factor,
                               dependent_idx, mode)

# Access results
spike_contribs = result['spikes']           # [n_timepoints, n_samples]
traj_contribs = result['trajectory']        # [n_samples]
```

5. Calculate Contributions (Expert Version)

Calculates both spike and integrated contributions using caller-provided work arrays for optimal performance.

Arguments:

- trajectories: 3D numpy array of shape (n_factors, n_samples, n_timepoints) in Fortran order
- i_factor: 0-based index of the independent variable
- dependent_idx: 0-based index of the dependent variable
- mode: Processing mode (1 = Normal, 2 = RAP)
- temp_factor_vector: 1D work array of shape (n_timepoints,)
- temp_dependent_vector: 1D work array of shape (n_timepoints,)

Returns:

- Dictionary containing:
 - spikes: Spike contributions [n_timepoints, n_samples] (read-only)
 - trajectory: Integrated contributions [n_samples] (read-only)

Listing 177: Expert Contributions Calculation with Work Arrays

```
result = tox_calc_contributions_expert(
    trajectories, i_factor, dependent_idx, mode,
    temp_factor_vector, temp_dependent_vector
)

# Access results (same structure as tox_calc_contributions)
spike_contribs = result['spikes']
traj_contribs = result['trajectory']
```

Processing Modes:

- mode = 1 (Normal): Returns cosine similarity for normal vectors
- mode = 2 (RAP): Returns arccos of cosine similarity for RAP projected vectors

Key Features of Python Interface:

- **Automatic Memory Management:** Handles Fortran memory layout and data type conversion automatically
- **Comprehensive Error Handling:** Converts Fortran error codes to Python exceptions
- **Memory Efficient:** Output arrays are marked as read-only to prevent accidental modification
- **Flexible Input:** Accepts both lists and numpy arrays with automatic conversion
- **Performance Optimized:** Expert versions allow pre-allocated work arrays for critical code paths

Usage Example:

Listing 178: Complete Contribution Analysis Workflow

```
import numpy as np

# Example data
n_factors, n_samples, n_timepoints = 5, 100, 50
trajectories = np.random.rand(n_factors, n_samples, n_timepoints).astype(
    np.float64)

# Single vector analysis
factor_vec = np.random.rand(n_timepoints)
dependent_vec = np.random.rand(n_timepoints)

# Compute trajectory contribution
traj_contrib = tox_trajectory_contribution(factor_vec, dependent_vec,
    mode=1)

# Compute spike contributions
spike_contribs = tox_spike_contribution(factor_vec, dependent_vec, mode=1)

# Full trajectory analysis
result = tox_calc_contributions(trajectories, i_factor=0,
    dependent_idx=1, mode=1)
spikes = result['spikes'] # [50, 100]
trajectory = result['trajectory'] # [100]

# Expert version with pre-allocated work arrays
temp_factor = np.empty(n_timepoints, dtype=np.float64)
temp_dependent = np.empty(n_timepoints, dtype=np.float64)
result_expert = tox_calc_contributions_expert(
    trajectories, i_factor=0, dependent_idx=1, mode=1,
    temp_factor, temp_dependent
)
```

1. Python Interface for Outlier Detection

The Python interface provides wrappers for the Fortran outlier detection routines, handling data type conversion, memory layout, and error checking automatically.

(a) Calculate Integrated Threshold (Expert Version)

Calculate empirical threshold for integrated contributions using pre-computed permutations for optimal performance.

Arguments:

- **contributions:** 1D numpy array of integrated contributions [n_samples] in Fortran order
- **percentile_val:** Percentile value for threshold (0.0–100.0)
- **permutation:** Pre-computed permutation indices for sorted contributions [n_samples] in Fortran order

Returns:

- **threshold:** Scalar threshold value

Listing 179: Integrated Threshold Calculation (Expert)

```
threshold = calc_integrated_threshold_expert(
    contributions, percentile_val, permutation
)
```

(b) **Calculate Integrated Threshold (Convenience Version)**

Calculate empirical threshold for integrated contributions with internal allocation and sorting.

Arguments:

- **contributions:** 1D numpy array of integrated contributions [n_samples] in Fortran order
- **percentile_val:** Percentile value for threshold (0.0–100.0)

Returns:

- **threshold:** Scalar threshold value

Listing 180: Integrated Threshold Calculation

```
threshold = calc_integrated_threshold(contributions, percentile_val)
```

(c) **Detect Integrated Outliers**

Identify outlier samples where integrated contributions exceed the empirical threshold.

Arguments:

- **contributions:** 1D numpy array of integrated contributions [n_samples] in Fortran order
- **threshold:** Scalar threshold value for outlier detection

Returns:

- **outlier_mask:** 1D boolean array indicating outlier samples [n_samples] (read-only)

Listing 181: Integrated Outlier Detection

```
outlier_mask = detect_outliers_integrated(contributions, threshold)
```

(d) **Calculate Spike Thresholds (Expert Version)**

Calculate empirical thresholds for spike contributions using pre-sorted permutations for optimal performance.

Arguments:

- **spike_contribs:** 2D numpy array of spike contributions [n_timepoints, n_samples] in Fortran order (rows=timepoints, columns=samples)
- **percentile_val:** Percentile value for threshold (0.0–100.0)
- **permutation:** Pre-computed permutation indices [n_samples, n_timepoints] in Fortran order (each column contains sorted indices for a timepoint)

Returns:

- **thresholds:** 1D numpy array of thresholds for each timepoint [n_timepoints] (read-only)

Listing 182: Spike Thresholds Calculation (Expert)

```
thresholds = calc_spike_thresholds_expert(  
    spike_contribs, percentile_val, permutation  
)
```

(e) **Calculate Spike Thresholds (Convenience Version)**

Calculate empirical thresholds for spike contributions with internal allocations and sorting.

Arguments:

- **spike_contribs:** 2D numpy array of spike contributions [n_timepoints, n_samples] in Fortran order (rows=timepoints, columns=samples)
- **percentile_val:** Percentile value for threshold (0.0–100.0)

Returns:

- **thresholds:** 1D numpy array of thresholds for each timepoint [n_timepoints] (read-only)

Listing 183: Spike Thresholds Calculation

```
thresholds = calc_spike_thresholds(spike_contribs, percentile_val)
```

(f) **Detect Spike Outliers**

Identify outliers in spike contributions by comparing against timepoint-specific thresholds.

Arguments:

- **spike_contribs**: 2D numpy array of spike contributions [*n_timepoints*, *n_samples*] in Fortran order (rows=timepoints, columns=samples)
- **thresholds**: 1D numpy array of thresholds for each timepoint [*n_timepoints*] in Fortran order

Returns:

- **outlier_mask**: 2D boolean array indicating outliers [*n_timepoints*, *n_samples*] (read-only)

Listing 184: Spike Outlier Detection

```
outlier_mask = detect_outliers_spike(spike_contribs, thresholds)
```

Key Features of the Python Interface:

- **Automatic Type Conversion**: Input arrays automatically converted to `np.float64` and `np.int32` with Fortran memory layout
- **Dimension Validation**: Input dimensions are validated before calling Fortran routines
- **Error Handling**: Fortran error codes are converted to Python exceptions via `check_err_code()`
- **Memory Efficiency**: Output arrays are marked as read-only to prevent accidental modification
- **Boolean Conversion**: C integer arrays (0/1) are automatically converted to Python boolean arrays

Usage Notes:

- Use **expert** versions when you have pre-computed permutations for better performance
- Use **convenience** versions for simpler workflows where internal allocation is acceptable
- All input arrays should be in **Fortran order** (column-major) for optimal performance
- Output arrays are **read-only** to maintain data integrity

2. **Complete Pipeline Functions**

The Python interface provides two complete pipeline functions for processing trajectories with integrated outlier detection workflows.

(a) **Process Trajectories with Per-Timepoint Percentiles**

Processes multiple trajectories with percentile thresholds calculated separately for each timepoint. This provides timepoint-specific sensitivity for spike contribution analysis.

Arguments:

- **trajectories**: 3D numpy array of shape (*n_factors*, *n_samples*, *n_timepoints*) in Fortran order
- **factor_mask**: 1D boolean array of shape (*n_factors*,) indicating which factors to process
- **dependent_idx**: 1-based index of the dependent variable (will be excluded from processing)
- **mode**: Processing mode (1 = Normal, 2 = RAP)
- **percentile**: Percentile value for threshold calculation (0.0–100.0)

Returns:

- Dictionary containing:
 - **integrated_contribs**: Integrated contributions [*n_samples*, *n_processed_factors*]
 - **spike_contribs**: Spike contributions [*n_timepoints*, *n_samples*, *n_processed_factors*]
 - **thresholds_integrated**: Thresholds for integrated contributions [*n_processed_factors*]
 - **thresholds_spike**: Thresholds for spike contributions [*n_timepoints*, *n_processed_factors*]
 - **outliers_integrated**: Outliers for integrated contributions [*n_samples*, *n_processed_factors*]

- (boolean)
- outliers_spike: Outliers for spike contributions [n_timepoints, n_samples, n_processed_factors]
(boolean)

Listing 185: Complete Trajectory Processing with Per-Timepoint Thresholds

```
results = tox_process_trajectories(
    trajectories, factor_mask, dependent_idx, mode, percentile
)

# Access results
integrated_contribs = results['integrated_contribs']
# [n_samples, n_processed]

spike_contribs = results['spike_contribs']
# [n_timepoints, n_samples, n_processed]

thresholds_integrated = results['thresholds_integrated']
# [n_processed]

thresholds_spike = results['thresholds_spike']
# [n_timepoints, n_processed]

outliers_integrated = results['outliers_integrated']
# [n_samples, n_processed] (bool)

outliers_spike = results['outliers_spike']
# [n_timepoints, n_samples, n_processed] (bool)
```

(b) **Process Trajectories with Global Percentile**

Processes trajectories using a single global percentile threshold for all timepoints in spike contributions. This provides consistent sensitivity across all timepoints.

Arguments:

- **trajectories:** 3D numpy array of shape (n_factors, n_samples, n_timepoints) in Fortran order
- **factor_mask:** 1D boolean array of shape (n_factors,) indicating which factors to process
- **dependent_idx:** 1-based index of the dependent variable (will be excluded from processing)
- **mode:** Processing mode (1 = Normal, 2 = RAP)
- **percentile:** Percentile value for threshold calculation (0.0–100.0)

Returns:

- Dictionary containing:
 - integrated_contribs: Integrated contributions [n_samples, n_processed_factors]
 - spike_contribs: Spike contributions [n_timepoints, n_samples, n_processed_factors]
 - thresholds_integrated: Thresholds for integrated contributions [n_processed_factors]
 - thresholds_spike: Thresholds for spike contributions [n_processed_factors] (1D array)
 - outliers_integrated: Outliers for integrated contributions [n_samples, n_processed_factors] (boolean)
 - outliers_spike: Outliers for spike contributions [n_timepoints, n_samples, n_processed_factors] (boolean)

Listing 186: Complete Trajectory Processing with Global Threshold

```

results = tox_process_trajectories_flat(
    trajectories, factor_mask, dependent_idx, mode, percentile
)

# Access results - note thresholds_spike is 1D in flat version
integrated_contribs = results['integrated_contribs']
# [n_samples, n_processed]

spike_contribs = results['spike_contribs']
# [n_timepoints, n_samples, n_processed]

thresholds_integrated = results['thresholds_integrated']
# [n_processed]

thresholds_spike = results['thresholds_spike']
# [n_processed] (1D)

outliers_integrated = results['outliers_integrated']
# [n_samples, n_processed] (bool)

outliers_spike = results['outliers_spike']
# [n_timepoints, n_samples, n_processed] (bool)

```

Key Differences Between Pipeline Functions:

- `tox_process_trajectories`: Uses **per-timepoint percentiles** for spike thresholds (`thresholds_spike` shape: `[n_timepoints, n_processed]`)
- `tox_process_trajectories_flat`: Uses **global percentile** for spike thresholds (`thresholds_spike` shape: `[n_processed]`)

Pipeline Features:

- **Automatic Factor Filtering**: Processes only factors marked in `factor_mask`, excluding the dependent variable
- **Memory Efficient**: All output arrays use Fortran memory layout and are marked as read-only
- **Error Handling**: Comprehensive error checking with automatic exception conversion
- **Boolean Output**: Outlier masks are automatically converted from C integers to Python boolean arrays
- **Dimension Validation**: Input dimensions and factor mask consistency are validated

Usage Example:

Listing 187: Typical Pipeline Usage

```

import numpy as np

# Example data: 10 factors, 100 samples, 50 timepoints
trajectories = np.random.rand(10, 100, 50).astype(np.float64)
factor_mask = np.array([True, True, False, True, True, False,
                        True, True, False])
dependent_idx = 5 # 1-based index
mode = 1 # Normal mode
percentile = 95.0

# Process with per-timepoint thresholds

```



```

results = tox_process_trajectories(
    trajectories, factor_mask, dependent_idx, mode, percentile
)

# Process with global threshold
results_flat = tox_process_trajectories_flat(
    trajectories, factor_mask, dependent_idx, mode, percentile
)

```

3. Python Core Contribution Analysis Functions

The Python interface provides comprehensive wrappers for trajectory contribution analysis, including EDF computation and various contribution calculation modes.

(a) Empirical Distribution Function (EDF) Computation

Computes the empirical cumulative distribution function for observed data values.

Arguments:

- **values:** Array of observed data values (list or numpy array)

Returns:

- Dictionary containing:
 - **unique_values:** Sorted unique data values (read-only numpy array)
 - **cdf_values:** Corresponding cumulative frequencies between 0 and 1 (read-only)
 - **n_unique:** Number of unique values found (int)

Listing 188: Empirical Distribution Function Computation

```

result = compute_edf(values)

# Access results
unique_vals = result['unique_values']
# Sorted unique values

cdf_vals = result['cdf_values']
# Cumulative distribution values

n_unique = result['n_unique']
# Number of unique values

```

(b) Empirical Distribution Function (Expert Version)

Computes EDF using pre-sorted permutation for optimal performance.

Arguments:

- **values:** Array of observed data values (list or numpy array)
- **perm:** Pre-sorted permutation indices (1-based Fortran style, must be sorted by values[perm])

Returns:

- Dictionary containing:
 - **unique_values:** Sorted unique data values (read-only numpy array)
 - **cdf_values:** Corresponding cumulative frequencies between 0 and 1 (read-only)
 - **n_unique:** Number of unique values found (int)

Listing 189: Expert EDF Computation with Pre-sorted Permutation

```

result = compute_edf_expert(values, perm)

# Access results (same structure as compute_edf)

```

```
unique_vals = result['unique_values']
cdf_vals = result['cdf_values']
n_unique = result['n_unique']
```

3.2.10 Python Utils

1. Python BST Interface

(a) Build BST Index

Python wrapper for building BST index from NumPy arrays. Note that indices are 1-based and need to be converted for correct usage in python.

Input:

- `values` `numpy.ndarray`: 1D array of float64 values

Output:

- `np.array`: Permutation indices (1-based Fortran indices)

```
# - 1 at the end converts from fortrons 1-based indexing to python's 0-based
indices = build_bst_index(values) - 1
```

(b) BST Range Query

Performs range queries on BST index. Accepts and returns Fortran 1-based indices.

Input:

- `values` `numpy.ndarray`: Original values array
- `indices` `numpy.ndarray`: BST index array from `build_bst_index` (1-based)
- `lower_bound` `float`: Lower bound of range (inclusive)
- `upper_bound` `float`: Upper bound of range (inclusive)

Output:

- Dictionary: (`matching_indices`, `count`) where indices are 1-based

```
# + 1 is added because of 0-based indexing. If the indices are already
# 1 based, this is not necessary
res = bst_range_query(values, indices + 1, lower_bound, upper_bound)
```

(c) Get Sorted Value

Note: This function is not implemented as a separate Python function. Use direct array indexing instead:

```
value = values[indices[position]] # position is 1-based
```

2. Python k-d Tree Interface

(a) Build k-d Tree Index

Python interface for constructing k-d trees. Requires Fortran-ordered arrays and returns 1-based indices.

Input:

- `points` `numpy.ndarray`: 2D array of points (`dimensions × n_points`) in Fortran order
- `dimension_order` `numpy.ndarray`: Dimension ordering (1-based Fortran indices, optional)

Output:

- `np.array`: Permutation indices (1-based Python indices)

```
kd_indices = build_kd_index(points, dimension_order)
```

(b) **Build Spherical k-d Tree**

Alias for `build_kd_index` with the same parameters and behavior.

Input: Same as `build_kd_index` **Output:** Same as `build_kd_index`

```
sphere_indices = build_spherical_kd(vectors, dimension_order)
```

(c) **Get Point from k-d Index**

Note: This function is not implemented as a separate Python function. Use direct array indexing instead:

```
point = points[:, kd_indices[position - 1]] # position is 1-based
```

3.3 R Pipeline

3.3.1 Overview

The R workflow is exposed through the `tensoromics.functions.R`, which interfaces via Rcpp in `tensoromics.functions.cpp` with Fortran. module. The functions provide a high-level interface that mirrors the Fortran pipeline.

3.3.2 Data Management

1. Data Reading and processing

- **Read Gene IDs from TSV File (R)**

The function reads gene identifiers from a tab-separated value (TSV) file, extracting specific gene IDs from a designated column while skipping header rows. This function is typically used as the first step in processing gene expression data to obtain the complete list of gene identifiers. Call the `read_gene_ids_from_tsv_file` function with the following arguments:

- `filename`: Character string of the filename
- `ngenes`: Number of gene IDs to read
- `gene_len`: Maximum length of each gene ID
- **(Optional)** `n_header_rows`: Number of header rows to skip
- **(Optional)** `gene_col`: Column index (1-based) of gene IDs in the file

Return list:

- `gene_ids`: Character vector of gene IDs read from the file
- `ierr`: Integer error code (0 if successful)

Listing 190: R Read Gene IDs from TSV File

```
read_gene_ids_from_tsv_file(filename, ngenes, gene_len,  
    n_header_rows = 1, gene_col = 1)
```

- **Read Expression Vectors (R)**

The function reads gene expression data from multiple tabular files (CSV/TSV format), extracting expression values for specified genes across multiple samples. This function consolidates expression data from multiple files into a single matrix. Call the `read_expression_vectors` function with the following arguments:

- `file_list`: Character vector of filenames
- `gene_ids`: Character vector of gene IDs to match
- `n_header_rows`: Number of header rows to skip in each file
- `gene_col`: Column index (1-based) of gene IDs in each file

- **value_cols**: Vector of column indices (1-based) of expression values in each file
- **n_samples**: Total number of samples (rows) to read across all files
- **(Optional) delimiter**: Delimiter used in the files (default: tab)

Return list:

- **expression_vectors**: Numeric matrix of dimensions ($n_samples \times n_genes$)
- **ierr**: Integer error code (0 if successful)

Listing 191: R Read Expression Vectors

```
read_expression_vectors(file_list, gene_ids, n_header_rows, gene_col,
                        value_cols, n_samples, delimiter = "\t")
```

• **Read OrthoFinder File (R)**

The function processes an OrthoFinder output file to map genes to their respective orthologous families. This function establishes the gene-family relationships essential for downstream orthology-based analyses. Call the `read_orthofinder_file` function with the following arguments:

- **filename**: Character string of the filename
- **gene_ids**: Character vector of gene IDs to match
- **n_families**: Number of unique gene families expected
- **family_len**: Maximum length of each family ID

Return list:

- **family_ids**: Character vector of family IDs read from the file
- **gene_to_fam**: Integer vector mapping each gene to its family index (0 = unassigned)
- **ierr**: Integer error code (0 if successful)

Note: If genes are not found in the gene IDs list a message will be printed but NO error code is thrown.

Listing 192: R Read OrthoFinder File

```
read_orthofinder_file(filename, gene_ids, n_families, family_len)
```

Complete R Data Loading Workflow Example:

Listing 193: R Complete Data Loading Workflow

```
# Step 1: Load gene identifiers
gene_result <- read_gene_ids_from_tsv_file(
  filename = "data/expression_data.tsv",
  ngenes = 50000,
  gene_len = 32,
  n_header_rows = 1,
  gene_col = 1
)
gene_ids <- gene_result$gene_ids

# Step 2: Load expression data from multiple samples
samples <- c("sample1.tsv", "sample2.tsv",
            "sample3.tsv", "sample4.tsv")

expression_result <- read_expression_vectors(
  file_list = samples,
  gene_ids = gene_ids,
  n_header_rows = 1,
```

```

    gene_col = 1,
    value_cols = 2,
    n_samples = 4
)
expression_vectors <- expression_result$expression_vectors

# Step 3: Load orthology information
family_result <- read_orthofinder_file(
  filename = "data/Orthogroups.tsv",
  gene_ids = gene_ids,
  n_families = 15000,
  family_len = 32
)
family_ids <- family_result$family_ids
gene_to_fam <- family_result$gene_to_fam

```

2. Data Filtering and Validation

• Filter Unassigned Genes (R)

The function filters out genes that are not assigned to any family (where `gene_to_fam = 0`). This is essential for cleaning data before orthology-based analyses, ensuring only genes with valid family assignments are processed. Call the `filter_unassigned_genes` function with the following arguments:

- `gene_ids`: Character vector of gene IDs
- `expression_vectors`: Numeric matrix of expression values ($n_samples \times n_genes$)
- `gene_to_fam`: Integer vector mapping each gene to its family index (0 = unassigned)

Return list:

- `gene_ids`: Filtered character vector of gene IDs
- `expression_vectors`: Filtered numeric matrix of expression values
- `gene_to_fam`: Filtered integer vector mapping each gene to its family index
- `n_genes_kept`: Number of genes kept after filtering

Listing 194: R Filter Unassigned Genes

```
filter_unassigned_genes(gene_ids, expression_vectors, gene_to_fam)
```

• Validate Gene to Family Mapping (R)

The function validates the gene-to-family mapping array, ensuring all family indices are within valid range and properly assigned. This prevents downstream errors from invalid family indices. Call the `validate_gene_to_family_mapping` function with the following arguments:

- `gene_to_fam`: Integer vector mapping each gene to its family index (0 = unassigned)
- `n_genes`: Number of genes
- `n_families`: Number of gene families

Listing 195: R Validate Gene to Family Mapping

```
validate_gene_to_family_mapping(gene_to_fam, n_genes, n_families)
```

• Validate Expression Data (R)

The function validates expression data, checking for NaN/Inf values and optionally verifying non-negative values. This ensures data quality before computational analyses. Call the `validate_expression_data` function with the following arguments:

- `expression_vectors`: Numeric matrix of expression values ($n_samples \times n_genes$)

- `n_genes`: Number of genes
- `n_samples`: Number of samples
- **(Optional)** `check_non_negative`: Logical flag to check for non-negative values (default: TRUE)

Listing 196: R Validate Expression Data

```
validate_expression_data(expression_vectors, n_genes, n_samples,
  check_non_negative = TRUE)
```

- **Validate Family Centroids (R)**

The function validates family centroids data, checking for NaN/Inf values and proper data structure. Centroids represent the central tendency of each gene family’s expression profile. Call the `validate_family_centroids` function with the following arguments:

- `family_centroids`: Numeric matrix of family centroids ($n_samples \times n_families$)
- `n_families`: Number of gene families
- `n_samples`: Number of samples

Listing 197: R Validate Family Centroids

```
validate_family_centroids(family_centroids, n_families, n_samples)
```

- **Validate Shift Vectors (R)**

The function validates shift vectors, ensuring data types are correct and structure matches expected patterns. Shift vectors represent the deviation of each gene from its family centroid. Call the `validate_shift_vectors` function with the following arguments:

- `shift_vectors`: Numeric matrix of shift vectors ($2 \times n_samples \times n_genes$)
- `expression_vectors`: Numeric matrix of expression values ($n_samples \times n_genes$)
- `family_centroids`: Numeric matrix of family centroids ($n_samples \times n_families$)
- `gene_to_fam`: Integer vector mapping each gene to its family index
- `n_samples`: Number of samples

Listing 198: R Validate Shift Vectors

```
validate_shift_vectors(shift_vectors, expression_vectors,
  family_centroids, gene_to_fam, n_samples)
```

- **Validate String Array Uniqueness (R)**

The function validates that all strings in an array are unique, using a hashset internally. Note: This may increase memory usage temporarily for large datasets. Call the `validate_string_array_uniqueness` function with the following arguments:

- `string_arr`: Character vector of gene IDs
- `n_strings`: Number of genes

Listing 199: R Validate String Array Uniqueness

```
validate_string_array_uniqueness(string_arr, n_strings)
```

- **Validate All Data (R)**

The function performs comprehensive validation of all data components in one call. This function executes all individual validations sequentially for complete data quality assurance. Call the `validate_all_data` function with the following arguments:

- `n_genes`: Number of genes
- `n_families`: Number of gene families

- `n_samples`: Number of samples
- `gene_ids`: Character vector of gene IDs
- `gene_family_ids`: Character vector of gene family IDs
- `gene_to_fam`: Integer vector mapping each gene to its family index (0 = unassigned)
- `expression_vectors`: Numeric matrix of expression values ($n_samples \times n_genes$)
- `family_centroids`: Numeric matrix of family centroids ($n_samples \times n_families$)
- `shift_vectors`: Numeric matrix of shift vectors ($2*n_samples \times n_genes$)

Listing 200: R Validate All Data

```
validate_all_data(n_genes, n_families, n_samples, gene_ids,
  gene_family_ids, gene_to_fam, expression_vectors,
  family_centroids, shift_vectors)
```

R Data Filtering and Validation Workflow Example:

Listing 201: R Data Filtering and Validation Workflow

```
# After loading data, filter unassigned genes
filter_result <- filter_unassigned_genes(gene_ids, expression_vectors,
  gene_to_fam)

gene_ids <- filter_result$gene_ids
expression_vectors <- filter_result$expression_vectors
gene_to_fam <- filter_result$gene_to_fam
n_genes_kept <- filter_result$n_genes_kept

# Run comprehensive validations
validate_string_array_uniqueness(gene_ids, n_genes_kept)
validate_string_array_uniqueness(family_ids, n_families)
validate_expression_data(expression_vectors, n_genes_kept,
  n_samples, check_non_negative = TRUE)

validate_gene_to_family_mapping(gene_to_fam, n_genes_kept, n_families)

# For complete analysis with centroids and shift vectors
validate_all_data(
  n_genes_kept, n_families, n_samples,
  gene_ids, family_ids, gene_to_fam,
  expression_vectors, family_centroids, shift_vectors
)
```

3.3.3 Data Integration

The R module provides comprehensive routines for data integration, based on Jensen-Shannon Divergence. This is very useful for cross-study data.

Key Features:

- **Calculate Jensen-Shannon Compatibility Score:** Score to evaluate how compatible two studies are.
- **Perform JSD Permutation Test:** Random sampling of the input data and recalculation of the JS score to test stability of the compatibility.
- **Family-Wise Score Contribution:** Calculates the impact of selected families on the calculated score.

1. Compute Gene Means

Compute per-gene mean expression values across replicates, handling NaN values appropriately.

Arguments:

- `expr`: numeric matrix of expression values (`replicates * genes`).

Returns:

- `numeric`: vector of per-gene mean expression values.

Listing 202: Per-Gene Mean Expression

```
means <- tox_compute_gene_means(expr)
```

2. Compute Signed Residuals

Compute signed residuals for each gene by subtracting the per-gene mean expression from each replicate.

Arguments:

- `expr`: numeric matrix (`replicates * genes`).
- `means`: numeric vector of per-gene mean expression values.

Returns:

- `numeric`: matrix of signed residuals (`replicates * genes`).

Listing 203: Signed Residual Computation

```
resid <- tox_compute_residuals(expr, means)
```

3. Pool Mean Expression Values

Pool per-gene mean expression values from two studies and compute reference points for neighborhood construction.

Arguments:

- `mean_S1`: numeric vector of mean expression values for study 1.
- `mean_S2`: numeric vector of mean expression values for study 2.
- `n_points`: integer number of reference points to compute.

Returns:

- `list`: { `n_pool`: integer number of valid pooled means,
 `x_star`: numeric vector of reference points }

Listing 204: Pooling Mean Expression Values

```
out <- tox_pool_means(mean_S1, mean_S2, n_points)

n_pool <- out$n_pool
x_star <- out$x_star
```

4. Pool Mean Expression Values (Expert)

Pool mean-expression values using pre-pooled and pre-sorted arrays. This expert interface bypasses internal pooling and sorting, allowing direct computation of reference points.

Arguments:

- `pooled_means`: numeric vector of pooled mean expression values.
- `pooled_perm`: integer vector giving the sorting permutation for `pooled_means`.
- `n_points`: integer number of reference points to compute.

Returns:

- `list`: { `n_pool`: integer number of valid pooled means,
 `x_star`: numeric vector of reference points }

Listing 205: Pooling Means (Expert Interface)

```
out <- tox_pool_means_expert(
  pooled_means,
  pooled_perm,
  n_points
)

n_pool <- out$n_pool
x_star <- out$x_star
```

5. Construct Neighborhoods

Construct neighborhood-based residual sets by selecting, for each reference point, the closest genes in mean-expression space. The neighborhood size is determined automatically unless a desired size is specified.

Arguments:

- `x_star`: numeric vector of reference mean-expression points.
- `n_pool`: integer number of pooled mean-expression values.
- `mean_S`: numeric vector of per-gene mean expression values.
- `resid_S`: numeric matrix of signed residuals (`replicates * genes`).
- `desired_n_neighbors`: integer desired neighborhood size (default 0).

Returns:

- `list`: { `neighborhood_residuals`: numeric array,
`neighborhood_indices`: integer matrix of gene indices }

Listing 206: Neighborhood Construction

```
out <- tox_construct_neighborhoods(
  x_star,
  n_pool,
  mean_S,
  resid_S,
  desired_n_neighbors = 0
)

neigh_res <- out$neighborhood_residuals
neigh_idx <- out$neighborhood_indices
```

6. Determine Shared Residual Range (Expert)

Compute the shared residual range R from a precomputed residual pool and its sorting permutation. This expert interface assumes that residuals have already been flattened, cleaned, and sorted externally.

Arguments:

- `residual_pool`: numeric vector of absolute residuals (NaNs removed).
- `residual_pool_perm`: integer vector giving the permutation that sorts `residual_pool`.
- `residual_range_quantile`: numeric scalar specifying the percentile (default 95.0).

Returns:

- `numeric`: shared residual range R .

Listing 207: Shared Residual Range (Expert)

```
R <- tox_determine_shared_residual_range_expert(
  residual_pool,
  residual_pool_perm,
```

```

    residual_range_quantile = 95.0
)

```

7. Determine Shared Residual Range

Compute the shared residual range R from two residual matrices. Both matrices are flattened, pooled, and the percentile threshold is applied to determine the symmetric range $[-R, R]$.

Arguments:

- `neighborhood_residuals_S1`: numeric matrix (`n_residuals * n_neighbors`).
- `neighborhood_residuals_S2`: numeric matrix (`n_residuals * n_neighbors`).
- `residual_range_quantile`: numeric scalar (default 95.0).

Returns:

- `numeric`: shared residual range R .

Listing 208: Shared Residual Range

```

R <- tox_determine_shared_residual_range(
  neighborhood_residuals_S1,
  neighborhood_residuals_S2,
  residual_range_quantile = 95.0
)

```

8. Build Residual Histograms

Build histogram bin counts and normalized probability mass functions (PMFs) for each neighbor. Residuals are binned using a shared residual range and a fixed number of bins.

Arguments:

- `neighborhood_residuals`: numeric matrix (`n_residuals * n_neighbors`).
- `shared_residual_range`: numeric scalar R .
- `n_bins`: integer number of histogram bins.

Returns:

- `list`: { `counts`: integer matrix (`n_neighbors * n_bins`),
`pmf`: numeric matrix (`n_neighbors * n_bins`),
`included_n_residuals`: integer vector (`n_neighbors`) }

Listing 209: Residual Histogram Construction

```

out <- tox_build_residual_histograms(
  neighborhood_residuals,
  shared_residual_range,
  n_bins
)

counts <- out$counts
pmf <- out$pmf
included <- out$included_n_residuals

```

9. Build Residual Histograms (Filtered)

Build histogram bin counts and normalized probability mass functions (PMFs) for each neighbor, using a shared residual range and a fixed number of bins. A logical or integer mask selects which neighbors contribute to each reference point's histogram.

Arguments:

- `neighborhood_residuals`: numeric matrix (`n_residuals * n_neighbors`).
- `shared_residual_range`: numeric scalar R .
- `n_bins`: integer number of histogram bins.

- `neighbor_mask`: logical or integer matrix (`n_neighbors * n_points`), where non-zero indicates inclusion.

Returns:

- **list**: { `counts`: integer matrix (`n_neighbors * n_bins`),
`pmf`: numeric matrix (`n_neighbors * n_bins`),
`included_n_residuals`: integer vector (`n_neighbors`) }

Listing 210: Filtered Residual Histogram Construction

```
out <- tox_build_residual_histograms_filtered(
  neighborhood_residuals,
  shared_residual_range,
  n_bins,
  neighbor_mask
)

counts <- out$counts
pmf <- out$pmf
included <- out$included_n_residuals
```

10. Compute Divergence per Reference Point

Compute the Jensen–Shannon divergence for each neighbor using two PMF matrices.

Arguments:

- `pmf_S1`: numeric matrix (`n_neighbors * n_bins`).
- `pmf_S2`: numeric matrix (`n_neighbors * n_bins`).

Returns:

- **numeric**: vector of length `n_neighbors` containing JSD values.

Listing 211: Per-Neighbor JSD

```
jsd <- tox_compute_divergence_per_reference_point(
  pmf_S1,
  pmf_S2
)
```

11. Compute Weighted Global Divergence

Compute the weighted global Jensen–Shannon divergence from per-neighbor divergences and included residual counts.

Arguments:

- `js_divergences`: numeric vector of length `n_neighbors`.
- `included_n_residuals_S1`: integer vector of length `n_neighbors`.
- `included_n_residuals_S2`: integer vector of length `n_neighbors`.

Returns:

- **list**: { `global_js_divergence`: numeric scalar,
`weights`: numeric vector (`n_neighbors`) }

Listing 212: Weighted Global JSD

```
out <- tox_compute_weighted_global_divergence(
  js_divergences,
  included_n_residuals_S1,
  included_n_residuals_S2
)
```

```
global_jsd <- out$global_js_divergence
weights <- out$weights
```

12. Permutation Test for Global JSD

Estimate how likely the observed global Jensen–Shannon divergence is to occur by chance under the null hypothesis that both studies are exchangeable. Residuals for each reference point are concatenated, shuffled, and reassigned to generate null-distribution JSD values.

Arguments:

- `neighborhood_residuals_S1`: numeric vector or array of residuals for study 1.
- `neighborhood_residuals_S2`: numeric vector or array of residuals for study 2.
- `global_jsd_observed`: numeric scalar, the observed global weighted JSD.
- `n_bins`: integer number of histogram bins.
- `shared_residual_range`: numeric scalar specifying the shared residual range.
- `n_permutations`: integer number of permutations.
- `random_seed`: integer seed for reproducibility.

Returns:

- `list`: { `jsd_null`: numeric vector of length `n_permutations`,
`p_value`: numeric scalar }

Listing 213: Permutation Test for Global JSD

```
out <- tox_gjct_permutation_test(
  neighborhood_residuals_S1,
  neighborhood_residuals_S2,
  global_jsd_observed,
  n_bins,
  shared_residual_range,
  n_permutations,
  random_seed = 42
)

jsd_null <- out$jsd_null
p_value <- out$p_value
```

13. Permutation Test for Global JSD (Filtered)

Estimate how likely the observed global Jensen–Shannon divergence is to occur by chance under the null hypothesis that both studies are exchangeable. Neighbor masks allow excluding specific neighbors for each reference point in each study during histogram construction.

Arguments:

- `neighborhood_residuals_S1`: numeric array (`n_reps * n_neighbors * n_points`) for study 1.
- `neighborhood_residuals_S2`: numeric array (`n_reps * n_neighbors * n_points`) for study 2.
- `global_jsd_observed`: numeric scalar, the observed global weighted JSD.
- `n_bins`: integer number of histogram bins.
- `shared_residual_range`: numeric scalar specifying the shared residual range.
- `n_permutations`: integer number of permutations.
- `neighbor_mask_S1`: logical or integer matrix (`n_neighbors * n_points`) for study 1.
- `neighbor_mask_S2`: logical or integer matrix (`n_neighbors * n_points`) for study 2.
- `random_seed`: integer seed for reproducibility.

Returns:

- `list`: { `jsd_null`: numeric vector of length `n_permutations`,
`p_value`: numeric scalar }

Listing 214: Filtered Permutation Test for Global JSD

```
out <- tox_gjct_permutation_test_filtered(
  neighborhood_residuals_S1,
  neighborhood_residuals_S2,
  global_jsd_observed,
  n_bins,
  shared_residual_range,
  n_permutations,
  neighbor_mask_S1,
  neighbor_mask_S2,
  random_seed = 42
)

jsd_null <- out$jsd_null
p_value <- out$p_value
```

14. Compute Family-Level JSD

Compute the family-level Jensen–Shannon divergence for a given paralog family. Residuals are filtered by family membership, histograms are constructed, per-point JSD values are computed, and a weighted global divergence is returned.

Arguments:

- `family_idx`: integer index of the gene family.
- `gene_to_family_S1`: integer vector of length `n_genes_S1` mapping genes in study 1 to families.
- `gene_to_family_S2`: integer vector of length `n_genes_S2` mapping genes in study 2 to families.
- `neighborhood_residuals_S1`: numeric array (`n_reps_S1 * n_neighbors * n_points`).
- `neighborhood_residuals_S2`: numeric array (`n_reps_S2 * n_neighbors * n_points`).
- `neighborhood_genes_S1`: integer matrix (`n_neighbors * n_points`).
- `neighborhood_genes_S2`: integer matrix (`n_neighbors * n_points`).
- `n_bins`: integer number of histogram bins.
- `shared_residual_range`: numeric scalar specifying the shared residual range.

Returns:

- `list`: { `js_divergences`: numeric vector (`n_points`),
`included_n_reps_S1`: integer vector (`n_points`),
`included_n_reps_S2`: integer vector (`n_points`),
`total_included_n_reps`: integer scalar,
`global_js_divergence`: numeric scalar,
`weights`: numeric vector (`n_points`),
`ierr`: integer error code }

Listing 215: Family-Level JSD

```
out <- tox_fjct_compute_jsd(
  family_idx,
  gene_to_family_S1,
  gene_to_family_S2,
  neighborhood_residuals_S1,
  neighborhood_residuals_S2,
  neighborhood_genes_S1,
  neighborhood_genes_S2,
  n_bins,
  shared_residual_range
)
```

```
jsd      <- out$js_divergences
global_jsd <- out$global_js_divergence
weights  <- out$weights
```

15. Compute Family-Level JSD (Expert)

Compute family-level Jensen–Shannon divergence using precomputed neighbor masks. This expert interface bypasses family-mapping logic and directly applies masks to select neighbors for each reference point.

Arguments:

- `neighborhood_residuals_S1`: numeric array (`n_reps_S1 * n_neighbors * n_points`).
- `neighborhood_residuals_S2`: numeric array (`n_reps_S2 * n_neighbors * n_points`).
- `neighbor_mask_S1`: logical or integer matrix (`n_neighbors * n_points`).
- `neighbor_mask_S2`: logical or integer matrix (`n_neighbors * n_points`).
- `n_bins`: integer number of histogram bins.
- `shared_residual_range`: numeric scalar specifying the shared residual range.

Returns:

- `list`: { `js_divergences`,
`included_n_reps_S1`,
`included_n_reps_S2`,
`total_included_n_reps`,
`global_js_divergence`,
`weights`,
`pmf_S1`,
`pmf_S2`,
`tmp_counts`,
`ierr` }

Listing 216: Family-Level JSD (Expert Interface)

```
out <- tox_fjct_compute_jsd_expert(
  neighborhood_residuals_S1,
  neighborhood_residuals_S2,
  neighbor_mask_S1,
  neighbor_mask_S2,
  n_bins,
  shared_residual_range
)

global_jsd <- out$global_js_divergence
weights    <- out$weights
```

16. Compute Contribution Scores

Compute per-family contribution scores from global JSD values and the number of included residuals per family. Contribution scores quantify how strongly each family contributes to the overall divergence.

Arguments:

- `global_js_divergences`: numeric vector of length `k_families`.
- `total_included_n_reps_per_f`: integer vector of length `k_families`.

Returns:

- `list`: { `support_weights`: numeric vector,

```
contribution_scores: numeric vector,
ierr: integer error code }
```

Listing 217: Contribution Score Computation

```
out <- tox_fjct_compute_contribution_scores(
  global_js_divergences,
  total_included_n_reps_per_f
)

scores <- out$contribution_scores
weights <- out$support_weights
```

3.3.4 Data Persistence

1. R Interface for Array Serialization and Deserialization

The R interface provides bindings to the Fortran serialization and deserialization routines, enabling efficient handling of multi-dimensional arrays. This interface supports R arrays, matrices, and vectors with automatic type conversion to compatible Fortran data types.

(a) General Architecture

The R package provides ten main functions for array handling:

- `tox_serialize_int_array(array, filename)` - Serialize N-dimensional integer arrays
- `tox_deserialize_int_array(filename)` - Deserialize N-dimensional integer arrays
- `tox_serialize_real_array(array, filename)` - Serialize N-dimensional real arrays
- `tox_deserialize_real_array(filename)` - Deserialize N-dimensional real arrays
- `tox_serialize_char_array(array, filename)` - Serialize N-dimensional character arrays
- `tox_deserialize_char_array(filename)` - Deserialize N-dimensional character arrays
- `tox_serialize_logical_array(array, filename)` - Serialize N-dimensional logical array
- `tox_deserialize_logical_array(filename)` - Deserialize N-dimensional logical array
- `tox_serialize_complex_array(array, filename)` - Serialize N-dimensional complex array
- `tox_deserialize_complex_array(filename)` - Deserialize N-dimensional complex array

(b) Array Functions

Serialization Functions:

- `array` (input): R array, matrix, or vector to be serialized

Type:

- Integer
- Numeric (double precision)
- Characters
- Logical
- Complex

Dimensions: Theoretically are N dimensions supported, but typical use cases should not exceed 5 dimensions

- `filename` (input): Name of the binary file to create

Type: character

Listing 218: R Array Serialization Example

```
tox_serialize_int_array(array, filename)
tox_serialize_real_array(array, filename)
tox_serialize_char_array(array, filename)
```

Deserialization Functions:

- **filename** (input): Name of the binary file to read
Type: character
- **Return Value:** Deserialized R array with original dimensions
Type: array, matrix, or vector
Data Type: Same as original

Listing 219: R Array Deserialization Example

```
restored_int <- tox_deserialize_int_array(filename)
restored_real <- tox_deserialize_real_array(filename)
restored_char <- tox_deserialize_char_array(filename)
```

(c) Implementation Details

- The R interface automatically converts between R data types and Fortran-compatible formats
- Integer vectors are converted to 32-bit integers
- Numeric vectors are converted to 64-bit floating point
- Character arrays handle variable-length strings with automatic padding
- All array dimensions are preserved during serialization/deserialization
- The interface handles both regular arrays and matrices seamlessly

Usage Example

Listing 220: Full example of serialization and deserialization in R

```
tox_serialize_int_array(arr1, fn("int1d.bin"))
restored <- tox_deserialize_int_array(fn("int1d.bin"))

# Real array example
arr2 <- matrix(runif(12), nrow = 3, ncol = 4)
tox_serialize_real_array(arr2, fn("real2d.bin"))
restored <- tox_deserialize_real_array(fn("real2d.bin"))

# Character array example
arr3 <- c("GENE1", "BRCA1", "TP53", "EGFR")
tox_serialize_char_array(arr3, fn("char1d.bin"))
restored <- tox_deserialize_char_array(fn("char1d.bin"))
```

(d) Compatibility Notes

- Requires R with appropriate Fortran bindings
- Supports standard R data types (integer, numeric, character, logical, complex)
- Preserves array attributes and dimensions during serialization
- Character arrays maintain content but may have padding removed upon deserialization

2. Create Zip Archive (Low-Level) (R)

The low-level function creates a zip archive from keys and filenames, directly calling the Fortran backend. Use this function for custom data serialization scenarios or when you need fine-grained

control over archive creation. Call the `create_zip_archive` function with the following arguments:

- `zip_filename`: Name of the zip file to create
- `keys`: Vector of keys for the manifest (identifiers for each file)
- `filenames`: Vector of filenames to include in archive

Listing 221: R Create Zip Archive (Low-Level)

```
create_zip_archive(zip_filename, keys, filenames)
```

3. Extract Zip Archive (R)

The function extracts a zip archive created by `create_zip_archive` and returns a list mapping data keys to extracted filenames. This is useful for accessing individual files from custom archives. Call the `extract_zip_archive` function with the following arguments:

- `zip_filename`: Path to the zip file to extract

Returns:

- List mapping data keys to extracted temporary filenames

Listing 222: R Extract Zip Archive

```
extract_zip_archive(zip_filename)
```

4. Save TOX Data (High-Level) (R)

The high-level function saves TOX data to a zip archive, handling validation, serialization, and archive creation for standard-conform tox gene data. This function automatically manages temporary files and cleanup. Call the `save_tox_data` function with the following arguments:

- `zip_filename`: Name of the zip file to create
- **(Optional)** `gene_ids`: Character vector of gene IDs
- **(Optional)** `gene_ids_name`: Filename for gene IDs in the archive
- **(Optional)** `expression_vectors`: Numeric matrix of expression values ($n_samples \times n_genes$)
- **(Optional)** `expression_vectors_name`: Filename for expression vectors in the archive
- **(Optional)** `gene_to_fam`: Integer vector mapping each gene to its family index (0 = unsigned)
- **(Optional)** `gene_to_fam_name`: Filename for gene to family mapping in the archive
- **(Optional)** `family_ids`: Character vector of family IDs
- **(Optional)** `family_ids_name`: Filename for family IDs in the archive
- **(Optional)** `family_centroids`: Numeric matrix of family centroids ($n_samples \times n_families$)
- **(Optional)** `family_centroids_name`: Filename for family centroids in the archive
- **(Optional)** `shift_vectors`: Numeric matrix of shift vectors ($2*n_samples \times n_genes$)
- **(Optional)** `shift_vectors_name`: Filename for shift vectors in the archive

Listing 223: R Save TOX Data

```
save_tox_data(zip_filename,
              gene_ids = NULL, gene_ids_name = NULL,
              expression_vectors = NULL, expression_vectors_name = NULL,
              gene_to_fam = NULL, gene_to_fam_name = NULL,
              family_ids = NULL, family_ids_name = NULL,
              family_centroids = NULL, family_centroids_name = NULL,
              shift_vectors = NULL, shift_vectors_name = NULL)
```

5. Read TOX Data (R)

The function reads data from a zip archive created by `save_tox_data`, extracting and deserializing the requested data components for standard-conform tox gene data. This function automatically handles file extraction and cleanup. Call the `read_tox_data` function with the following arguments:

- `zip_filename`: Name of the zip file to read from
- **(Optional)** `gene_ids`: If not NULL, will attempt to read gene IDs from archive
- **(Optional)** `expression_vectors`: If not NULL, will attempt to read expression vectors from archive
- **(Optional)** `gene_to_fam`: If not NULL, will attempt to read gene to family mapping from archive
- **(Optional)** `family_ids`: If not NULL, will attempt to read family IDs from archive
- **(Optional)** `family_centroids`: If not NULL, will attempt to read family centroids from archive
- **(Optional)** `shift_vectors`: If not NULL, will attempt to read shift vectors from archive

Return list:

- `gene_ids`: Loaded gene IDs vector (if requested)
- `expression_vectors`: Loaded expression vectors matrix (if requested)
- `gene_to_fam`: Loaded gene-to-family mapping vector (if requested)
- `family_ids`: Loaded family IDs vector (if requested)
- `family_centroids`: Loaded family centroids matrix (if requested)
- `shift_vectors`: Loaded shift vectors matrix (if requested)

Listing 224: R Read TOX Data

```
read_tox_data(zip_filename,
              gene_ids = NULL,
              expression_vectors = NULL,
              gene_to_fam = NULL,
              family_ids = NULL,
              family_centroids = NULL,
              shift_vectors = NULL)
```

R Data Archiving Workflow Examples:

Listing 225: R Standard TOX Data Archiving

```
# Save complete TOX dataset
save_tox_data(
  zip_filename = "complete_tox_dataset.zip",
  gene_ids = gene_ids,
  gene_ids_name = "gene_ids.bin",
  expression_vectors = expression_vectors,
  expression_vectors_name = "expression.bin",
  gene_to_fam = gene_to_fam,
  gene_to_fam_name = "gene_to_fam.bin",
  family_ids = family_ids,
  family_ids_name = "family_ids.bin",
  family_centroids = family_centroids,
  family_centroids_name = "centroids.bin",
  shift_vectors = shift_vectors,
  shift_vectors_name = "shift_vectors.bin"
)
```

```

# Save partial dataset
save_tox_data(
  zip_filename = "basic_data.zip",
  gene_ids = gene_ids,
  gene_ids_name = "gene_ids.bin",
  expression_vectors = expression_vectors,
  expression_vectors_name = "expression.bin"
)

# Read data back with selective loading
loaded_data <- read_tox_data(
  zip_filename = "complete_tox_dataset.zip",
  gene_ids = TRUE,
  expression_vectors = TRUE,
  gene_to_fam = TRUE
)

```

Listing 226: R Custom Archive Creation and Extraction

```

# Create custom archive with non-standard data
create_zip_archive(
  zip_filename = "custom_data.zip",
  keys = c("3d_tensor", "results"),
  filenames = c("tensor_data.bin", "analysis_results.bin")
)

# Extract and access custom archive
file_mapping <- extract_zip_archive("custom_data.zip")
cat("Files in archive:\n")
print(file_mapping)
# Output: $3d_tensor = "tensor_data.bin", $results = "analysis_results.
          bin"

```

Listing 227: R Mixed Standard and Custom Data Archiving

```

# Serialize custom arrays
tox_serialize_real_array(custom_array, "custom_data.bin")
tox_serialize_int_array(another_array, "another_data.bin")

# Create mixed archive with both standard and custom data
create_zip_archive(
  zip_filename = "mixed_archive.zip",
  keys = c("standard_expression", "custom_feature",
            "analysis_results"),
  filenames = c("expression.bin", "custom_data.bin",
                "another_data.bin")
)

# Later, extract and process
mapping <- extract_zip_archive("mixed_archive.zip")
expression_data <- tox_deserialize_real_array(
  mapping[["standard_expression"]])
custom_features <- tox_deserialize_real_array(

```

```
mapping[["custom_feature"]])
results <- tox_deserialize_int_array(
  mapping[["analysis_results"]])
```

Important Notes for R Archive Functions:

- All archive functions automatically handle temporary file cleanup
- The `save_tox_data` function validates data dimensions before serialization
- Both `save_tox_data` and `read_tox_data` use the standard TOX data keys: "gene_ids", "expression", "gene_to_family", "family_ids", "family_centroids", "shift_vectors"
- Archives created with R functions are compatible with Python and Fortran implementations
- Error codes are automatically checked and converted to informative error messages

3.3.5 Data Processing

1. Normalization of Expression Values

The function runs a complete normalization pipeline on gene expression data. It first applies standard deviation normalization, then quantile normalization, followed by replicate averaging, and finally a log2 transformation. The result is returned in an array, which contains the fully normalized expression values. Call the `tox_normalization_pipeline` function with the following arguments:

- `input_matrix`: Numeric expression vector matrix with size `genes × tissues`
- `group_s`: Integer vector: start column index for each replicate group (1-based)
- `group_c`: Integer vector: number of columns per replicate group

Return:

- Numeric matrix: $\log_2(x+1)$ normalized expression

Listing 228: R Normalization Pipeline

```
tox_normalization_pipeline(input_matrix, group_s, group_c)
```

Alternatively you can calculate each step of the normalization pipeline manually by calling the following functions in sequence:

(a) Normalize by Standard Deviation

The function normalizes each gene's expression vector using `sqrt(mean(x^2))` across tissues (not classical standard deviation). Call the `tox_normalize_by_std_dev` function with the following arguments:

- `input_matrix`: A numeric matrix with genes as rows and tissues as columns

Returns:

- A normalized numeric matrix with the same dimensions and names as the input

Listing 229: R Standard Deviation Normalization

```
tox_normalize_by_std_dev(input_matrix)
```

(b) Quantile Normalization

The function performs quantile normalization of a gene expression matrix (F42-compliant) and computes average expression per rank across tissues. Call the `tox_quantile_normalization` function with the following arguments:

- `input_matrix`: A numeric matrix with genes as rows and tissues as columns

Returns:

- A quantile-normalized numeric matrix with the same dimensions and names as the input

Listing 230: R Quantile Normalization

```
tox_quantile_normalization(input_matrix)
```

(c) Calculate Tissue Averages

The function calculates tissue averages by averaging replicates within each group. For each group of tissue replicates, it computes the average expression per gene. The input matrix is column-major, flattened as a 1D array. Call the `tox.calculate_tissue_averages` function with the following arguments:

- **df**: A data frame or matrix with genes as rows and tissue replicates as columns

Returns:

- A data frame with genes as rows and averaged tissues as columns

Listing 231: R Tissue Averages Calculation

```
tox_calculate_tissue_averages(df)
```

(d) Log2 Transformation

The function applies $\log_2(x + 1)$ transformation to each element of the input matrix. This is computed via $\log(x + 1) / \log(2)$ for compatibility with WebAssembly (WASM). Call the `tox.log2_transformation` function with the following arguments:

- **input_matrix**: A numeric matrix with genes as rows and tissues as columns

Returns:

- A numeric matrix with log2-transformed expression values, preserving the same dimensions and names as the input.

Listing 232: R Log2 Transformation

```
tox_log2_transformation(input_matrix)
```

Additionally there are two helper functions to improve the data quality and prevent calculation errors:

(a) Diagnose Data Quality

The function checks your gene expression matrix for common data quality issues, such as missing values (NA), infinite values, NaNs, zeros, and negative values. It provides a summary of these problems, lists affected genes, and reports basic statistics like minimum, maximum, and mean values. Use this function to quickly assess the quality of your data before running normalization or analysis steps. Call the `tox.diagnose_data_quality` function with the following arguments:

- **input_matrix**: A numeric matrix with genes as rows and tissues as columns
- (Optional) **show_details**: Logical indicating whether to show detailed information (default is TRUE)

Returns:

- A list with diagnostic information about the data quality

Listing 233: R Diagnose Data Quality

```
tox_diagnose_data_quality(input_matrix, show_details)
```

(b) Clean Data for Normalization

The function prepares your gene expression matrix for normalization by handling problematic values. It can remove or impute NA, NaN, and Inf values, filter out genes with all zero values, and optionally convert very small values to zero. You can choose different strategies for

dealing with missing data, ensuring your matrix is clean and ready for downstream analysis. Call the `tox_clean_data_for_normalization` function with the following arguments:

- `df_matrix`: A numeric matrix with genes as rows and tissues as columns
- (Optional) `remove_all_zero_genes`: Logical, whether to remove genes that are all zeros (default = TRUE)
- (Optional) `na_strategy`: Strategy for handling NA values: "remove_genes", "remove_samples", "impute_zero", "impute_mean" (default = "remove_genes")
- (Optional) `min_expression_threshold`: Minimum expression value to consider (values below this become 0 and default is 0)
- (Optional) `convert_small_to_zero`: Logical, whether very small numbers should be converted to zero (default = FALSE)

Returns:

- A cleaned matrix ready for normalization

Listing 234: R Clean Data for Normalization

```
tox_clean_data_for_normalization(df_matrix, remove_all_zero_genes,
                                na_strategy, min_expression_threshold, convert_small_to_zero)
```

2. Calculate Fold Change

If fold change calculation is needed, this function can be used after normalization. The function calculates $\log_2$ fold changes between condition and control columns. For each control-condition pair, this function computes the $\log_2$ fold change by subtracting the expression value in the control column from the corresponding value in the condition column, for all genes. Call the `tox_calculate_fold_changes` function with the following arguments:

- `df`: A data frame with genes as rows and tissues/conditions as columns
- `control_pattern`: A string pattern to detect control columns
- `condition_patterns`: A character vector with patterns to detect condition columns

Returns:

- A data frame with genes as rows and $\log_2$ fold change values as columns

Listing 235: R Fold Change Calculation

```
tox_calculate_fc_by_patterns(df, control_pattern, condition_patterns)
```

3.3.6 Family Based Analysis

1. Calculation of family Centroids

The function computes the centroid (mean expression vector) for each gene family in a gene expression dataset. It iterates over all families, selects the relevant genes for each family and calculates the mean vector for each family. There are two possible modes to use. To use all genes for calculation, select `all` mode. If you want to specify relevant genes for calculation, select `orthologs` mode. It is important that you provide a logical `ortholog_set` array when using `orthologs` mode. Call the `tox_group_centroid` function with the following arguments:

- `expression_vectors`: Expression vector matrix with size `n_axes` $\times$ `n_genes`
- `gene_to_family`: Gene-to-family mapping array with the length of `n_genes` (1-based indexing)
- `n_families`: Total number of gene families
- `mode`: Calculation mode ("all" or "orthologs")

- (Optional) `ortholog_set`: Logical array indicating if a gene is part of the calculation subset (only used in `orthologs` mode)

Return:

- `centroid_matrix`: Matrix of calculated family centroid vectors with size `n_axes × n_families`

Listing 236: R Calculation of Family Centroids

```
tox_group_centroid(expression_vectors, gene_to_family, n_families, mode,
  ortholog_set)
```

(a) **Calculate Mean Vector**

Alternatively you can calculate a single mean vector for a given set of vectors by calling the `mean_vector` function directly with the following arguments:

- `expression_vectors` `real(real64)`: Expression vector matrix with size `n_axes × n_genes`
- `gene_indices` `integer(int32)`: An array containing the column indices of the selected genes in `expression_vectors`

Return:

- `centroid`: Calculated mean vector (centroid) with length of `n_axes`

Listing 237: R Mean Vector Calculation

```
tox_mean_vector(expression_vectors, gene_indices)
```

2. Compute Shift Vector Field

Calculates the shift vector field for each gene expression vector based on its family centroid. The shift vector represents the difference between the gene expression vector and its corresponding family centroid.

Mathematical Definition:

$$\text{shift_vector} = \text{expression_vector} - \text{family_centroid}$$

Arguments:

- `expression_vectors`: Matrix of size `(n_axes_genes × n_vectors)` where each column is a gene expression vector
- `family_centroids`: Matrix of size `(n_axes_centroids × n_families)` where each column is a family centroid vector
- `gene_to_centroid`: Integer vector of length `n_vectors` containing 1-based family indices mapping each gene to its centroid

Returns:

- `shift_vectors`: Matrix of size `(2*n_axes_genes × n_vectors)` containing centroids (rows 1:n_axes_genes) and shift vectors (rows (n_axes_genes+1):(2*n_axes_genes))

Listing 238: R Shift Vector Field Computation

```
shift_vectors <- tox_compute_shift_vector_field(expression_vectors,
  family_centroids, gene_to_centroid)
```

3. Calculate Distance to Centroid

For each gene in the gene expression matrix, the function computes the Euclidean distance to its corresponding family centroid and returns a numeric vector of distances from each gene to its family centroid. Call the `tox_distance_to_centroid` function with the following arguments:

- `genes`: Gene expression matrix with size of `(n_genes × d)`

- **centroids**: Family centroid matrix with size of (**n_families** × **d**)
- **gene_to_fam**: Gene-to-family mapping integer vector with the length of **n_genes** (0 = no family, >0 = family index)
- **d**: Gene expression matrix dimension as integer

Listing 239: R Distance to Centroid Calculation

```
tox_distance_to_centroid(genes, centroids, gene_to_fam, d)
```

(a) **Calculate Euclidean Distance**

Alternatively you can calculate a single Euclidean distance between two vectors manually by calling **tox_euclidean_distance** function with the following arguments:

- **vec1**: First vector
- **vec2**: Second vector

Both vectors must have same length!

Listing 240: R Euclidean distance calculation

```
tox_euclidean_distance(vec1, vec2)
```

4. Detect Gene Outliers

The function first computes family scaling, then calculates relative distance indices (RDI) and identifies outliers based on the specified percentile threshold. It returns a list of components containing the outlier boolean values and intermediate results. Call the **tox_detect_outliers** function with the following arguments:

- **distances**: Numeric vector of distances for each gene to its centroid
- **gene_to_fam**: Gene-to-family mapping integer vector mapping with the same length as the distances vector (0 = no family, >0 = family index)
- **n_families**: Integer number of families
- **(Optional) percentile**: Percentile threshold value for outlier detection (default is 95.0)

Return list:

- **is_outlier**: Logical vector indicating whether each gene is an outlier
- **loess_x**: Numeric vector of x coordinates used as reference points for LOESS smoothing (median distance)
- **loess_y**: Numeric vector of y coordinates used as reference points for LOESS smoothing (standard deviation)
- **loess_n**: Integer vector with numbers of genes in each family used to compute the corresponding median and standard deviation

Listing 241: R Gene Outlier Calculation

```
tox_detect_outliers(distances, gene_to_fam, n_families, percentile)
```

Alternatively you can calculate each step of **tox_detect_outliers manually with the following functions:**

(a) **Compute Family Scaling**

The function calculates a scaling factor for each gene family by computing the median and standard deviation of gene-to-centroid distances within each family, then applies LOESS smoothing to these values. Call the **tox_compute_family_scaling** function with the following arguments:

- **distances**: Numeric vector of distances for each gene to its centroid

- **gene_to_fam**: Gene-to-family mapping integer vector mapping with the same length as the distances vector (0 = no family, >0 = family index)
- **n_families**: Integer number of families

Return dictionary:

- **dscale**: Numeric vector of scaling factors per family
- **loess_x**: Numeric vector of x coordinates used as reference points for LOESS smoothing (median distance)
- **loess_y**: Numeric vector of y coordinates used as reference points for LOESS smoothing (standard deviation)
- **indices_used**: Integer vector of indices of families that have been included in the calculation

Listing 242: R Compute Family Scaling

```
tox_compute_family_scaling(distances, gene_to_fam, n_families)
```

(b) **Compute Family Scaling (Expert Version)**

This function is the expert version of `tox_detect_outliers` and requires pre-allocated work arrays for maximum performance and control. It can be used when you need fine-grained control over memory allocation or you are calling this function many times in a tight loop. Call the `tox_compute_family_scaling_expert` function with the following arguments:

- **distances**: Numeric vector of distances for each gene to its centroid
- **gene_to_fam**: Gene-to-family mapping integer vector mapping with the same length as the distances vector (0 = no family, >0 = family index)
- **n_families**: Integer number of families
- **perm_tmp**: Permutation integer vector for sorting with the same length as distances
- **stack_left_tmp**: Stack integer vector for sorting with the same length as distances
- **stack_right_tmp**: Stack integer vector for sorting with the same length as distances
- **family_distances**: Work integer vector for sorting with the same length as distances

Return list:

- **dscale**: Numeric vector of scaling factors per family
- **loess_x**: Numeric vector of x coordinates used as reference points for LOESS smoothing (median distance)
- **loess_y**: Numeric vector of y coordinates used as reference points for LOESS smoothing (standard deviation)
- **indices_used**: Integer vector of indices of families that have been included in the calculation
- **perm_tmp**: Permutation vector for sorting
- **stack_left_tmp**: Stack vector for sorting
- **stack_right_tmp**: Stack vector for sorting
- **family_distances**: Work vector for sorting

Listing 243: R Compute Family Scaling (Expert Version)

```
tox_compute_family_scaling_expert(distances, gene_to_fam, n_families,
    perm_tmp, stack_left_tmp, stack_right_tmp, family_distances)
```

(c) **Compute Relative Distance Index (RDI)**

The function calculates the Relative Distance Index (RDI) for each gene by dividing its Euclidean distance to the family centroid by the scaling factor of its family. Call the `tox_compute_rdi` function with the following arguments:

- **distances**: Numeric vector of distances for each gene to its centroid
- **gene_to_fam**: Gene-to-family mapping integer vector mapping with the same length as the distances vector (0 = no family, >0 = family index)
- **dscale**: Numeric vector of scaling factors per family

Return list:

- **rdi**: Numeric vector containing the RDI value for each gene
- **sorted_rdi**: Numeric vector containing the RDI values sorted in ascending order

Listing 244: R Compute Relative Distance Index (RDI)

```
tox_compute_rdi(distances, gene_to_fam, dscale)
```

(d) **Identify Outliers**

The function identifies outliers based on the top percentile of RDI values. Call the `tox_identify_outliers` function with the following arguments:

- **rdi**: Numeric vector containing the RDI value for each gene
- **(Optional) percentile**: Percentile threshold number for outlier detection (default is 95.0)

Return list:

- **is_outlier**: Logical vector indicating whether each gene is an outlier
- **threshold**: Actual threshold value used for outlier detection

Listing 245: R Identify Outliers

```
tox_identify_outliers(rdi, percentile)
```

5. **Compute Normalized Tissue Versatility:**

The function calculates a normalized tissue versatility score for selected gene expression vectors. It is based on the angle between each gene expression vector and the space diagonal. Versatility is normalized to [0, 1], where 0 means uniform expression 1 means expression in only one axis. Call the `tox_compute_tissue_versatility` function with the following parameters:

- **expression_vectors**: Numeric matrix array with size `n_axes × n_vectors`
- **vector_selection**: Logical vector with length of `n_vectors` to specify which vectors should be included in the calculation
- **axes_selection**: Logical vector with length of `n_axes` to specify which axes should be included in the calculation

Return list:

- **tissue_versatilities**: Numeric vector containing the calculated tissue versatilities for selected vectors
- **tissue_angles_deg**: Numeric vector containing the calculated angles in degrees for selected vectors
- **n_selected_vectors**: Integer number of vectors processed
- **n_selected_axes**: Integer number of axes used in calculation

Listing 246: R Tissue Versatility Calculation

```
tox_calculate_tissue_versatility(expression_vectors, vector_selection,
                                axis_selection)
```

3.3.7 R RAP Projection and Analysis Functions

The R interface provides user-friendly wrappers for RAP (Relative Axes Projection) analysis, including vector projection, angle calculation, and axis contribution analysis.

R Implementation Features:

- **Automatic Dimension Handling:** Automatically infers dimensions from input data
- **Flexible Mask Input:** Accepts both logical and integer masks for vector/axis selection
- **Smart Defaults:** Provides sensible defaults for optional parameters
- **Comprehensive Validation:** Validates input dimensions, types, and consistency
- **Error Conversion:** Converts Fortran error codes to informative R errors

Validation Checks:

- Input vectors must be numeric and have same dimensions
- Matrices must have consistent dimensions between origins and targets
- Selection masks must have appropriate lengths
- For dimensions > 3 , selected axes must be valid and unique
- All input validation occurs in R before Fortran calls

1. Clock Hand Angle Between Vectors

Calculates the signed rotation angle between two RAP-projected and normalized vectors. Automatically handles 2D, 3D, and higher-dimensional cases.

Arguments:

- **v1:** First normalized vector in RAP space (numeric vector)
- **v2:** Second normalized vector in RAP space (numeric vector)
- **selected_axes_for_signed:** Indices of 3 axes for directionality calculation (default: `c(1, 2, 3)`), ignored if dimension ≤ 3)

Returns:

- Signed angle between vectors in radians $[-\pi, \pi]$ (numeric)

Listing 247: Signed Angle Between Vectors

```
angle <- tox_clock_hand_angle_between_vectors(v1, v2,
  selected_axes_for_signed = c(1, 2, 3))
```

2. Clock Hand Angles for Shift Vectors

Computes signed rotation angles between corresponding pairs of RAP-projected and normalized vectors (e.g., expression centroids and paralogs).

Arguments:

- **origins:** Matrix of origin vectors ($n_dims \times n_vecs$) - e.g., centroids
- **targets:** Matrix of target vectors ($n_dims \times n_vecs$) - e.g., paralogs
- **vecs_selection_mask:** Logical vector indicating which vector pairs to compute (default: all TRUE)
- **selected_axes_for_signed:** Indices of 3 axes for directionality calculation (default: `c(1, 2, 3)`)

Returns:

- Vector of signed angles in radians $[-\pi, \pi]$ (numeric vector)

Listing 248: Batch Angle Calculation for Vector Pairs

```
angles <- tox_clock_hand_angles_for_shift_vectors(
  origins, targets,
  vecs_selection_mask = NULL,
  selected_axes_for_signed = c(1, 2, 3))
```

3. Relative Axis Contributions from Shift Vector

Computes fractional contribution of each axis to a RAP-projected and normalized shift vector. Values range [0,1] and sum to 1.

Arguments:

- **vec**: RAP-projected and normalized shift vector (numeric vector)

Returns:

- Numeric vector of relative axis contributions (sums to 1)

Listing 249: Shift Vector Axis Contributions

```
contributions <- relative_axes_changes_from_shift_vector(shift_vector)
```

4. Relative Axis Contributions from Expression Vector

Computes fractional contribution of each axis to a RAP-projected and normalized expression vector. Values range [0,1] and sum to 1.

Arguments:

- **vec**: RAP-projected and normalized expression vector (numeric vector)

Returns:

- Numeric vector of relative axis contributions (sums to 1)

Listing 250: Expression Vector Axis Contributions

```
contributions <- relative_axes_expression_from_expression_vector(
  expression_vector
)
```

5. Omics Vector RAP Projection

Projects selected expression vectors onto the RAP constructed from a selected set of axes.

Arguments:

- **vecs**: Matrix with expression vectors ($n_axes \times n_vecs$)
- **vecs_selection_mask**: Logical or integer mask for vectors to project
- **axes_selection_mask**: Logical or integer mask for axes to include in RAP

Returns:

- Matrix of projected vectors ($n_selected_axes \times n_selected_vecs$)

Listing 251: Vector RAP Projection

```
projections <- omics_vector_RAP_projection(vecs, vecs_selection_mask,
  axes_selection_mask)
```

6. Omics Field RAP Projection

Projects selected vector fields (e.g., shift vectors) onto the RAP constructed from a selected set of axes. Automatically computes shift vectors as differences between origins and targets.

Arguments:

- **vecs**: Matrix with vector fields ($2 \times n_axes \times n_vecs$) - first n_axes rows: origins, last n_axes rows: targets
- **vecs_selection_mask**: Logical or integer mask for vectors to project
- **axes_selection_mask**: Logical or integer mask for axes to include in RAP

Returns:

- Matrix of projected vectors ($n_selected_axes \times n_selected_vecs$)

Listing 252: Vector Field RAP Projection

```
projections <- omics_field_RAP_projection(vecs, vecs_selection_mask,
  axes_selection_mask)
```

3.3.8 Paralog Analysis

Placeholder

3.3.9 Trajectory Contribution Analysis

Placeholder

3.3.10 R Utils

1. R BST Interface

(a) Build BST Index

R interface for building BST index. Uses R's native 1-based indexing.

Input:

- **x** numeric: Vector of real values

Output:

- **integer**: Permutation indices (1-based R indices)

```
ix <- build_bst_index(x)
```

(b) BST Range Query

Performs range queries on BST index. Returns a list with components.

Input:

- **x** numeric: Original values vector
- **ix** integer: BST index vector from `build_bst_index` (1-based)
- **lo** numeric: Lower bound of range (inclusive)
- **hi** numeric: Upper bound of range (inclusive)

Output:

- **list**: Contains indices and count components

```
result <- bst_range_query(x, ix, lo, hi)
indices <- result$indices # 1-based indices
count <- result$count
```

(c) Get Sorted Value

Retrieves value from sorted position using R 1-based indexing.

Input:

- **x** numeric: Original values vector
- **ix** integer: BST index vector (1-based)
- **position** integer: Position in sorted order (1-based)

Output:

- **numeric**: Value at specified position

```
value <- get_sorted_value(x, ix, position)
```

2. R k-d Tree Interface

(a) Build k-d Tree Index

R interface for k-d tree construction. Input matrix dimensions are rows=dimensions, columns=points.

Input:

- **X** matrix: Matrix of points (nrow = dimensions, ncol = n_points)
- **dim_order** integer: Dimension ordering (1-based, optional)

Output:

- **integer:** Permutation indices (1-based R indices)

```
kd_ix <- build_kd_index(X, dim_order)
```

(b) Build Spherical k-d Tree

R interface for spherical k-d tree construction with separate implementation.

Input:

- **V matrix:** Matrix of vectors (nrow = dimensions, ncol = n_vectors)
- **dim_order integer:** Dimension ordering (1-based, optional)

Output:

- **integer:** Permutation indices (1-based R indices)

```
sphere_ix <- build_spherical_kd(V, dim_order)
```

(c) Get Point from k-d Index

Retrieves point from k-d tree using R matrix indexing.

Input:

- **X matrix:** Original points matrix
- **kd_ix integer:** k-d tree index vector (1-based)
- **position integer:** Position in k-d tree order (1-based)

Output:

- **numeric:** Point values as vector

```
point <- get_kd_point(X, kd_ix, position)
```